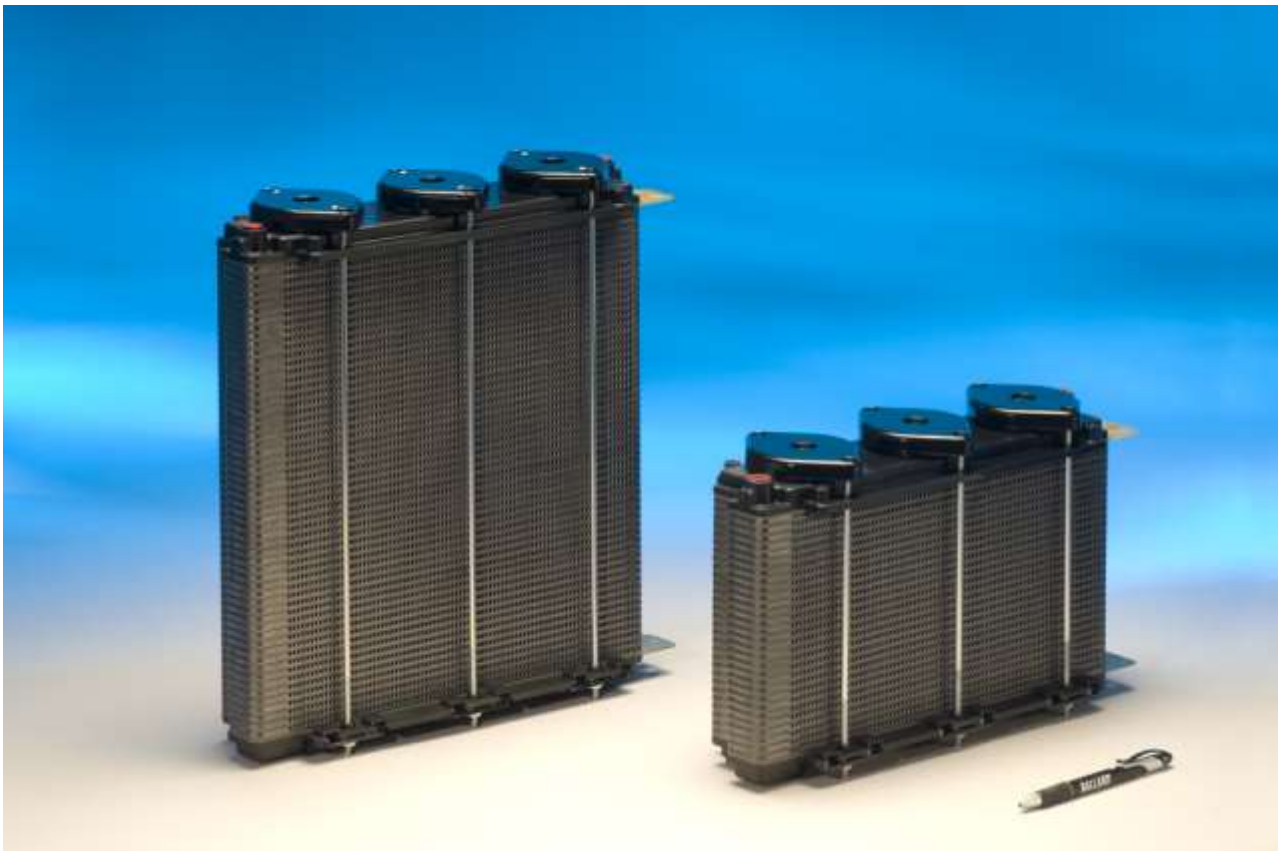
BALLARD

Putting fuel cells to work

FCgen®-1020ACS Fuel Cell Stack
FCvelocity®-1020ACS Fuel Cell Stack



Product Manual and Integration, Handling, Transportation and Storage Guide

Trademark: FCgen®-1020ACS/FCvelocity®-1020ACS

Part Number: 5127049-5127063, 5133354, 5133801-5133812, 5134318

Document Number: MAN5100319-0G

Date: Oct. 31, 2019

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NOTE: YOU MUST READ THIS BEFORE ATTEMPTING ANY PROCEDURE DESCRIBED IN THIS MANUAL

There are inherent risks in the operation of a fuel cell stack that could result in death or bodily injury or loss or damage to property. It is therefore necessary for the customer to take all prudent safety precautions in the installation, use, and maintenance of this device. Those who install, operate, and service the device must be technically qualified and experienced in working with electrical equipment, compressed gases, and hydrogen. This manual includes certain safety guidelines and recommendations; however, this manual is not intended to cover all situations. The customer is responsible for determining the suitability of the customer's particular design or application, and for ensuring the safe operation, maintenance, and storage of the device and any systems into which it is integrated. Ballard Power Systems, Inc. ("Ballard") cannot be responsible for the use of the device in ways, or as part of systems, that deviate from the operating recommendations in the manual. The customer is responsible for being aware of, and complying with, all applicable laws and regulations. Ballard has limited its liability for damages incurred by the customer or its personnel in the contract documents pursuant to which the device is provided to the customer. Please refer to those documents for additional information.

The customer must read this entire manual before attempting any procedure described in any part of the manual. Failure to follow any instruction or recommendation could result in death or bodily injury or loss or damage to property.

Ballard intends for the device to be installed, operated, and serviced only by technically qualified and experienced people who understand the principles of fuel cell technology, are aware of the safe operating limits of fuel cells, and are familiar with the risks posed operating a fuel cell stack. Ballard relies on customers to use their own knowledge and experience when installing, operating, and servicing the device. Ballard also relies on the people using this manual to be familiar with common terms and abbreviations used in the manual.

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Table of Contents

1.0 Safety 12
1.1 General Safety 12
1.2 High Temperature and High Pressure Safety 12
1.3 High Voltage Safety 13
1.4 Hydrogen Safety 14
1.5 Stack Fire Safety 15
1.6 Asphyxiation Safety 16

2.0 General Features and Specifications 17
2.1 Standard Stack Sizes 18
2.2 Stack Physical Characteristics 18
2.3 Basic Performance 21
2.4 Basic Operating Specifications 23

3.0 Integrating the 1020ACS into a System 27
3.1 Stack Enclosure / Stack Mounting 28
3.2 Oxidant/Coolant Subsystem Configuration 36
3.3 Fuel Subsystem Configuration 39
3.4 Electrical Configuration 41
3.5 Configurations for Multiple Stacks 46

4.0 Air Stream 49
4.1 Air Stream Configuration 49
4.2 Oxidant Specification 50
4.3 Coolant Specification 54
4.4 Air Stream Component Selection 58

5.0 Fuel Stream 61
5.1 Fuel Stream Configuration 61
5.2 Fuel Specification 62
5.3 Anode Fuel Purge 65
5.4 Anode Pressure Drop 69
5.5 Purge Exhaust 69
5.6 Anode Component Selection 70
5.7 Fuel Leaks 72

6.0 Stack Performance 75
6.1 Voltage, Current, and Power Range 75
6.2 Stack Performance 76
6.3 Stack Fuel Consumption and Fuel Efficiency 80
6.4 Stack Efficiency 82
6.5 Stack Performance Degradation 83
6.6 Tips to Maximize Stack Life 88
6.7 Stack Durability Estimates 89

7.0 Carbon Corrosion & Fuel Starvation 92
7.1 Fuel Cell Reaction During Normal Operation 92
7.2 Carbon Corrosion Currents 92

The information contained in this manual is confidential and/or proprietary business data.
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7.3 Carbon Corrosion Currents – Start-up 94
7.4 Carbon Corrosion Currents – Fuel Leak 95
7.5 Carbon Corrosion Currents – Leaky Purge Valve 96
7.6 Carbon Corrosion Currents – Leaky Supply Valve 96
7.7 Fuel Starvation 97

8.0 Stack Operation 98
8.1 Standby 98
8.2 Start-Up 101
8.3 Warm-Up 105
8.4 Run 111
8.5 Shutdown 114
8.6 Air-Starves and Current Pulses 118
8.7 Stack Alarms 126
8.8 Stack Variables to Log 129

9.0 Stack Sizing 131

10.0 Shipping and Storage 132
10.1 Preparation for Storage 132
10.2 Shipping and Storage Ambient Environment 132
10.3 Shipping and Storage Shock and Vibration 133
10.4 Shipping and Storage Air Quality 133

11.0 Troubleshooting 134
11.1 Diagnostic Tools 134
11.2 Troubleshooting 140
11.3 Contacting Ballard for Further Assistance 142

Appendix A: Useful Fuel Cell Equations 143
A.1 References 143
A.2 Overall Fuel Cell Equations 143
A.3 Hydrogen Equations 143
A.4 Air Equations 145
A.5 Coolant Equations 146

Appendix B: Thermistor Temperature vs. Resistance Curve 147

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Table of Figures

Figure 1: A Schematic of the 1020ACS Stack and the Associated Balance of Plant 17
Figure 2: 1020ACS Features 18
Figure 3: 1020ACS Features 19
Figure 4: Stack Dimensions 20
Figure 5: Beginning-of-Life Stack Performance 21
Figure 6: Air Quality Specification 24
Figure 7: Fuel Quality Specification 25
Figure 8: Acceptable Stack Orientations 29
Figure 9: Unacceptable Stack Orientations 30
Figure 10: Stack Mounting Features 32
Figure 11: Compression Hardware 32
Figure 12: Potential Mounting Strategy 1 33
Figure 13: Potential Mounting Strategy 2 33
Figure 14: Ballard Lab Enclosure 35
Figure 15: Oxidant/Coolant Subsystem Configuration in a Typical System 36
Figure 16: Airflow Direction and Stack Thermistor(s) 37
Figure 17: Thermistor(s) Wiring 38
Figure 18: Fuel Subsystem Configuration in a Typical System 39
Figure 19: Acceptable Fuel Subsystem Materials 40
Figure 20: Electrical Configuration in a Typical System 41
Figure 21: Bus Plates and Lugs 42
Figure 22: CVM Connections in Ballard's Lab 43
Figure 23: Spring-Pin CVM Connections 43
Figure 24: Power Resistor 45
Figure 25: Current Pulse Board 45
Figure 26: No Power Electronics 45
Figure 27: Oxidant/Coolant Subsystem Configurations for Multiple Stacks 46
Figure 28: Fuel Subsystem Configuration for Multiple Stacks 47
Figure 29: Multiple Stacks in Series 47
Figure 30: Multiple Stacks in Parallel 48
Figure 31: Multiple Stacks in Parallel, Multi-Input Power Electronics 48
Figure 32: Oxidant/Coolant Stream Configuration 49
Figure 33: Oxidant Flow Direction – Stack Label 52
Figure 34: Oxidant Quality Specification – Chemical 53
Figure 35: Oxidant Quality Specification – Particulate 53
Figure 36: Optimum Stack Temperature 55
Figure 37: Stack Cathode Outlet Temperature Delta Requirements 57
Figure 38: Stack Cathode Outlet Temperature 57
Figure 39: Fan Curve for ebm-papst W1G180-AB31-01 60
Figure 40: Fuel Stream Configuration 61
Figure 41: Fuel Inlet and Outlet Ports 63
Figure 42: Fuel Quality Specification 64
Figure 43: Measuring Purge Volume 66
Figure 44: Cell to Cell Voltage Variability Indicates a Purge is Required 67
Figure 45: 6000 Amp·Sec Purge Interval 68

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Figure 46: 2300 Amp·Sec Purge Interval 68
Figure 47: Purge Exhaust – Into Air Stream 69
Figure 48: Accumulator in the Ballard Lab 71
Figure 49: Stack Fuel Leaks – After Shutdown 73
Figure 50: Beginning-of-Life Stack Performance 77
Figure 51: Measuring Stack Voltage across the Bus Plates 77
Figure 52: Bus Plate Voltage Loss 77
Figure 53: Ballard Polarization Curve Generation 78
Figure 54: End-of-Life (EOL) Stack Performance 79
Figure 55: Electricity and Heat Production in a Fuel Cell 82
Figure 56: PtO_x Layer Growth as a Function of Cell Voltage 84
Figure 57: Ballard Lab Test Cycle 89
Figure 58: 1020ACS Durability at 65.3A 90
Figure 59: Normal Fuel Cell Operation 92
Figure 60: Mixture of Air and H_2 Gas in the Anode 92
Figure 61: Air and H_2 Gas on the Anode Create Two Cells 93
Figure 62: Mixture of Air and H_2 Gas in the Anode 93
Figure 63: H_2/Air Front During Start-Up Purge 94
Figure 64: Corrosion Currents Resulting from a Fuel Leak 95
Figure 65: Cell Voltage After Shutdown for a Leaky Stack 95
Figure 66: Corrosion Currents Resulting from a Leaky Purge Valve 96
Figure 67: Corrosion Currents Resulting from a Leaky Supply Valve 96
Figure 68: Start-Ups at Different Ambient RHs 101
Figure 69: Start-Up Procedure – 56 cell stack 103
Figure 70: Start-Up Purge Load – Cell Reversal 104
Figure 71: Thermal Performance during a Start-Up in -5°C Ambient 106
Figure 72: Bridging Power at Start-Up 107
Figure 73: Start-Up Time and Bridging Energy Requirements 107
Figure 74: Current Ramp 108
Figure 75: Stack Voltage Limited to 0.5 V During Start-Up ..... **Error! Bookmark not defined.**
Figure 76: 1020ACS Purge Valve and Stack Current – 56 cell Stack 112
Figure 77: 1020ACS Thermal – 56 cell stack 113
Figure 78: The Anode Pressure After a Shutdown 114
Figure 79: Cell Voltages During Shutdown 115
Figure 80: Air-Starve Reaction 119
Figure 81: Air Starve Procedure 122
Figure 82: Stack Performance Before and After an Air-Starve 123
Figure 83: Current Pulse Profile 124
Figure 84: Stack Being Pulsed 125
Figure 85: Low Stack Voltage Alarm for a 56 cell Stack 127
Figure 86: Stack Damage 134
Figure 87: Cells Out of Alignment 134
Figure 88: Heat Damage to MEAs 135
Figure 89: Burn Marks on the Bus Plates 135
Figure 90: Stack Damage 135
Figure 91: Measure Stack Voltage 136
Figure 92: Measure Stack Current 136
Figure 93: Measure Cell Voltage 137

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Figure 94: Check Stack Thermistor(s) 138
Figure 95: Ballard Lab Leak Test Set-Up 139

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Symbols Used in Manual



WARNINGS in this manual indicate actions that may result in an accident, which could cause immediate bodily injury or loss of life or could cause damage to equipment or other property.



CAUTIONS in this manual indicate actions that may result in an accident, which could cause damage to equipment or other property and which could cause bodily injury if not promptly detected and remedied.



ELECTRICAL HAZARDS in this manual indicate actions that pose a risk of high voltage or electrical shock that may result in an accident, which could cause bodily injury or loss of life or could cause damage to equipment or other property.

Disclaimer

The fuel cell stack may be integrated in a variety of stationary and motive applications, as reflected in the dual product names. This manual is applicable to both the FCgen-1020ACS product for stationary applications and the FCvelocity-1020ACS product for motive applications. This document does not attempt to address all possible integration options. The customer is responsible for determining the suitability of the various integration options for the customer's particular needs, and for ensuring the safe operation, maintenance, and storage of the device and any systems into which it is integrated. It is the responsibility of the customer to ensure that his or her design is free from claims of patent infringement. Ballard cannot be responsible for any injury or damage caused by the use of its products in ways, or as part of systems, not expressly approved by Ballard.

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Glossary

Term	Description
ASL	Above Sea Level
A	Ampere
BOL	Beginning of Life
ºC	degree Celsius
cc	cubic centimetre
CVM	Cell Voltage Monitoring
EMF	Electromotive Force (voltage)
EOL	End of Life
kW	kiloWatt
kPa	kiloPascal (absolute unless otherwise specified)
IEC	International Electrotechnical Commission
mm	millimetre
ms	millisecond
mV	milliVolt
NOPL	Non-Operating Performance Loss
OCV	Open Circuit Voltage
PEM	Proton-Exchange Membrane
PM2.5	Particles up to 2.5 micrometers in diameter
PM10	Particles up to 10 micrometers in diameter
ppm	parts per million
slpm	standard litre per minute, evaluated at 1 atmosphere and 0°C
stack	Fuel cell stack, generally referring to the 1020ACS fuel cell stack
V	Volt

Scope of Manual

This manual is a guide for the customer to use in integrating the 1020ACS into their system and successfully operating it. Product specifications, expected product performance, required system inputs, electrical interfaces and acceptable operating conditions are included in the manual. This manual will provide guidance for operation within the bounds of the specification. Please contact Ballard Power Systems for questions relating to the operation outside the stated specification.

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1.0 Safety



ELECTRICAL HAZARD: Fuel cell stacks generate high voltage. Obey ALL warnings, cautions, and safety instructions. Failure to do so may result in electrical shock leading to personal injury or death.

The installer and operator of the fuel cell stack must be technically qualified and experienced in handling electrical equipment, compressed gases, and hydrogen.

1.1 General Safety



1. **The fuel cell stack may contain residual voltage when not operating. If possible wait 20 minutes so it bleeds down completely and check the voltage before handling with voltmeter.**
2. **Keep all guards, screens, and electrical enclosures in place when the system is operating.**
3. **The fuel cell stack should not be used or stored in wet or damp conditions.**
4. **Remove jewellery, watches, rings, and metal objects on clothing that can cause short circuits when working with the fuel cell stack or system.**

1.2 High Temperature and High Pressure Safety



1. **The fuel cell stack can reach a temperature of 70°C or higher if operated outside the specification. Avoid touching exposed components during or shortly after operation.**
2. **The fuel cell stack and associated system use pressurized gases, which can be hazardous. Use caution and ensure circuits are de-pressurized before opening any lines or fittings.**
3. **The fuel cell stack is assembled under high compression. Do not attempt to disassemble the stack.**

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1.3 High Voltage Safety



1. **The fuel cell stack generates up to 80 VDC (open circuit voltage) for an 80-cell stack (maximum ~1 V/cell). Always ensure that the Stack Power HV+ and HV- terminals are connected to an appropriate load prior to operation.**
2. **Current leakage from the stack can also occur if there is inadequate isolation elsewhere in the electrical system and the stack is not fully isolated from that portion of the electrical system. The inadequate isolation could occur elsewhere in the fuel cell module or external to the fuel cell module. This leak path can be minimized by ensuring all electrical equipment and wiring in the fuel cell module is adequately isolated and by ensuring that the fuel cell module electrical buses are isolated from the application electrical system.**
3. **The fuel cell stack has a potential for current leakage across clearances and creepage distances. The fuel cell stack installation should be designed with adequate clearances and creepage distances as listed in applicable standards.**
4. **Stack power connection cables must be appropriately sized to suit the application for voltage, current and insulation temperature limits. Cables must have suitable voltage rating, current carrying capacity, and insulation temperature rating, depending on the end-users' specific application and operating environment.**
5. **Exercise caution when routing the Stack Power Cables. In particular, ensure that no other electrical cables are routed in between the physical loop formed by the Fuel Cell Stack power terminals, the HV+ and HV- and the load power terminals. Electrical current may be induced through cables routed between the fuel cell stack power terminals.**
6. **Exercise caution when working with the stack. Residual reactants within the stack can rapidly develop a charge, even when there is no fuel flow and the stack has been short-circuited. A reading of zero volts across the entire stack does not guarantee that all cells are uncharged.**
7. **All non-current carrying metal parts on the stack must be either electrically bonded or considered electrically live. In the case that non-current carrying parts are considered live, the system integrator must provide a means to: 1) isolate the module's compression hardware from the integrator's system, 2) prevent people or objects from touching the live parts 3) provide a label/marking to warn service personnel that the compression hardware may be electrically live.**
8. **Be sure that all electrical connections and connectors are properly installed and connected with proper torque. Do not over-torque, as this**

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can damage the stack.

High Voltage Safety (*continued*)



9. **Avoid hazardous voltage situations that could result from unsafe conditions such as, but not limited to, the following:**
   - A. **Improper grounding**
   - B. **Accumulation of foreign material or debris between live stack parts and hardware that could lead to loss of isolation or reduction in creepage/clearance.**
   - C. **Handling electrical leads or devices with wet hands or on wet ground**
   - D. **Frayed electrical leads**
   - E. **Improper connection or re-connection of the terminal leads**
   - F. **Short circuits**
   - G. **Back-feed from energized normal and emergency power sources.**

1.4 Hydrogen Safety



1. **Hydrogen is a colourless, odourless, highly flammable gas.**
2. **Hydrogen must be sited and handled in accordance with applicable regulations and the gas supplier's recommendations.**
3. **Hydrogen is non-toxic, but can cause asphyxiation by displacing the oxygen in the air. There are no warning symptoms before unconsciousness results.**
4. **Hydrogen molecules are smaller than any other gas, making hydrogen more difficult to contain. It can diffuse through many materials considered airtight. Fuel lines, non-welded connections, and non-metal seals such as gaskets, O-rings, pipe thread compounds and packings present potential leakage or permeation sites. Furthermore, hydrogen's small molecule size results in high buoyancy and diffusivity, so leaked hydrogen will diffuse and become diluted quickly. Stack hydrogen leak rates will generally increase with stack lifetime.**
5. **The responsibility for leak detection and the mitigation of combustible leaks rests with the customer. Hydrogen leaks emanating from the fuel cell stack can be readily detected by means of a hydrogen detector, which can trigger warnings well before the hydrogen/air mixture reaches a flammable concentration.**

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1.5 Stack Fire Safety

WARNING

1. **Operation of the fuel cell stack in a manner that is significantly outside specification may result in negative cell voltages in one or more cells, which can lead to open flame at the stack.**
2. **Specifically, the following conditions may result in negative cell voltages:**
   - **A. Fuel Starvation: Operation with insufficient purge, especially the initial start-up purge; or long periods of over-cooled operation.**
   - **B. Membrane Dehydration: Operation above maximum cell temperature rating. This can be caused by:**
     - **Operating without sufficient coolant flow.**
     - **Operation with one or more severely degraded cells in an otherwise healthy stack.**
3. **To prevent these conditions, stack operation should be automatically stopped in the event of loss of fuel or coolant flow using either a voltage or stack temperature signal. Furthermore, detection of negative cell voltages, and a system response to prevent stack fire due to negative cell voltages should form an integral part of the system safety strategy. Two methods are recommended:**
   - **C. Measure individual cell voltages or voltages of groups of cells. This allows for a direct measurement of the potential hazard; however, it will involve a more complicated interface for measuring cell voltages. A recommended safe shutdown limit if this method is implemented is to shut down instantly (<1 sec delay) if a single cell reaches a negative cell voltage of –5 V. To avoid stack degradation due to negative cell voltage, a limit of 0 V is more suitable.**
   - **D. Measure overall stack voltage. A low stack voltage limit can be set based on the value of negative cell voltage to be detected. This method becomes less effective for larger stack sizes. As a starting point, it is recommended that an instant system shutdown (<1 sec delay) be implemented if stack voltage drops below Vn-10 V (where Vn = nominal stack voltage at a given current). This should detect a minimum single-cell voltage reversal of about –10 V.**
4. **Stack operation after a fire poses the potential additional hazard of delayed ignition of the external fuel leak in subsequent restart. This risk can be minimized by prompt detection and immediate shutdown in the event of a stack fire to minimize the resulting fuel leak. Mitigation of this potential hazard may include preventing restart following a stack fire failure or on detection of large fuel leaks, or appropriate enclosure design for overpressure relief.**

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Stack Fire Safety (*continued*)

WARNING

5. **The cells in the stack contain platinum catalyst, which may react if exposed to certain chemicals. The stack should not be exposed to any chemicals not mentioned in this application guide, and used stacks should be returned to Ballard for proper disposal.**

1.6 Asphyxiation Safety

WARNING

The fuel cell stack consumes O_2 while operating. If operating the stack in poorly ventilated, small enclosures, care must be taken that O_2 concentrations do not drop below safe levels.

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2.0 General Features and Specifications

1. The 1020ACS fuel cell stack is a proton exchange membrane (PEM) fuel cell stack designed to provide stable electrical power while operating over a wide range of operating and environmental conditions.
2. Compared to other fuel cell systems, the 1020ACS requires a simple and inexpensive Balance-of-Plant (BOP) to operate the stack.
   - **A.** The 1020ACS's coolant and oxidant requirements are met with a fan that moves air through the stack. The air is used to cool the stack as well as to provide oxidant for the fuel cell reaction. No external humidification of the air is required.
   - **B.** The 1020ACS is designed to be used in a dead-ended configuration using dry hydrogen, with no external humidification required.
3. The figure below is a simplified schematic of the stack and required balance-of plant.

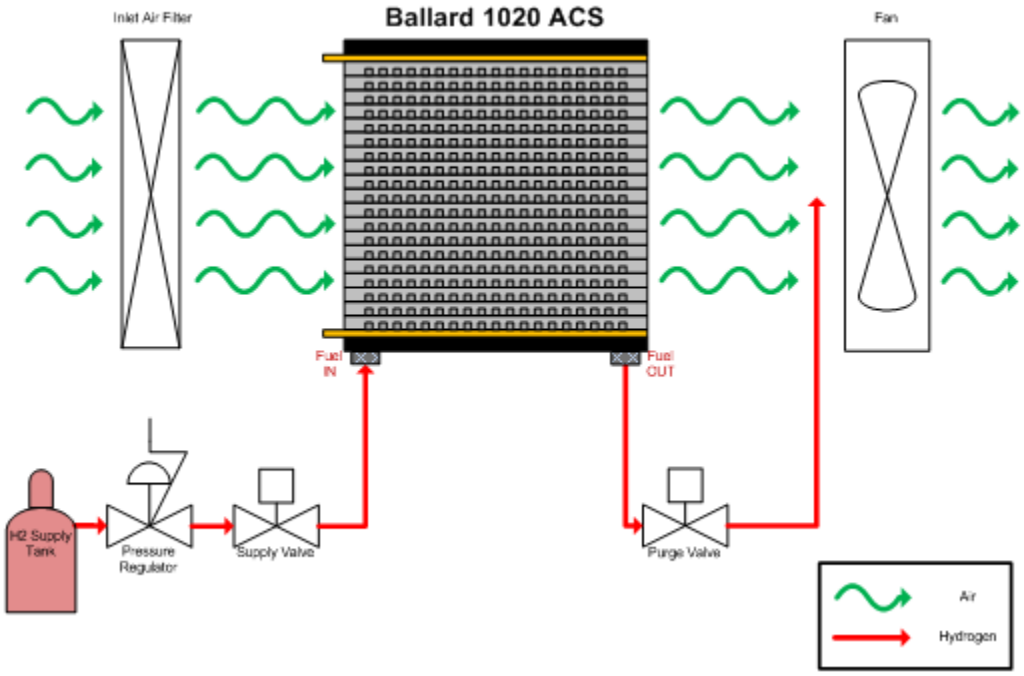


Figure 1: A Schematic of the 1020ACS Stack and the Associated Balance of Plant

See Section 3.0, Page 27 for more information on the integrating the stack into a system.

2.1 Standard Stack Sizes

Cell Count	P/N (v2.1)	P/N (v2.2)
10	5127049	5133801
12	5127050	5134318
13	5127051	5133802
18	5127052	5133803
20	5127053	5133804
28	5127054	5133805
36	5127055	5133806
39	-	5127056

Cell Count	P/N (v2.1)	P/N (v2.2)
46	5127057	5133807
56	5127058	5133808
64	5127059	5133809
66	5127060	5133810
70	5127061	5133811
73	5127062	5133812
80	5127063	5133354

2.2 Stack Physical Characteristics

The figure below show the main physical features of the 1020ACS stack.

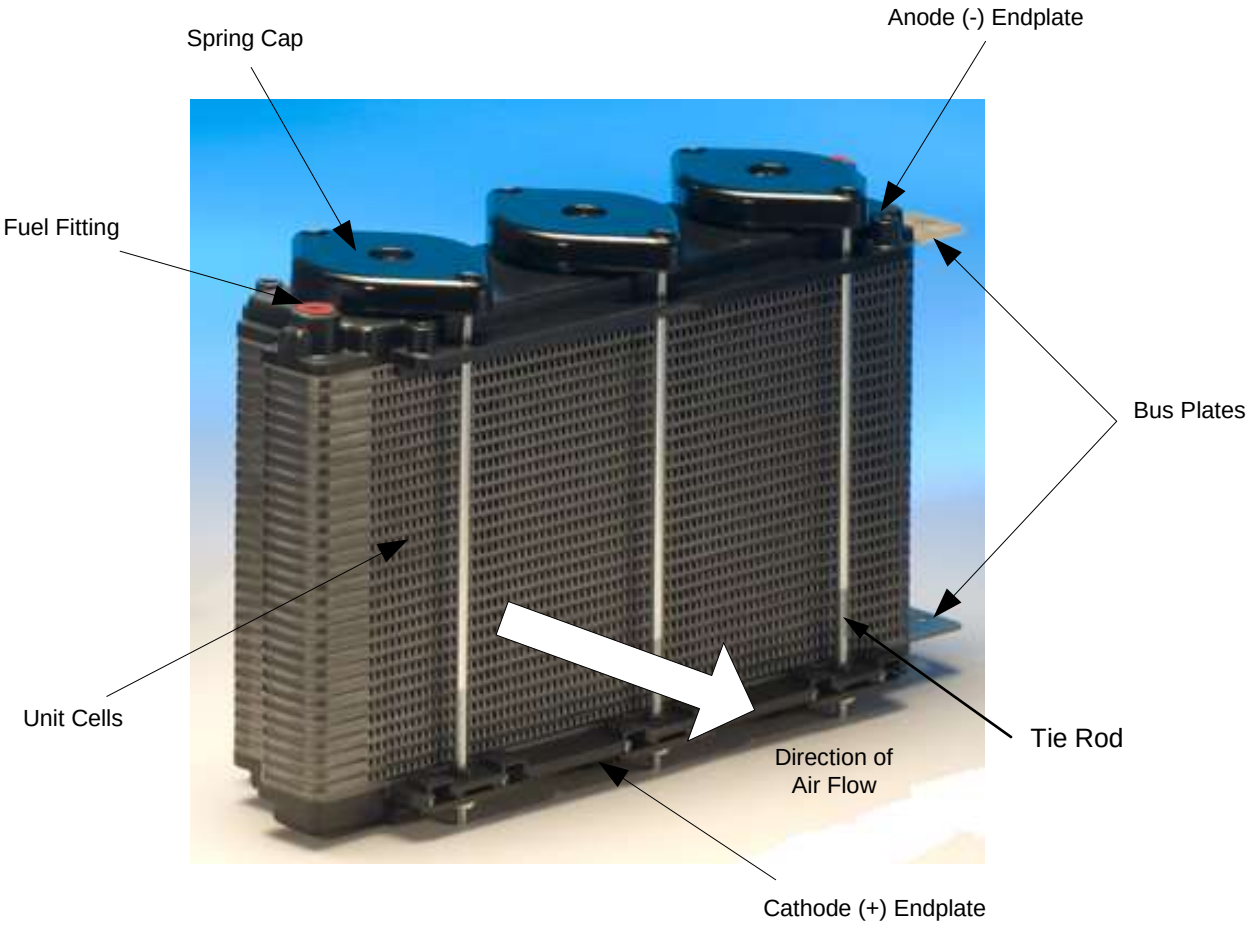


Figure 2: 1020ACS Features

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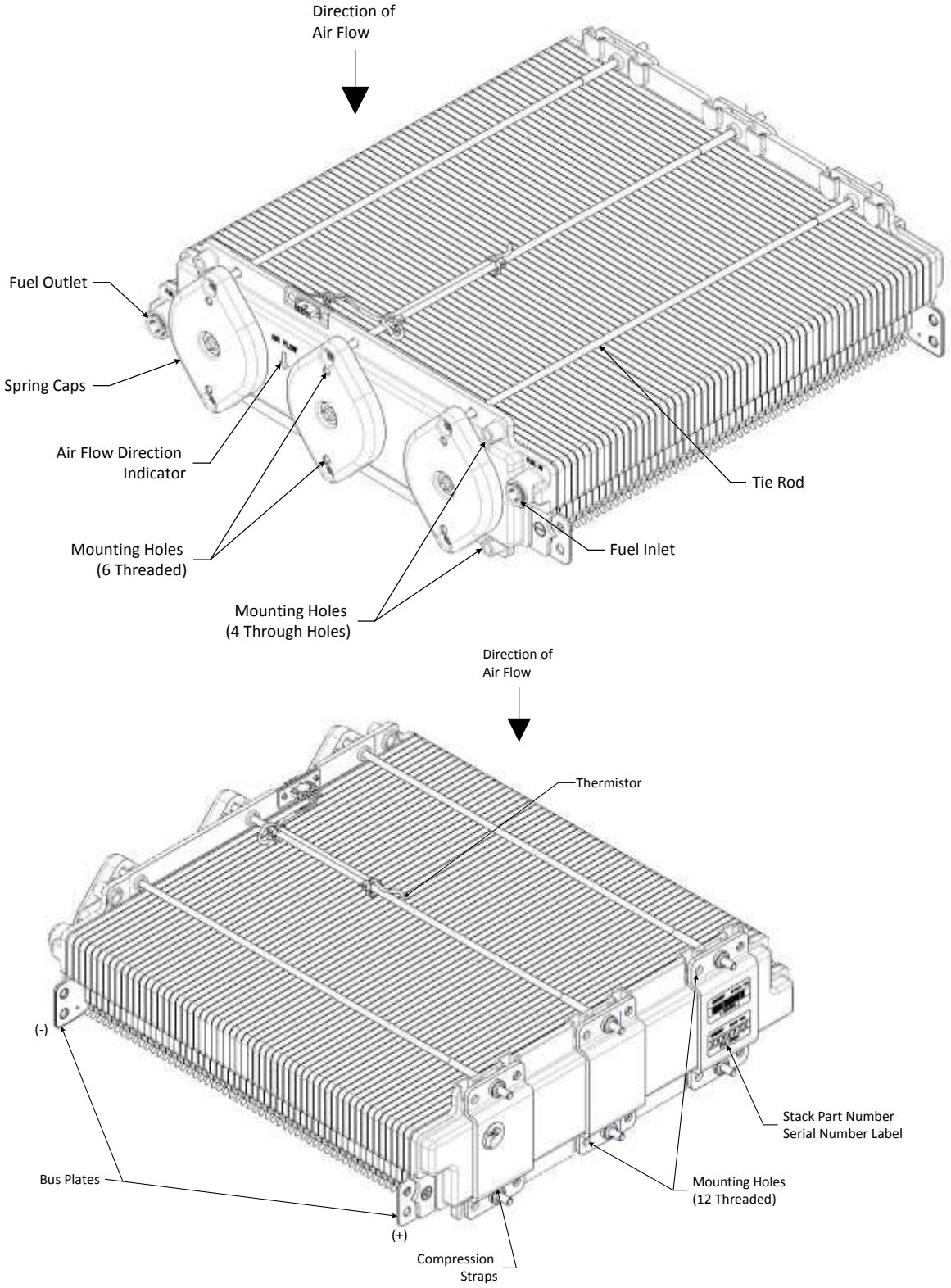


Figure 3: 1020ACS Features

Note on stack Serial Number label:
SN only = new stack
SN + "U" = Used – stack has been repaired (some but not all MEAs replaced, or other repair)
SN + "R" = Refurbished (all MEAs replaced to BOL)
SN + "R" + "U" = refurbished stack subsequently repaired

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2.2.1 Stack Dimensions

All dimensions are nominal. Refer to the stack interface drawing (DRW5117682) for details. CAD models of the stack are available from Ballard.

Width:	$351mm$
Length:	$103mm$
Height:	$Number\ of\ cells \times 5.5mm\ +\ 55.4mm$

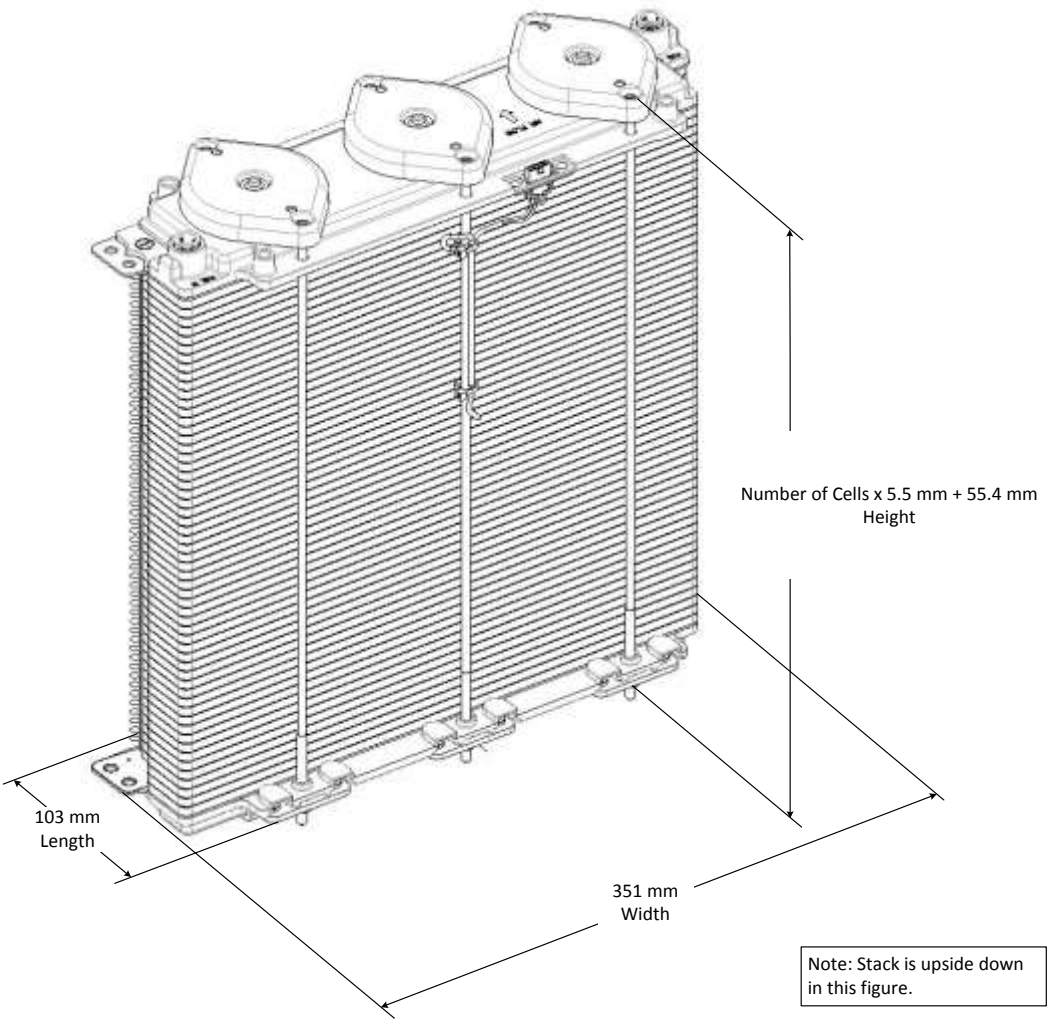


Figure 4: Stack Dimensions

2.2.2 Stack Mass

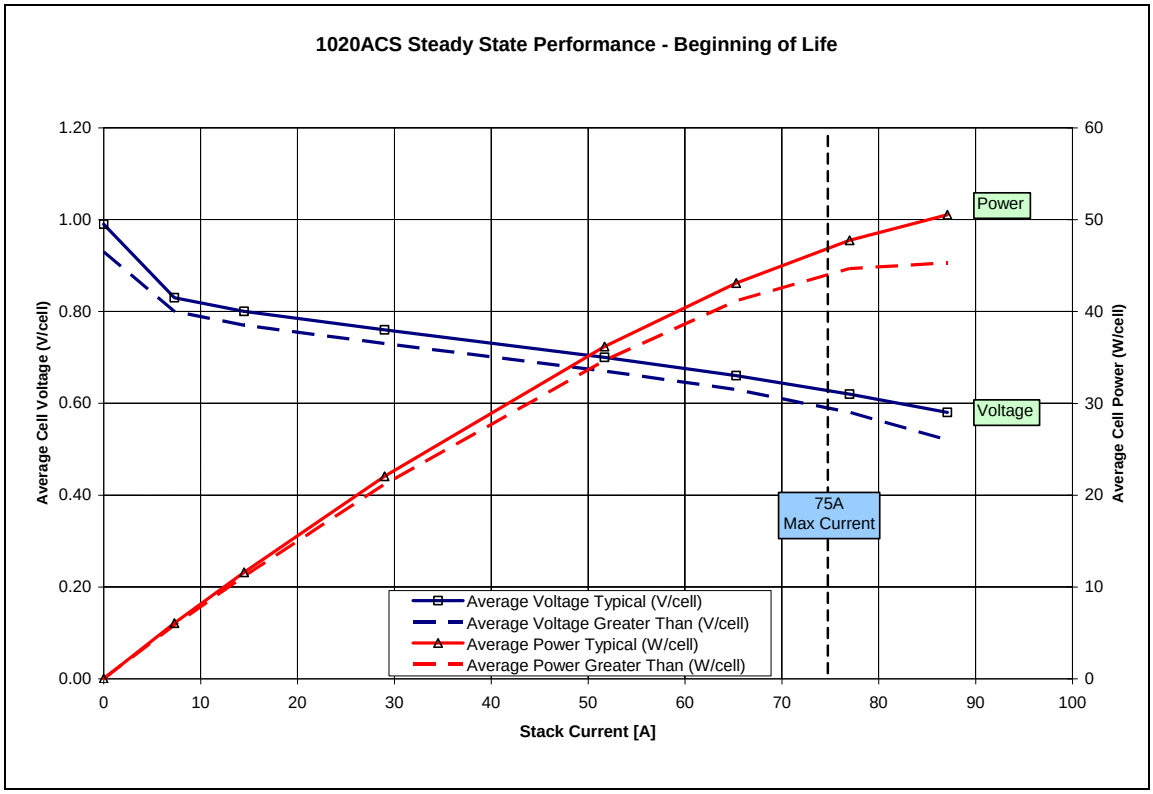
Mass:	$Mass = Number\ of\ cells \times 0.16kg\ +\ 2.1kg$

2.3 Basic Performance

The following is a brief overview of stack performance. Subsequent sections of the manual should be consulted for more detailed information on stack performance.

2.3.1 Stack Performance – Beginning-of-Life

The stack performance at beginning-of-life (BOL) for a fully conditioned stack is presented below. *See Section 6.0, Page 75 for more information on the stack performance.*



Note: The polarization curve was generated using the procedure described in *Section 6.2.3, Page 78*, and the stack was conditioned before the polarization curve was generated.

Stack Current (A)	0	7.3	14.5	29.0	51.7	65.3	77.0	87.1
Avg. Cell Voltage Greater Than (V/cell)	0.93	0.80	0.77	0.73	0.67	0.63	0.58	0.52
Avg. Power Greater Than (W/cell)	0	5.8	11.2	21.2	34.6	41.1	44.7	45.3
Avg. Cell Voltage, Typical (V/cell)	0.99	0.83	0.80	0.76	0.70	0.66	0.62	0.58
Avg. Power, Typical (W/cell)	0	6.1	11.6	22.0	36.2	43.1	47.7	50.5

Figure 5: Beginning-of-Life Stack Performance

2.3.2 Stack Life Estimates

1. Testing in the Ballard lab has shown a stack running on the 1020ACS Ballard Lab Standard Test Cycle has a mean lifetime of 1650 on-off cycles (45 minute run time per air-air start) before EOL cell voltage is reached. *See Section 6.7.1, Page 89 for the definition of the Ballard Lab Standard Test Cycle*. *See Section 6.7, Page 89 for more information on stack durability.*

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2.4 Basic Operating Specifications

Basic operating specifications are presented below. More detailed specifications can be found in subsequent sections of the manual.

2.4.1 Current, Voltage, and Power Range

Current	**Range:** 0 A to 75 A.
Voltage	**Range:** 0.5 V/cell to 1 V/cell
Power	**Range:** $P_{stack} = V_{cell} \times I_{stack} \times n$ where $n =$ number of cells

See Section 6.1, Page 75 for more information on stack current, voltage, and power ranges.

2.4.2 Stack Temperature

Stack Temperature	

Stack Current (A)	0	7.3	14.5	29.0	51.7	65.3	77.0	87.1
Maximum (°C)	52	55	57	62	70	75	75	75
Optimum (°C)	26	30	34	41	53	61	67	72
Minimum (°C)	6	10	14	21	33	41	47	65

Start-up Temp	**Required:** greater than -10°C

See Section 4.3, Page 54 for more information on stack temperature.

See Section 8.1.2, Page 99 for more information on stack temperature on start-up.

2.4.3 Oxidant/Coolant Stream (Air)

Inlet Temp	**Required:** -20°C to 52°C	**Optimal:** 10°C to 40°C
Altitude	**Required:** -400 m to 7600 m	**Optimal:** 0 m (Sea Level)
Inlet Humidity	**Required:** 0% RH to 100% RH (Non-Condensing)	
Flow	Flow is a function of stack current and inlet air temp. *See Section 4.3.2, Page 55.*	
Stoichiometry	Greater than 20 at steady-state.	

See Section 4.2, Page 50 for more information on the air stream specification.

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2.4.4 Oxidant Quality

The stack is capable of meeting its performance and durability specification if the air stream meets the *nominal* air quality specifications in the table below. The stack should never be exposed to oxidant that exceeds *peak* contamination levels.

Chemical	Nominal	Peak
Sulfur Dioxide (SO_2)	< 0.01 ppm	<0.18 ppm
Nitrogen Monoxide (NO)	< 0.025 ppm	<0.20 ppm
Nitrogen Dioxide (NO_2)	< 0.05 ppm	<0.25 ppm
Volatile Organic Compounds (e.g. Benzene C6H6, Toluene C7H8)	< 0.008 ppm	<0.050 ppm
Hydrogen Sulfide	< 0.04 ppm	<0.04 ppm
Ammonia	< 0.01 ppm	<0.1 ppm
Ozone	< 1 ppm	<1 ppm
Carbon Monoxide	< 5 ppm	<32 ppm
Carbon Dioxide	< 1% vol.	N/A

Particulate Size	Nominal	Peak
Coarse Particles (PM 10)	< 90 µg/m^3	N/A
Fine Particles (PM 2.5)	< 15 µg/m^3	N/A

Figure 6: Air Quality Specification

See Section 4.2.6, Page 52 for more information on oxidant quality.

2.4.5 Fuel Stream

Inlet Temp	**Required:** -15°C to 65°C
Pressure	**Required:** 0.16 barg to 0.56 barg **Optimal:** 0.36 barg
Inlet Humidity	**Required:** 0% RH (dry hydrogen)
Flow	**Required:** 0.559 slpm/cell (At 75 A, more flow required during purge)
Stoichiometry	**Required:** 1.0 for dead-ended operation, 1.07 including required periodic purge

See Section 5.2, Page 62 for more information on the fuel stream specification.

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2.4.6 Fuel Purge

Periodically, the anode gas needs to be exchanged with fresh hydrogen from upstream of the stack. This is accomplished by opening the purge valve for a specified time to allow fresh hydrogen from upstream of the stack to displace the anode gas.

Start-Up Purge	**Start-Up Purge Duration:** 200 ms **Start-Up Purge Volume:** 20 mL/cell (See Note 1) **Start-Up Purge Load:** Keep average cell voltage below 0.80 V.
Run-Time Purge	**Run-Time Purge Duration:** Less than 500 ms **Run-Time Purge Volume:** 20 mL/cell (See Note 1) **Run-Time Purge Interval:** 2300 Amp·Seconds

Note 1: *The purge volume should also include the tubing volume between the supply valve and stack and purge valve*

See Section 5.3, Page 65 for more information on the fuel purge.

2.4.7 Fuel Quality

The stack is capable of meeting its performance and durability specification if the H_2 fuel is commercial Grade 3.5 hydrogen (99.95% H_2 concentration). All stack durability testing at Ballard is performed using 99.999% pure (Industrial Grade 5) hydrogen.

Chemical	Nominal	Peak
Total inert gases (99.95% H_2 min)	< 500 ppm	<19000 ppm
Water	< 5 ppm	<19000 ppm
Total hydrocarbons	< 2 ppm	< 2 ppm
Oxygen	< 5 ppm	< 5 ppm
Helium	< 300 ppm	< 300 ppm
Nitrogen, Argon	< 200 ppm	< 4000 ppm
Carbon dioxide	< 2 ppm	< 2 ppm
Carbon monoxide	< 0.2 ppm	< 10 ppm
Total sulfur compounds	< 0.004 ppm	< 0.004 ppm
Formaldehyde	< 0.01 ppm	< 0.01 ppm
Formic acid	< 0.2 ppm	< 0.2 ppm
Ammonia	< 0.1 ppm	< 0.1 ppm
Total halogenated compounds	< 0.05 ppm	< 0.05 ppm

Figure 7: Fuel Quality Specification

See Section 5.2.7, Page 64 for more information on the fuel quality.

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2.4.8 Emissions

Liquid Water	Minimal amounts of liquid water may collect at the fuel outlet.
H_2 Gas	**Start-Up Purge:** 20 mL/cell in 200 ms (~6.0 slpm/cell)
	Run-Time Purge: 20 mL/cell in 500 ms or less (~2.4 slpm/cell)
	External Fuel Leak: Less than 5sccm/cell at 0.36 barg H_2 fuel pressure at BOL

See Section 5.3, Page 65 for more information on fuel purges and Section 5.7, Page 72, for more information on fuel leaks.

2.4.9 Shock and Vibration

Tested to:	**1.** NEBS GR-63-CORE **A.** Transportation Vibration Test **B.** Zone 4 Earthquake Test **C.** Office Vibration Test
	2. Vibration: UL2267
	3. Shock: 15 g, 11 ms, 1000 cycles (based on IEC-68-2-27, Ea and IEC-68-2-29, Eb)

2.4.10 Storage and Shipping

Temperature Range	**Required:** -40°C to 70°C (with duration limits)
Freeze-Thaw Cycles	**Required:** Up to 473 (the limit reflects the maximum cycles tested at Ballard)
Humidity	**Required:** 0% RH to 100% RH (Non-Condensing)
Shock and Vibration	Designed to withstand normal shipping shock and vibration in standard Ballard packaging.

See Section 10.0, Page 132 for more information on stack storage requirements.

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3.0 Integrating the 1020ACS into a System

This section outlines how to physically integrate the stack into a system. When integrating the 1020ACS into a system, there are four major subsystems that need to be considered:

Stack Enclosure / Stack Mounting	
Function	**1.** Restrain the stack in the proper orientation. **2.** Provide ducting to allow the fan to effectively move air through the stack. **3.** Prevent unintended or unauthorized contact with the stack by the end user.

For more information on the enclosure and stack mounting, see Section3.1.1, Page 28.

Oxidant / Coolant Subsystem	
Function	**1.** Deliver oxidant to cathode for the fuel cell reaction. **2.** Deliver coolant to the stack to maintain the stack at the proper temperature.

For more information on the air subsystem configuration in a system, see Section 3.2, Page 36.

Fuel Subsystem	
Function	**1.** Deliver hydrogen fuel to the stack at the appropriate pressure and flow. **2.** Periodically remove impurities that have built up on the anode with a purge at the correct purge flow rate.

For more information on the fuel subsystem configuration in a system, see Section 3.3, Page 39.

Electrical, Power Electronics, Energy Storage	
Function	**1.** Draw power from the stack at the specified power, current, or voltage. **2.** Deliver the required power to the system.

For more information on the electrical system, see Section 3.4, Page 41

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3.1 Stack Enclosure / Stack Mounting

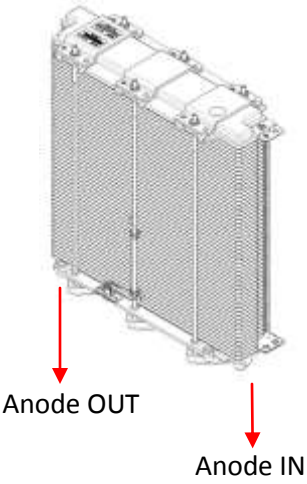
3.1.1 Stack Orientations

3.1.1.1 Acceptable Stack Orientations Requirements	
When installed in the system, the stack must be oriented to allow liquid water to drain out of the anode outlet header, preventing liquid water from pooling in the anode.	
Requirements	**1.** The anode ports should either be either pointing downwards or level. **2.** The anode inlet port must be higher than the anode outlet port.

3.1.1.2 Acceptable Stack Orientations

Acceptable stack orientations are shown in the figure below. Mounting the stack at an angle is also possible, as long as the two above requirements are followed.

1

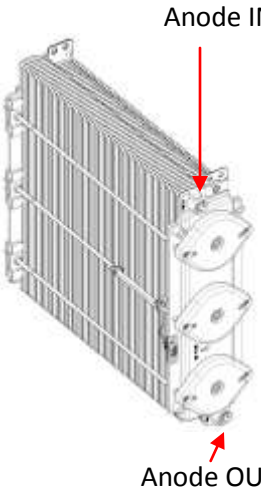


BALLARD RECOMMENDED ORIENTATION

The anode ports are both facing downwards

✓

2

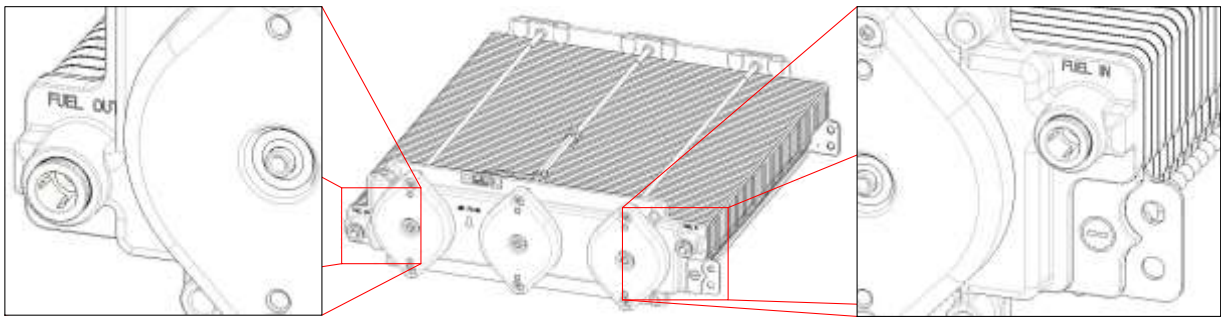


The anode inlet port is higher than the anode outlet port[1]

✓

NOTE 1: If this orientation must be used, there is a Cell Retaining Clip accessory (Part #5125697) that can be acquired from Ballard to provide additional stack anode end cell support. Please contact Ballard if you have any questions.

3



The anode inlet port is higher than the anode outlet port[2]

✓

NOTE 2: If this orientation must be used, an adequate frame to support the stack cells must be provided in the system design. Please contact Ballard if you have any questions.

Figure 8: Acceptable Stack Orientations

3.1.1.3 Unacceptable Stack Orientations

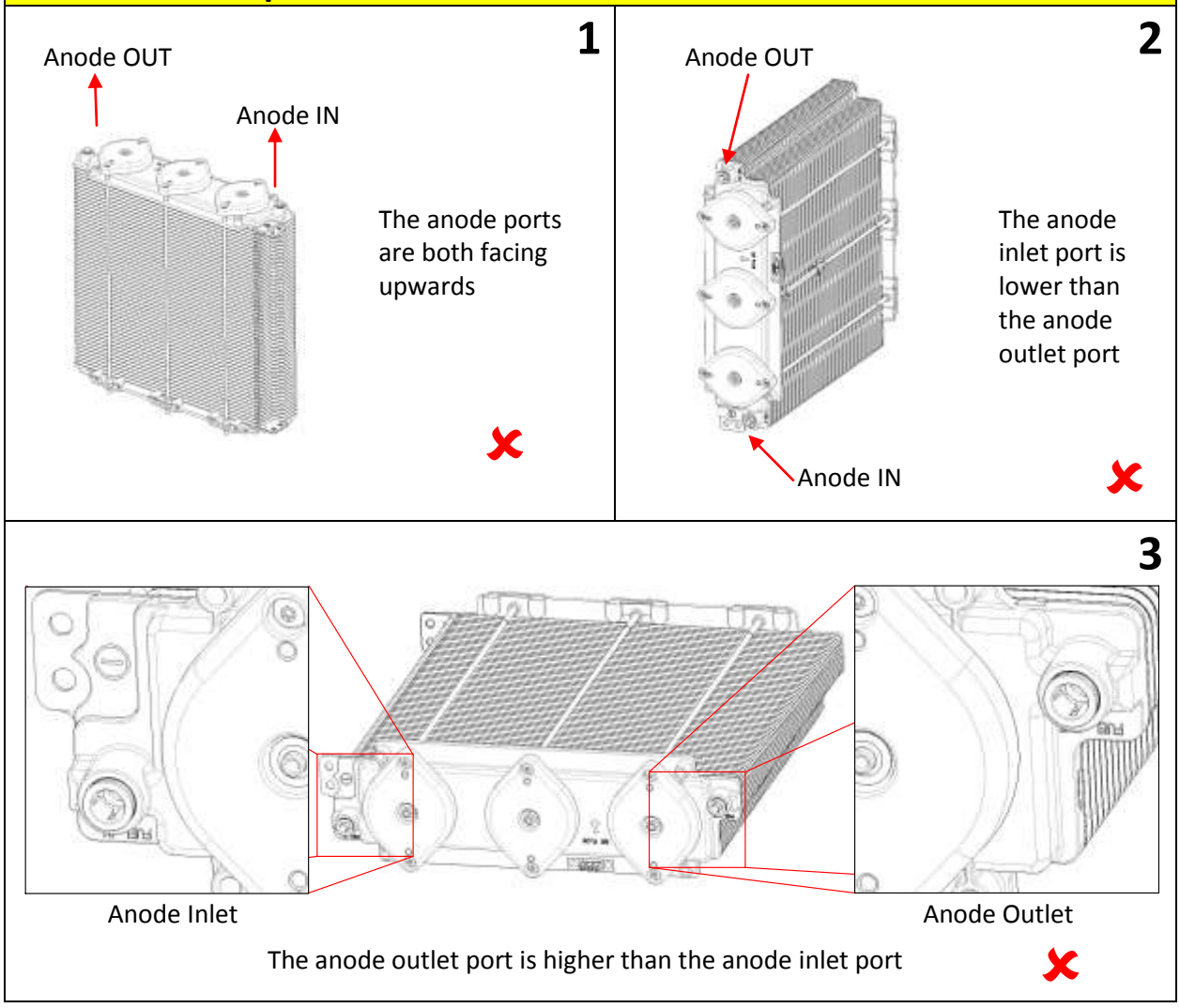


Figure 9: Unacceptable Stack Orientations

3.1.2 Stack Mounting

3.1.2.1 Stack Mounting Requirements

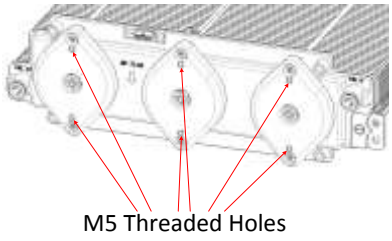
The stack should be securely mounted in the system to prevent damage if the system is exposed to shock and vibration. *See Section 2.4.9, Page 26 for more information on shock and vibration*

Requirements	**1.** The stack should not be subjected to any twisting or bending moments, or be used as a load-bearing member for other system components. **2.** The cells must be allowed to freely move with thermal expansion and creep: **A.** Thermal Expansion: Over a 70°C increase in temperature, the cells will enlarge by 0.015 mm/cell. **B.** Creep: The cells will creep -0.05 mm/cell over the life of the stack. (The cells will shrink under compression) **3.** The mounting strategy must accommodate stacks that fall within the distribution of stack heights for a given cell count. Refer to the stack interface drawing (DRW5117682) for details on stack height distribution.

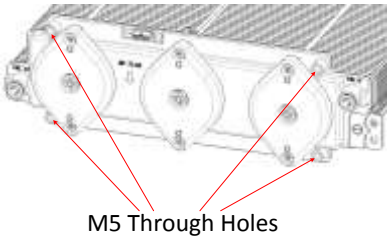
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3.1.2.2 Stack Mounting Features

6 x M5 threaded holes in the end-caps



4 x M5 through-holes on the spring-cap-end-hardware



12 x M5 threaded holes in the compression-straps

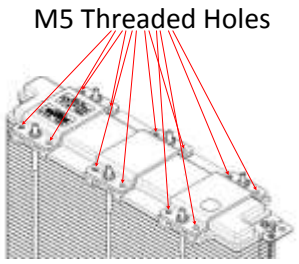
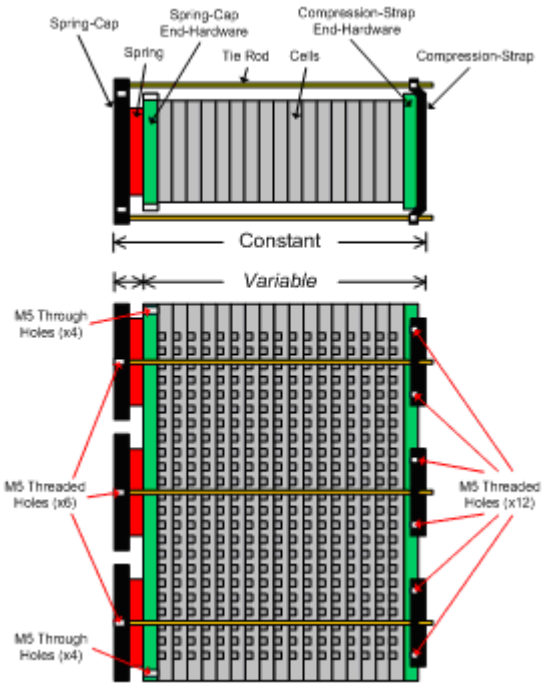


Figure 10: Stack Mounting Features

3.1.2.3 Compression Hardware



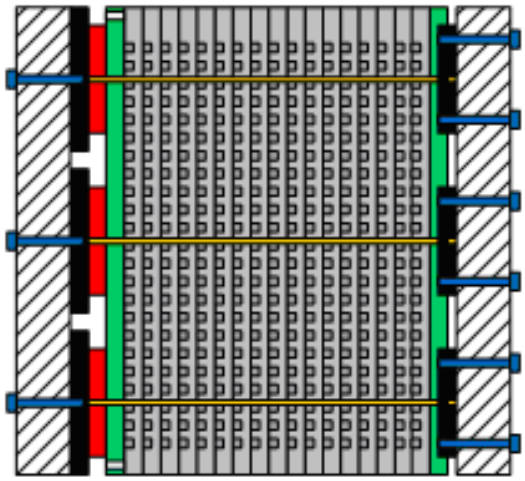
1. The springs in the spring caps allow the cells to freely expand and contract.
2. The distance between the compression-straps and the spring-cap-end-hardware is variable.
3. The distance between the compression-straps mounting holes and the spring-caps mounting holes is constant.
4. On the compression strap end of the stack, the stack should be rigidly mounted using the using the threaded holes in the compression straps.
5. On the spring cap end of the stack, the stack should be either:
   - **A.** Rigidly mounted using the threaded holes in the compression straps.
   - **B.** Supported using the through holes in the end-hardware.

Figure 11: Compression Hardware

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3.1.3 Potential Stack Mounting Strategies

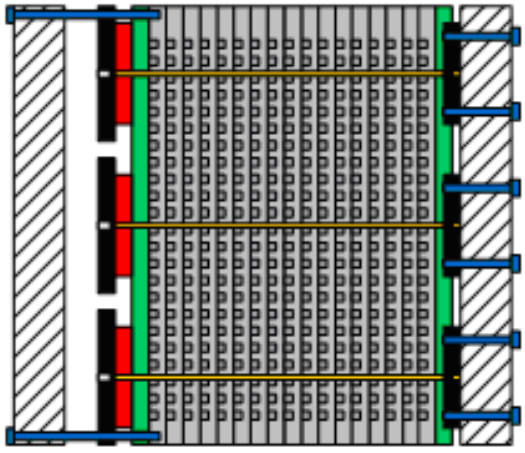
3.1.3.1 Potential Mounting Strategy #1



1. **Preferred method for high shock and vibration environments.**
2. **Compression-Strap End:** Rigidly mount the stack using the threaded holes in the compression-straps.
3. **Spring-Cap End:** Rigidly mount the stack using the threaded holes in the spring caps.
4. Mounting strategy must be adaptable for the distribution of stack sizes. *See Section 3.1.2.1*.

Figure 12: Potential Mounting Strategy 1

3.1.3.2 Potential Mounting Strategy #2



1. **Compression-Strap End:** Rigidly mount the stack using the threaded holes in the compression-straps.
2. **Spring-Cap End:** Use non-threaded 5mm rod through the through-holes in the end hardware to support the stack.

Figure 13: Potential Mounting Strategy 2

Ballard strongly recommends that the stack integrator test the suitability of their application-specific mounting strategy by subjecting the system to a shock and vibration test.

3.1.4 Stack Enclosure

3.1.4.1 Stack Enclosure

Requirements	**1.** The stack enclosure must include ducting to allow the fan to effectively move air through the stack. **2.** The geometry of the enclosure ducting should deliver uniform airflow at a uniform temperature to the cathode channels, resulting in a uniform stack temperature. *See Section 4.3.3, Page 56 for more information on the uniform stack temperature requirement.* **3.** The airflow through the stack must be in the proper direction. *See Section 4.2.5, Page 52 for more information on airflow direction.* **4.** If an inlet air filter is required for the application, the enclosure ducting should only allow filtered air to flow through the stack. **5.** Any electrically conductive enclosure components must not come into contact with stack cells or bus plates. *See Section 3.4.3, Page 44 for more information on electrical isolation.* **6.** The enclosure must prevent any foreign objects from contacting the stack and potentially electrically shorting the stack to the enclosure. **7.** The enclosure should protect the stack from environmental pollutant such as dust, oils, etc.
Considerations	**1.** The distance required between the stack and the fan is a function of the fan and the ducting geometry. As described in *Section 4.3.3, Page 56,* the stack air-outlet face temperature should be measured to verify the uniformity of the coolant flow. **2.** Weather-stripping (closed-cell foam, 1-side adhesive) is recommended to seal the stack against the enclosure.

3.1.4.2 Acceptable Stack Enclosure Materials

Requirements	**1.** The enclosure should be constructed from materials that do not chemically damage the stack. **2.** Do ***NOT*** use the following material in the system: **A.** Materials that off-gas (usually indicated by an odour) **B.** Wet Paint (any wet paint should be given sufficient time to dry before the stack is installed in the system) **3.** If there are questions regarding a material's suitability, contact Ballard.

⚠ CAUTION **Materials that off-gas, usually indicated by an odour, will damage the stack. If a material has an odour, keep it away from the stack.**

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3.1.4.3 Ballard Lab Enclosure

1. The figure below is a photograph of the enclosure used in the Ballard lab. It is made from polycarbonate, and is adjustable to accommodate different stack sizes. The foam used to seal the stack against the polycarbonate is closed-cell foam.
2. The enclosure is designed for ebm-pabst W1G180-AB31-01 fans.
   a. Stack sizes of 45 cells or less require 1 fan.
   b. Stack sizes greater than 45 cells require 2 fans.
3. Drawings of the Ballard lab enclosure are available from Ballard.
   a. (DRW5109521 – 1 fan)
   b. (DRW5109533 – 2 fan)
4. The distance between the stack and fan is oversized in the Ballard enclosure.

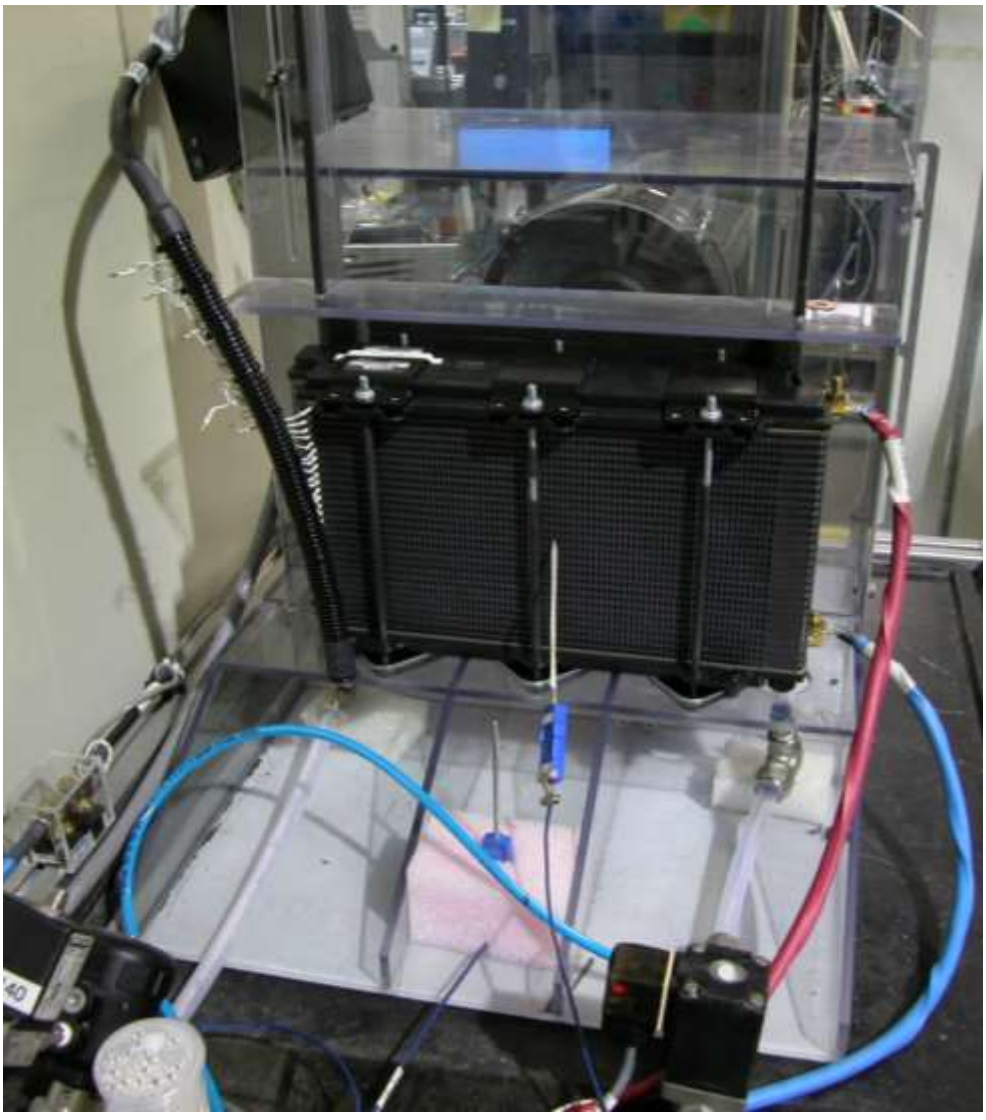


Figure 14: Ballard Lab Enclosure

3.2 Oxidant/Coolant Subsystem Configuration

3.2.1 Oxidant / Coolant Subsystem Components

1 **Inlet Air Filter (Application Dependent)** – Filters the inlet air for both chemical and particulate contaminants.

2 **Fan** – Moves air through the stack to provide oxygen for the fuel cell reaction as well as to cool the stack. (*See Section 4.4.1, Page 58*)

3 **Stack Thermistor –** Measures stack temperature to control coolant flow.(*See Section 3.2.4, Page 37*)

4 **Ambient Temperature Sensor (Recommended)** – Measures ambient temperature.

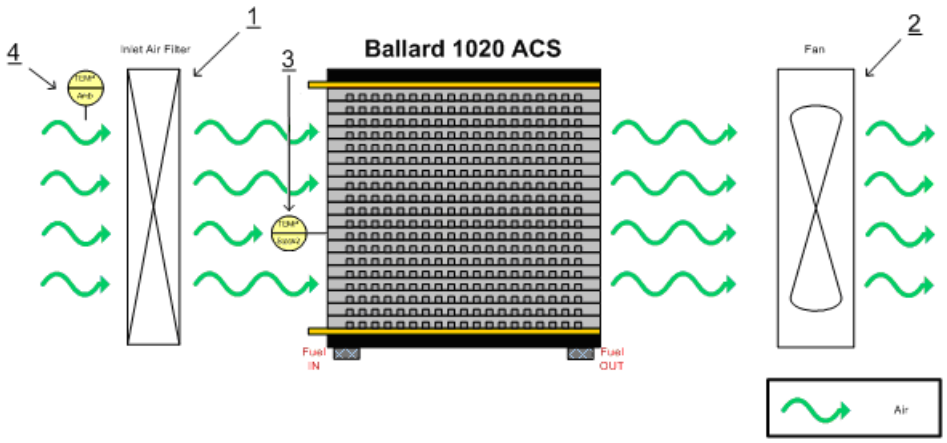


Figure 15: Oxidant/Coolant Subsystem Configuration in a Typical System

See Section 4.0, Page 49 for more information on the air stream.

3.2.2 Oxidant / Coolant Subsystem Stack Connections

Ducting is attached to the stack cathode inlet and outlet to deliver coolant/oxidant air.

Requirements	**1.** Materials that contact the stack plates must be nonconductive to prevent electrical shorting. **2.** The airflow through the stack must be in the proper direction. *See Section 4.2.5, Page 52 for more information on proper airflow direction.*
Considerations	**1.** The seal between the stack and the ducting should be tight to prevent air leakage. **2.** Foam strips (weather-stripping-type material) can be used to form a seal between the stack and the ducting. Use closed-cell foam, 1-side adhesive.

3.2.3 Acceptable Oxidant / Coolant Subsystem Materials

Materials that can safely interact with the oxidant / coolant stream are the same as the acceptable stack enclosure materials. *See Section 3.1.4.2, Page 34 for acceptable materials.*

3.2.4 Stack Thermistor(s)

1. Each 1020ACS stack includes a stainless steel single thermistor assembly that are permanently attached to the stack to measure stack temperature.
2. The stack temperature is used to control stack coolant airflow. For more information on stack temperature and coolant flow, *see Section 4.3, Page 54*.
3. The stack thermistor also used to detect stack-temperature alarm conditions. For more information on stack temperature alarms, *see Section 8.7.1, Page 126*.
4. The Temperature vs. Resistance curve for the thermistor(s) element can be found in *Appendix B:, Page 147.*
5. During calibration, the stack temperature measured by the system should be confirmed using an external thermocouple, as described in *Section 11.1.8, Page 138*.

Requirements	
	1. The thermistor electrically isolated from the stack, and should only be electrically connected to the system control electronics. **2.** The resistance of the stack thermistor must be measured by the system control electronics. This resistance must then be converted to temperature, using the Temperature vs. Resistance curve in *Appendix B:, Page 147.*

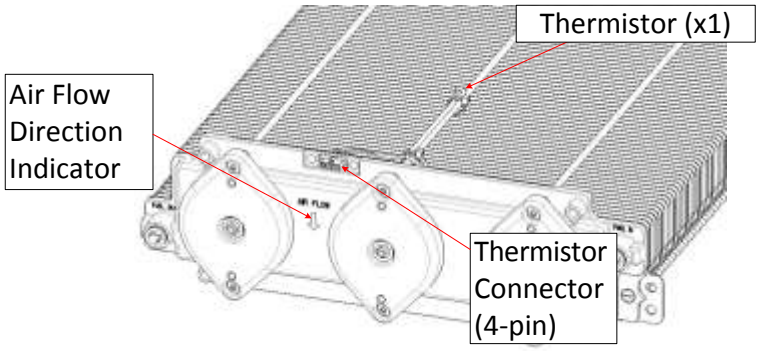


Figure 16: Airflow Direction and Stack Thermistor

3.2.4.1 Thermistor Wiring

Stack Thermistor 4-Pin Connector (Ballard Supplied) and 4-Pin Mating Connector (Customer Supplied)	
Stack Connector	Molex 4-Pin
System Connector Housing	Molex P/N 43645-0400
System Connector Terminal	Molex P/N 43030-0002

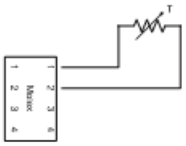


Figure 17: Thermistor Wiring

A Temperature vs. Resistance curve for the thermistor(s) can be found in *Appendix B:, Page 147.*

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3.3 Fuel Subsystem Configuration

3.3.1 Fuel Subsystem Components

1 **H_2 Supply** – Stores hydrogen fuel.

2 **Supply Pressure Sensor** – Measures the pressure of the H_2 Supply.

3 **Pressure Regulator** – Regulates the supply pressure down to the required anode pressure. (*See Section 5.6.1, Page 70*)

4 **Supply Valve** – Controls hydrogen flow to stack. (*See Section 5.6.3, Page 70*)

5 **Anode Pressure Sensor** – Measures the stack anode pressure. (*See Section 5.6.2, Page 70*)

6 **Purge Valve** – Controls hydrogen flow out of the stack. (*See Section 5.6.4, Page 71*)

7 **Purge Outlet** – The location of the purge gas outlet. (*See Section 5.5, Page 69*)

Pressure Relief Valve (not shown, optional) – The pressure relief valve is between the pressure regulator and the supply valve. It protects the stack if the pressure regulator fails and exposes the stack to high anode hydrogen pressure. (*See Section 5.6.5, Page 71*)

Accumulator (not shown, optional) – The accumulator is a vessel between the pressure regulator and the supply valve. It is an H_2 reservoir that helps to maintain anode pressure during a purge, resulting in a more effective purge. (*See Section 5.6.6, Page 71*)

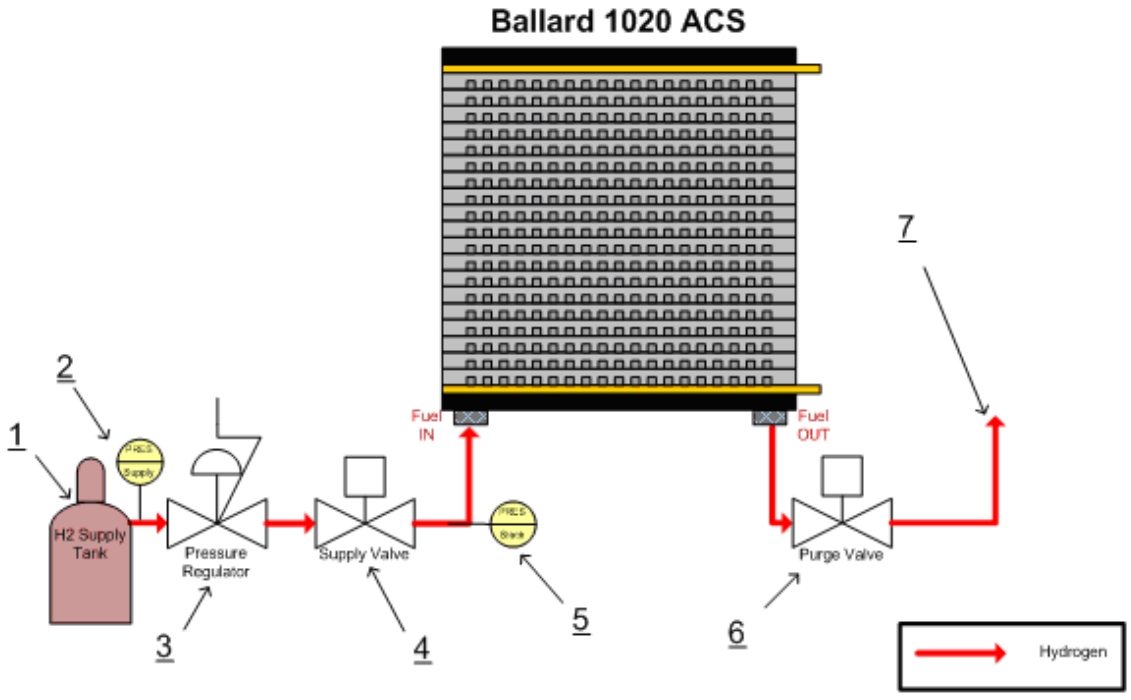


Figure 18: Fuel Subsystem Configuration in a Typical System

See Section 5.0, Page 61 for more information on the fuel stream.

3.3.2 Fuel Subsystem Stack Connections

Tubing delivers dry hydrogen to the stack anode inlet and tubing removes impurities from the stack anode outlet.

Requirements

1. The inlet and outlet connections must be made to the proper stack port. *See Section 5.2.6, Page 63.*
2. The anode inlet and outlet ports accept 5/16” (8 mm) OD tubing.
   Stack fitting: John Guest PCM2808S “Push-In”, 5/16” (8 mm) OD.
3. The tubing must be inserted 19 mm into the fitting from the top of the fitting (17 mm from the face of the end-plate) to ensure a proper seal.
4. Teflon tubing is recommended.
5. Metal tubing should not be used to connect to the stack because:
   A. The metal is electrically conductive and could electrically interact with the stack. The metal tubing can contact the bus plate, electrically connecting the stack to the fuel supply.
   B. The metal could contaminate the stack. (Stainless steel corrodes in a wet H_2 environment)
6. Side-load or pull forces on the tubing should not exceed 20 N. Rather than bending tubing to change direction, an elbow is recommended.

Considerations

1. Anode tubing lengths should be minimized. During the start-up purge, the gas in anode volume, including the tubing volume, needs to be exchanged with fresh H_2 gas from upstream of the stack. Reducing the tubing volume reduces the flow requirements. *See Section 5.3, Page 65.*
2. Plumbing downstream of the stack should slope downwards to ensure liquid water does not collect in the tubing.
3. After shutdown, water can collect in the anode tubing. In below-freezing ambient temperatures, there is potential for the water in the tubing to freeze, blocking gas flow. Heat tracing can be used to prevent the water in the tubing from freezing.

3.3.3 Acceptable Fuel Subsystem Materials

Hydrogen gas exiting the stack will be humidified, and minor quantities of liquid water may be present. The allowable material requirements are therefore different for upstream and downstream of the stack.

Dry or Humid Hydrogen

(both upstream and downstream of stack)

- Teflon
- PVDF
- Neoprene
- Glass
- Low sulphur content EPDM
- Viton
- Aluminum

Dry Hydrogen

(upstream of stack only)

- Brass
- Copper
- Stainless steel
- Carbon steel
- Zinc
- Buna N

Figure 19: Acceptable Fuel Subsystem Materials

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3.4 Electrical Configuration

3.4.1 Power Electronics, Energy Storage, and Electrical Components

The following schematic if for a typical back-up power system:

1 **FC Contactor** – Electrically isolates the stack from the power electronics. A contactor is recommended because applying a voltage to the stack or drawing current from the stack after shutdown can cause irreversible damage.

2 **DC-DC Converter** – Converts the stack voltage to energy storage voltage as well as controls the current drawn from the stack. Also known as power electronics.

3 **Energy Storage** – Provides power to the balance of plant and/or customer load while the stack starts-up. The stack can take on the order of minutes to reach rated power. The energy storage is usually either batteries or super-caps, and the energy storage should be oversized. *See Section 8.3.2, Page 106 for more information on bridging energy requirements.*

4 **Load Contactor** – Electrically isolates the system from the load.

5 **System Controller** – Used to control the stack and system, including I/O.

6 **Auxiliary Power Supply** – Used to power the fan, controller, valves, contactors, and other electronics and actuators in the system.

7 **Stack Current Sensor** – Measures the stack current.

8 **Stack Voltage Sensor** – Measures the stack voltage.

9 **Cell Voltage Monitor (CVM) (optional)** – Measure the individual cell voltages. Recommended for development testing and initial prototypes. Not required for final products. *See Section 3.4.2.4, Page 43*.

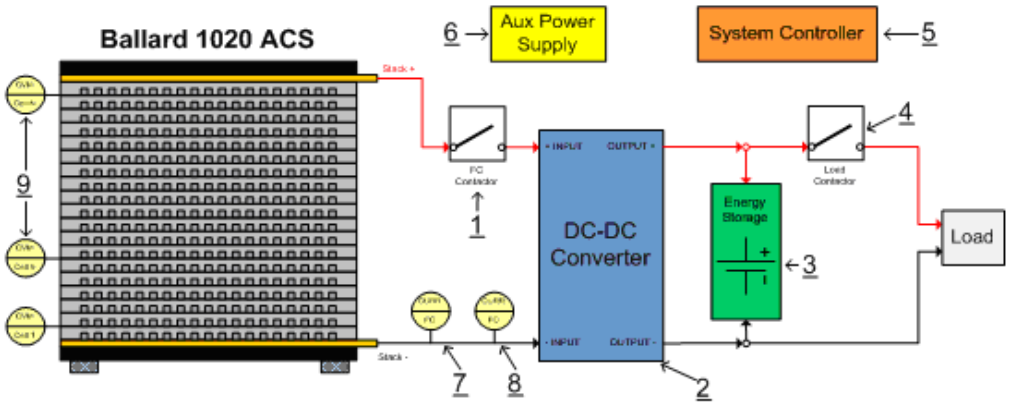


Figure 20: Electrical Configuration in a Typical System

3.4.2 Electrical Connections

3.4.2.1 Electrical Stack Connections – Bus Plates

The stack delivers electrical energy to the system through the bus plates. Cable lugs are used to electrically connect the bus plates to the rest of the system.

Requirements	
	1. The positive power lead must be connected to the positive bus plate tab. The positive bus plate is on the compression-strap end of the stack, on the opposite end of the stack as the anode inlet and outlet. **2.** The negative power lead must be connected to the negative bus plate tab. The negative bus plate is on the spring-cap end of the stack, on the same end of the stack as the anode inlet and outlet **3.** All connections must be made to the gold plated side of the bus plates. **4.** M6 hardware should be used to connect the lug to the bus plates, torqued to 5 N·m.

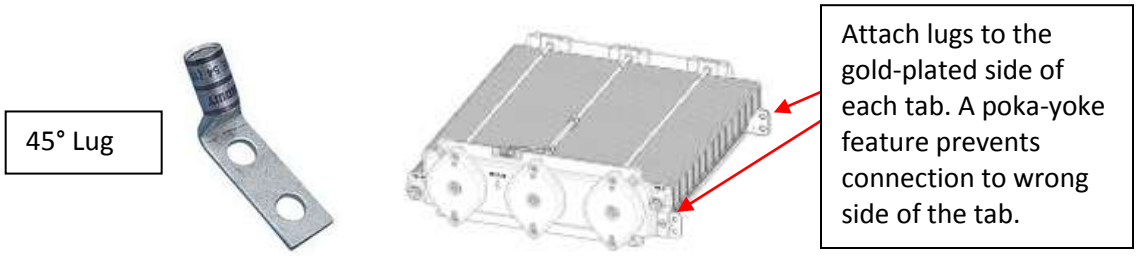


Figure 21: Bus Plates and Lugs

3.4.2.2 Electrical Stack Connections – Recommended Lugs

Manufacturer	Straight	45°	90°
Panduit	LCD6-14A-L	LCD6-14AH-L	LCD6-14AF-L
Thomas & Betts	54205	54205UF	54205UB

3.4.2.3 Electrical Stack Connections – Ripple Currents

The power electronics have the potential to impose ripple currents on the stack.

Requirements	
	1. The Balance-of-Plant must not impose a ripple current on the 1020ACS stack that is greater than 10% of the average stack current. **2.** Ripple currents below ~400 Hz may cause the stack to be starved of reactants at each current peak. **3.** Ripple currents above ~100 kHz may cause the stack to radiate EMI. **4.** The Balance-of-Plant must never apply a voltage or impose a reverse polarity on the 1020ACS stack.

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3.4.2.4 Electrical Stack Connections – Cell-Voltage-Monitoring

Cell-voltage-monitoring (CVM) measures individual cell voltages. A CVM is not required, but is recommended during system development. The CVM is very useful for diagnosing stack problems, especially in a laboratory set-up.

Requirements	**1.** To measure an individual cell voltage, an electrical connection needs to be made to the plate on either side of the cell. For an N cell stack, N+1 connections are required to measure every cell voltage. The positive connection of cell N is the same connection as the negative connection of cell N+1. **2.** The CVM connections must be positioned to not cause an electrical plate to plate short. **3.** Each CVM channel should be capable of measuring a 0.0 V to 1.0 V signal. In the Ballard lab, the CVM channels measure from -2.0 V to 2.0 V.
Considerations	**1.** The plate is highly conductive and is therefore at a constant voltage, allowing the CVM connection to be made anywhere on the plate. However, CVMs are traditionally placed on the anode outlet end of the stack, opposite the bus plate tabs.

3.4.2.5 CVM Connections in Ballard's Lab

1. In the Ballard lab, CVM connections are made by inserting a Molex crimp into the cathode channel of each plate.

2. The crimp is Molex P/N 08-56-0110. Each crimp must be modified by bending a part of the crimp to prevent sharp edges from damaging the cell.

3. Care must be taken to ensure that any probes inserted into the cooling channel do not scratch or damage the exposed MEA surface. Scratches can cause anode leaks, resulting in premature performance degradation and stack failure.

4. Other techniques, such as using spring-loaded-test-pins (pogo pins), have also been used successfully.

5. Each measured cell requires a differential DAQ input channel. If the stack is large enough, channel-to-channel isolation may be required.



Figure 22: CVM Connections in Ballard's Lab

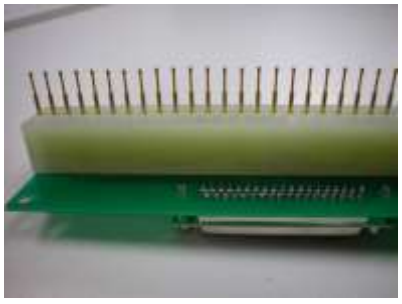


Figure 23: Spring-Pin CVM Connections

3.4.3 Electrical Isolation

1. Ballard recommends installing the 1020ACS stack as a floating electrical system. Under some circumstances it may be acceptable to connect one of the buses to earth. Contact Ballard for details.
2. Since the stack will normally be integrated as a floating electrical system, a means should be provided of draining any potential static discharge from the stack. This can be practically achieved by adding a high-impedance resistance to earth on either the negative or positive stack voltage terminals.



Non-current carrying metal parts of the compression hardware must be either:

A. electrically bonded,

B. OR, in the event that such parts are not bonded, treat the metal parts as electrically live. In this case, the system integrator must provide a means to: 1) isolate the module's compression hardware from the integrator's system, and 2) prevent people or objects from touching the live parts 3) provide a label/marking warning the service personnel that the compression hardware may be electrically live.

3. The maximum stack voltage for electrical isolation design of the system is the stack Open Circuit Voltage (OCV), as described in *Section 6.1.2, Page 75*. The 1020ACS stack is designed with sufficient internal isolation to satisfy a voltage withstand test at 620 V in a clean, low humidity, sea level environment (this satisfies the requirements of IEC 62282-2 for a working voltage of up to 80 V).
4. The primary current leakage paths for the 1020ACS stack are those typical of any high-voltage electrical equipment. Theoretically, there is also a potential for carbon tracking at the reactant outlets- that is, creation of a potential current leakage path through carbon that sheds from stack anode or cathode and deposits on anode or cathode exhaust lines. While carbon deposits have been noted on the anode exhaust line of the 1020ACS stack operating in the dead-ended configuration, carbon tracking has not been observed. Nonetheless, Ballard recommends that the integrator ensure there is enough creepage distance between the anode exhaust fitting and the purge valve.



The fuel cell stack has a potential for current leakage across clearances and creepage distances. The fuel cell stack installation should be designed with adequate clearances and creepage distances as listed in applicable standards

5. There is a greater possibility of a carbon track forming at the anode outlet if the stack is operated in an anode flow-through configuration with humidified anode inlet gas (the anode inlet gas is humidified if anode recirculation is used.)
6. The stack withstand voltage may need to be derated if the 1020ACS stack is to operate in a polluted environment or at higher altitudes.

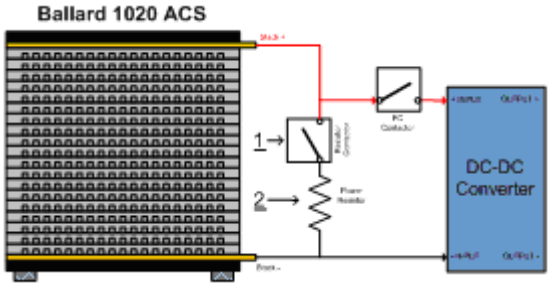
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3.4.4 Alternative Electrical Configurations

3.4.4.1 Power Resistor

Two potential uses for a **power resistor** are:

1. **Load-on-Start to increase stack life**. (*See Section 8.2.3, Page 103*)
2. **Air-Starve to improve stack performance**. (*See Section 8.6, Page 118*)



1 **Power Resistor Contactor** – Electrically isolates the power resistor from the stack.

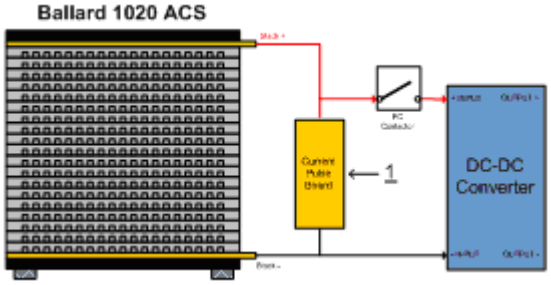
2 **Power Resistor** – Draws load from the stack at a constant resistance.

Figure 24: Power Resistor

3.4.4.2 Current Pulse Board

The purpose of a **current pulse board** is:

1. **Briefly short-circuit the stack, improving stack performance**. (*See Section 8.6, Page 118*)



1 **Current Pulse Board** – Shorts the stack in a controlled manner.

Figure 25: Current Pulse Board

3.4.4.3 No Power Electronics

1. In this configuration, the stack is connected directly to the energy storage through a contactor, and a DC-DC converter is not used.
2. When the contactor is closed, the stack voltage is fixed at the energy storage voltage, and the stack current is a function of the stack voltage.

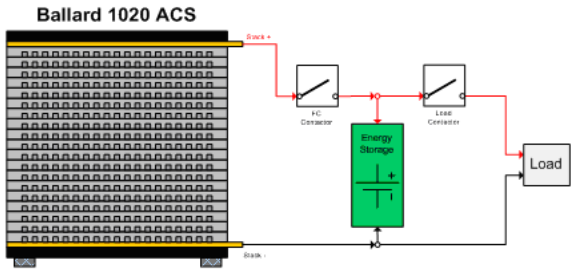


Figure 26: No Power Electronics

3.5 Configurations for Multiple Stacks

If the total system power output requirement cannot be met with a single 1020ACS stack, multiple 1020ACS stacks can be used. If the application does not require more than one stack, this section can be omitted. *See Section 9.0, Page 131 for more information on stack sizing.*

3.5.1 Oxidant/Coolant Subsystem Configurations for Multiple Stacks

1. A dedicated fan for each stack provides individual temperature control for each stack. This is the preferred and recommended approach.
2. A single fan for multiple stacks is possible, but more development effort is required to ensure even airflow to both stacks. The stacks should be in parallel with respect to the airflow, not series. If the airflow is not even between stacks, the stack temperatures will be different temperatures, and the stacks need to be operated at different currents. *See Section 4.3.2, Page 55 for more information on optimum stack temperature.*

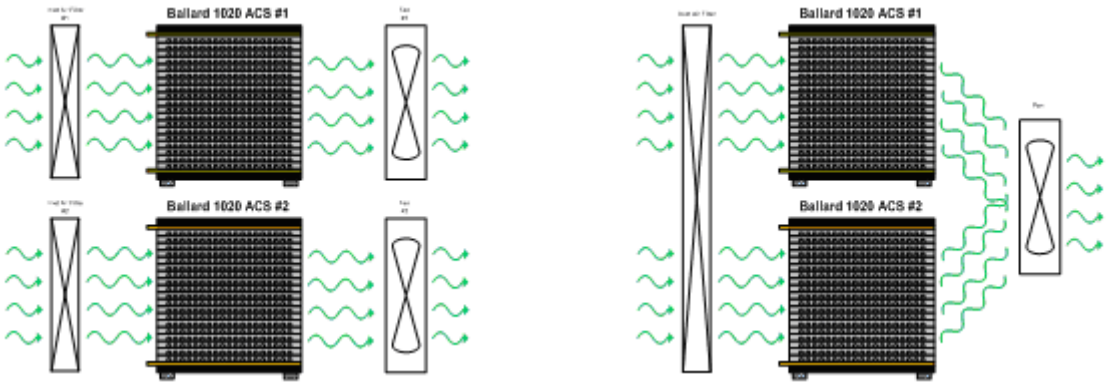


Figure 27: Oxidant/Coolant Subsystem Configurations for Multiple Stacks

3.5.2 Fuel Subsystem Configurations for Multiple Stacks

1. A dedicated anode stream is required for each stack to balance fuel flows between stacks and avoid fuel starvation and under-purging.
2. The stack should be purged at different times to prevent the anode pressure from drooping if both stacks are purged at the same time. *See Section 5.3, Page 65 for more information on anode purging.*
3. If independent stack operation is not required, a single supply valve may be used.

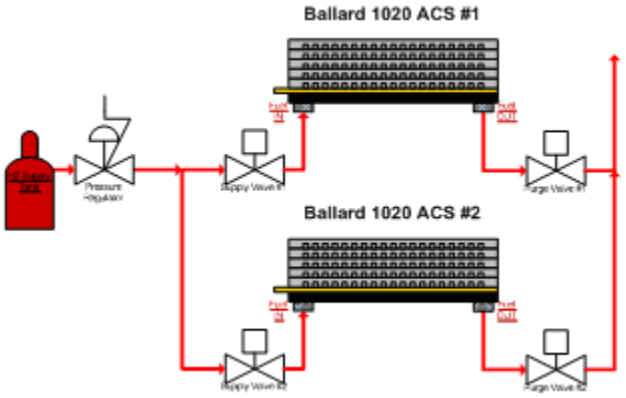


Figure 28: Fuel Subsystem Configuration for Multiple Stacks

3.5.3 Electrical Configurations for Multiple Stacks

Stacks can be configured in series, in parallel, or in parallel with multiple-input power electronics.

3.5.3.1 Multiple Stacks in Series

Advantages

1. Larger output voltage.
2. Same current drawn for both stacks.
3. Single input power electronics.

Disadvantages

1. Easy to put cells or stack in reversal (load drawn without reactants).
2. Single stack operation not possible.
3. Cell-to-cell variability within the stack is exaggerated due to variability in operating conditions.
4. Extensive testing is required to ensure a robust system.
5. Additional work is required for certification due to isolation issues.

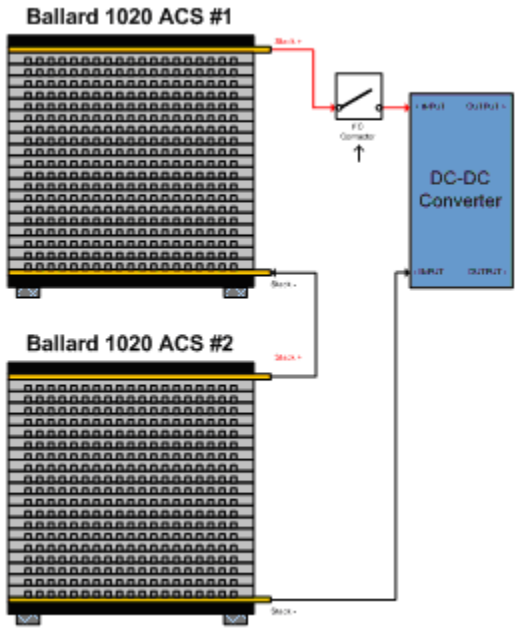


Figure 29: Multiple Stacks in Series

WARNING

Operating multiple stacks in series increases the chance of occurrence and the severity of cell reversals. Contact Ballard for more information on how to operate multiple stacks in series.

If the customer application requires two or more stacks connected in a series configuration, please contact Ballard for more information.

3.5.3.2 Multiple Stacks in Parallel

Advantages

1. Single stack operation possible.
2. Single input power electronics.

Disadvantages

1. Extra BOP components required to start-up stack independently.
2. The stack currents will not be equivalent because the stacks will perform differently.

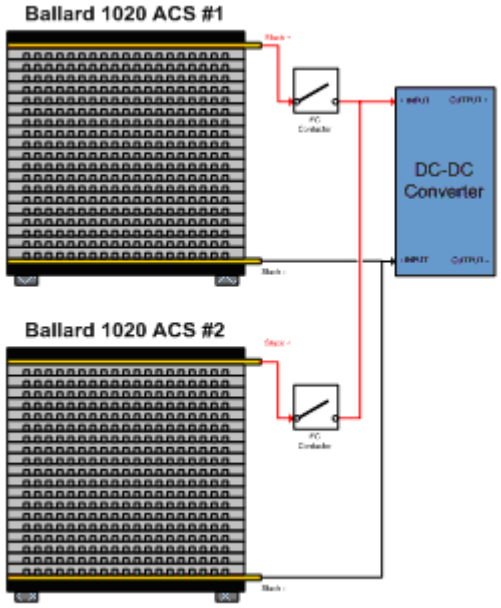


Figure 30: Multiple Stacks in Parallel

3.5.3.3 Multiple Stacks in Parallel, Multi-Input Power Electronics

Advantages

1. Single stack operation possible.
2. Independent current draw from each stack.

Disadvantages

1. Extra BOP components required to start-up stack independently.
2. Dual-input DC-DC converter is required.

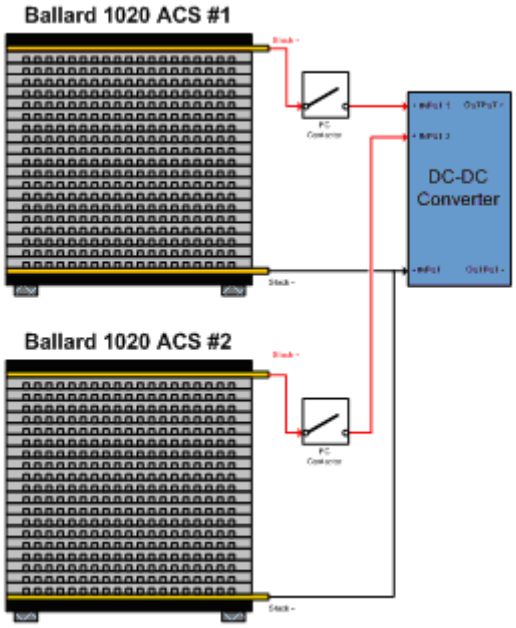


Figure 31: Multiple Stacks in Parallel, Multi-Input Power Electronics

4.0 Air Stream

The 1020ACS's open cathode design requires airflow through the stack. The air stream's two major functions are:

1. **Provide oxidant for the fuel cell reaction** – O_2 gas is required by the stack to complete the fuel cell reaction and create electrical energy. The O_2 gas is consumed from the air stream.
2. **Provide coolant for the stack** – Stack operation result in the production of waste heat that must be removed to maintain the stack at the proper temperature. Heat is rejected from the stack into the air stream.

The air stream specification is broken into two parts: the *Oxidant Specification*, which specifies the composition of the oxidant flow for stack operation, and the *Coolant Specification*, which specifies the amount of airflow needed to cool the stack and maintain it at the proper temperature.

In most situations, the coolant flow requirement far exceeds the oxidant flow requirement, and therefore the coolant flow requirement is used to control the airflow through the stack.

See Section 3.2, Page 36 for information on air stream system considerations.
See Section A.4, Page 145 and Section A.5, Page 146 for oxidant and coolant equations

4.1 Air Stream Configuration

The air stream is configured as shown in the figure below.

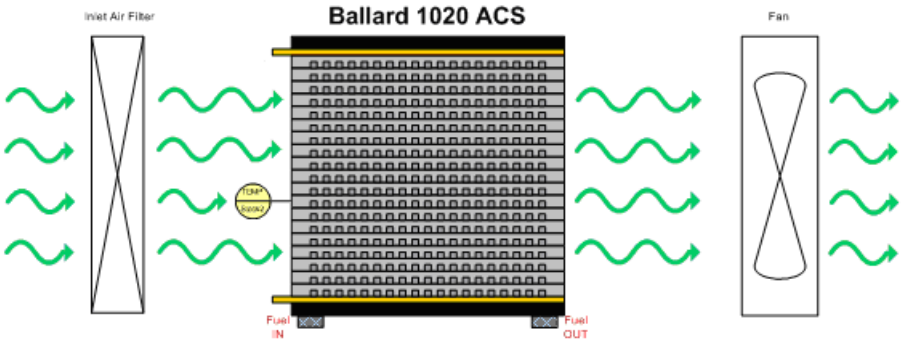


Figure 32: Oxidant/Coolant Stream Configuration

The key components in the air stream are the cooling fan, the stack thermistor(s), and an optional inlet air filter. More uniform airflow through the stack is achieved if the fan is downstream of the stack, pulling air through the stack, than if the fan is upstream of the stack, pushing air through the stack. *See Section 4.3.3, Page 56 for more information on coolant flow distribution.*

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4.2 Oxidant Specification

4.2.1 Oxidant Inlet Temperature

Specification	**Required:** -20°C to 52°C **Optimal:** 10°C to 40°C
Low Oxidant Inlet Temperature	**1.** At low oxidant inlet temperatures, a cooling fan at its minimum set-point tends to delivers too much coolant flow, overcooling the stack. *See Section 4.3.2, Page 55 for more information on cooling flow.* **2.** At low oxidant inlet temperatures, large temperature gradients exist within a cell between the air inlet and the air outlet, resulting in: **A.** **Derated performance** - A 1020ACS stack performs optimally when the entire stack is at the optimum stack temperature. *See Section 4.3.3, Page 56 for more information on stack temperature distribution.* **B.** **Cell flooding in overcooled regions of the stack** - *See Section 6.5.2, Page 86 for more information flooding and cell reversals.* **C.** **Localized freezing in overcooled regions of the stack (worst case)** If the stack freezes while running, the result is usually severe damage to the stack. **3.** At low oxidant inlet temperatures, a minimum stack current is required because at low stack currents, the stack might not produce enough heat to maintain temperature, causing the stack voltage to crash. **4.** Stack temperatures above -10°C are required to start-up the stack. *See Section 8.1.2, Page 99 for more information on cold starts.*
High Oxidant Inlet Temperature	**1.** At high oxidant inlet temperatures, the cooling fan may not be able to deliver enough flow to cool the stack to its optimum temperature. *See Section 4.3.2, Page 55 for more information on cooling flow requirements.*

4.2.2 Oxidant Pressure / Altitude

The coolant/oxidant air is close to atmospheric pressure during stack operation. Rather than specifying air pressure, the operating altitude is specified because it is a more useful parameter.

Specification	**Optimal:** 0 m (Sea Level) **Acceptable:** -400 m to 7600 m
High Altitude Operation	**1.** At high altitude, there is a performance derate resulting from the reduction in cathode pressure. *See Section 6.5.1, Page 83 for more information on altitude.* **2.** At high altitude, the optimum stack operating temperature will shift slightly lower.

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4.2.3 Oxidant Relative Humidity

Specification	**Acceptable:** 0% RH to 100% RH (Non-Condensing)
Low RH Operation	**1.** The stack is designed to run over the full range of environmental relative humidities. **2.** During operation, the stack generates enough water to sufficiently hydrate the membrane. **3.** Stack performance and start-up characteristics are degraded by low oxidant RH. **A.** Less than 5% is significant. **B.** Less than 2% is severe.

See Section 8.1.3, Page 100 for more information on low RH stack start-ups.

4.2.4 Oxidant Stoichiometry / Airflow

Specification

Stoichiometry	Performance Loss
Greater than 50	No performance Loss
20 to 50	Some performance loss
Less than 20	Significant performance loss
Less than 10	Will not operate above ~5 A (*see 4, below*)

1. When applied to the oxidant flow through a fuel cell, the *stoichiometric rate* is defined as the required airflow to provide the exact number of oxygen molecules to complete the fuel cell reaction for a given current. The *stoichiometry* (or *stoich*) is defined as the total airflow divided by the stoichiometric rate flow. *See Section A.4, Page 145 for stoichiometry equations.*

2. Except for cold ambient-air situations, the cooling airflow requirement exceeds the oxidant stoichiometry requirement, resulting in the stack being at a stoichiometry that does not cause performance loss.

3. Additionally, if the cooling fan is sized properly to provide adequate cooling flow at high temperatures, the fan at its minimum set-point usually provides enough oxidant flow to avoid performance loss. A stoich of 20 should be used when setting the minimum fan command during start-up.

4. If the stack is operating without a cooling fan, natural convection through the stack can sustain a current of no more than 5 A.

A. The amount of current that can be drawn is dependent on ducting geometry and environmental conditions.

B. Variable cell performance should be expected, and cells must not be allowed to fuel-starve and go into reversal. *See Section 6.5.2.1, Page 86 for more information on fuel starves and cell reversals.*

C. Ensure that the stack does not overheat during operation without a fan. Operating the stack for prolonged periods without coolant will result in a performance decrease.

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4.2.5 Oxidant Flow Direction

The oxidant flow direction must be in the direction as labeled on the stack end plate, as shown in the figure to the right.

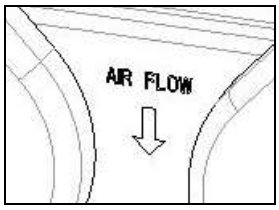


Figure 33: Oxidant Flow Direction – Stack Label

4.2.6 Oxidant Quality – Chemical

1. The stack is capable of meeting its performance and durability specification if the air stream meets the *nominal* air quality specifications in the table below.
2. The stack can function with higher levels of oxidant chemical contamination than *nominal*, but performance and durability may be degraded. If operating above nominal contamination levels, performance loss is expected. Depending on the type of chemical contamination, the performance loss may be recoverable.
3. Operation with oxidant that exceeds *peak* contamination levels should be avoided.
4. In general, the most damaging oxidant contaminants are sulphur-containing compounds, such as SO_2 or H_2S, which can cause irreversible stack performance loss.
5. Performance loss associated with volatile-organic-compounds (VOCs), such as benzene and toluene, is in most cases irreversible.
6. Nitrogen Oxides (NO_x) are commonly-occurring pollutants that can also cause significant performance degradation, but testing has shown that NO_x performance loss is reversible.
7. Reversible performance loss can be recovered though air-starves, current pulses, and standard start-up and shutdown cycles. *See Section 8.6, Page 118 for more information on recovering reversible performance loss.*
8. Performance degradation resulting from oxidant contamination is non-linear. If the contaminant concentration is constant, the initial performance degradation rate will high, but the rate decreases as exposure continues. The performance loss will eventually reach a limit determined by contaminant concentration, and over periods of greater than 4hrs, the performance degradation rate will be approximately linear.
9. For most applications, pollution-based lifetime performance degradation will not be significant, so chemical filtration should not be required.
10. If a chemical filter is required, the filter should be capable of removing contaminants such that the oxidant air entering the stack is at or below *nominal* levels in below table. The average airflow through the filter will be required to properly size the filter. *See Section 4.4.1, 58 for more information on stack airflow requirements.*

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4.2.7 Oxidant Quality – Chemical Specification

Chemical	Nominal	Peak
Sulphur Dioxide (SO_2)	< 0.01 ppm	<0.18 ppm
Nitrogen Monoxide (NO)	< 0.025 ppm	<0.20 ppm
Nitrogen Dioxide (NO_2)	< 0.05 ppm	<0.25 ppm
Volatile Organic Compounds (e.g. Benzene C6H6, Toluene C7H8)	< 0.008 ppm	<0.050 ppm
Hydrogen Sulphide	< 0.04 ppm	<0.04 ppm
Ammonia	< 0.01 ppm	<0.1 ppm
Ozone	< 1 ppm	<1 ppm
Carbon Monoxide	< 5 ppm	<32 ppm
Carbon Dioxide	< 1% vol.	N/A

Figure 34: Oxidant Quality Specification – Chemical

4.2.8 Oxidant Quality – Particulate

1. Similarly to chemical contaminants, operating at particulate contamination levels below those specified in the below table will guarantee stack performance over the full product lifetime. However, testing has demonstrated that the stack is quite robust to particulate contamination.
2. The particle sizes in the air stream that tended to adhere on to the cathode GDL were between 0.5μm and 30μm. Therefore, any filtration for the air stream should focus on particle sizes from 0.5 to 30μm.
3. Particulate contamination at a concentration of up to 90 mg/m³ can be expected to result in a performance loss of only ~3% at 65 A over a 1500 hour lifetime. This is based on an observed experimental performance loss of ~20 mV/cell (3%) at 65.3 A after 160 hours of particulate exposure at a concentration of 1.1 mg/m³. Voltage loss due to particulate contamination becomes worse at higher currents.
4. In some applications, coarse filtration of the inlet air is required to prevent large foreign objects, such as leaves, lint, etc., from obstructing flow into the stack.

4.2.9 Oxidant Quality – Particulate Specification

Particulate Size	Nominal	Peak
Coarse Particles (PM 10)	< 90 $\mu g/m^3$	N/A
Fine Particles (PM 2.5)	< 15 $\mu g/m^3$	N/A

Figure 35: Oxidant Quality Specification – Particulate

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4.3 Coolant Specification

4.3.1 Coolant Flow – Optimum Stack Temperature - Theory

1. Waste heat created by the stack is rejected to coolant air flowing through the stack.
2. The stack optimum temperature (T_{opt}) is a function of stack current. The coolant flow rate is controlled by comparing the stack temperature with a set-point.
3. For a given stack current, the stack voltage will be at a maximum at T_{opt}.
4. Water is created in the stack though the fuel cell reaction. The higher the current, the more water is produced. Water production is proportional to stack current.
5. Water is removed from the stack by the coolant air flowing through the stack. The higher the air temperature, the more water is removed. Water removal is proportional to stack temperature.
6. For a fixed current, adjusting the stack temperature adjusts the amount of water in the stack.
7. If the stack is operated at temperatures above T_{opt}, the MEA will dry out and become more electrically resistive, reducing stack performance. At extreme temperatures, the membrane can become damaged, resulting in irreversible performance loss.
8. If the stack is operated at temperatures below T_{opt}, the MEA will be over-humidified, resulting in the presence of liquid water in the cells. Too much water can cause anode "flooding", where the water blocks reactants from reaching the catalyst layer. This can cause irreversible performance loss though localized fuel starvation and cell reversals. *See Section 6.5.2, Page 86 for more information on flooding and cell reversals.*
9. T_{opt} will slightly change with altitude, ambient temperature, ambient RH, stack-to-stack variability, etc. However, if the stack is maintained at T_{opt}, as described in *Section 4.3.2, Page 55*, the stack will function properly.
10. The coolant flow needs to be delivered uniformly to the stack coolant channels, as described in *Section 4.3.3, Page 56*.
11. At low inlet-air temperatures, the fan at its minimum set-point will often deliver too much coolant flow, overcooling the stack. *See Section 8.4.4, Page 113 for more information fan operation.*



The fuel cell stack can reach temperatures of 70°C or higher if operated outside the specification. Avoid touching the stack during or shortly after operation.



Operating at temperatures higher than 75°C may result in stack damage and/or cell reversals due to cell dehydration. Stack current should be reduced before the stack reaches this critical temperature, and operation of the fuel cell stack should be stopped if the stack temperature exceeds 75°C.

4.3.2 Coolant Flow – Optimum Stack Temperature

T_{opt} changes with oxidant inlet temperature, RH, and altitude. The values given below were chosen for the suitability over a wide range of conditions. *See Section 8.7.1, Page 126 for more information on maximum and minimum stack temperature alarm values.*

Specification

Optimum Temperature

Optimum: $Topt = 0.53 \cdot i + 26.01$ i = stack current

Maximum Temperature

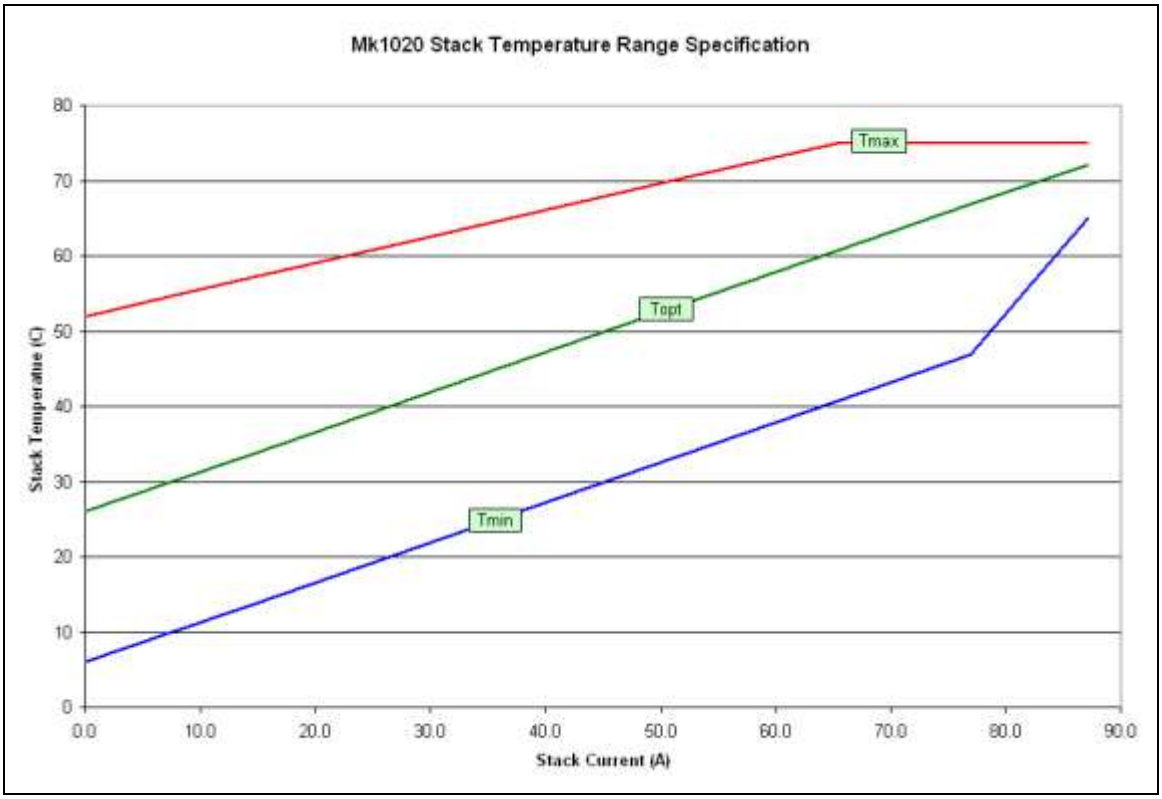
Continuous: See Below Table, 60 sec delay

Instantaneous: 75°C, 5 sec delay

Minimum Temperature

Continuous: See Below Table, 60 sec delay

Instantaneous: -10°C, 5 sec delay



Stack Current (A)	0	7.3	14.5	29.0	51.7	65.3	77.0	87.1
T_{max} (°C)	52	55	57	62	70	75	75	75
T_{opt} (°C)	**26**	**30**	**34**	**41**	**53**	**61**	**67**	**72**
T_{min} (°C)	6	10	14	21	33	41	47	65

Figure 36: Optimum Stack Temperature

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4.3.3 Coolant Flow – Distribution

1. For optimal stack performance, the stack should be at a uniform temperature. Since stack performance is a function of temperature, a uniform stack temperature ensures uniform stack performance in all regions of the stack.
2. However, due to air cooling and stack geometry, temperature gradients will always be present in the stack. Stack temperature gradients should be measured by measuring the temperature on the air outlet face of the stack.
3. In most situations, the centre of the air-outlet face of the stack will be at a higher temperature than the edges. The stack produces constant heat throughout its interior, but on the edges, there is more cooling, resulting in lower temperatures on the edges.

Specification	**Temperature Difference from End Cells to Middle of Stack: Less than ~8°C** **Temperature Difference along Length of Cell: Less than ~6°C** *At 65.3A and ~20°C Ambient Temperature, on the Stack Air-Outlet Face*

1. Temperature deviations significantly larger than the above specification are an indicator of non-uniform airflow distribution.
2. See *Section 11.1.9, Page 138* for a description of how to measure the stack air-outlet face temperature.
3. When designing the airflow ducting, care must be taken to ensure there are not any heat sources upstream of the stack that would unevenly heat the inlet air. *See Section 3.1.4, Page 34 for more information on air ducting.*
4. Regions of the stack that receive less airflow will be at a higher temperature. If the over-temperature region is not near the control thermocouple, the potential exists for the temperature in these regions to go above 75°C, causing membrane dehydration, cell reversals, permanent stack damage, and potentially stack fire.
5. If the system delivers uneven coolant flow distribution, ensure that the stack maximum temperature location is no more than 5°C hotter than the stack thermistor(s) location. Ideally, the thermistor(s) should be located at the hottest location.
6. Airflow distribution is affected by ducting geometry, fan placement, and the fan selected.
7. More uniform airflow is achieved if the fan is placed downstream of the stack, pulling air through the stack, rather than placing the fans upstream of the stack and blowing air through the stack.
8. Figure 37, below, graphically displays the stack air-outlet temperature requirement. Figure 38 shows the air-outlet temperature of a stack running at 65.3A.

WARNING

Non-uniform airflow can lead to regions of the stack operating at temperatures higher than the control thermistor(s) temperature. This may result in cell reversals due to cell dehydration, stack damage, and stack fire.

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Coolant Flow – Distribution (*continued*)

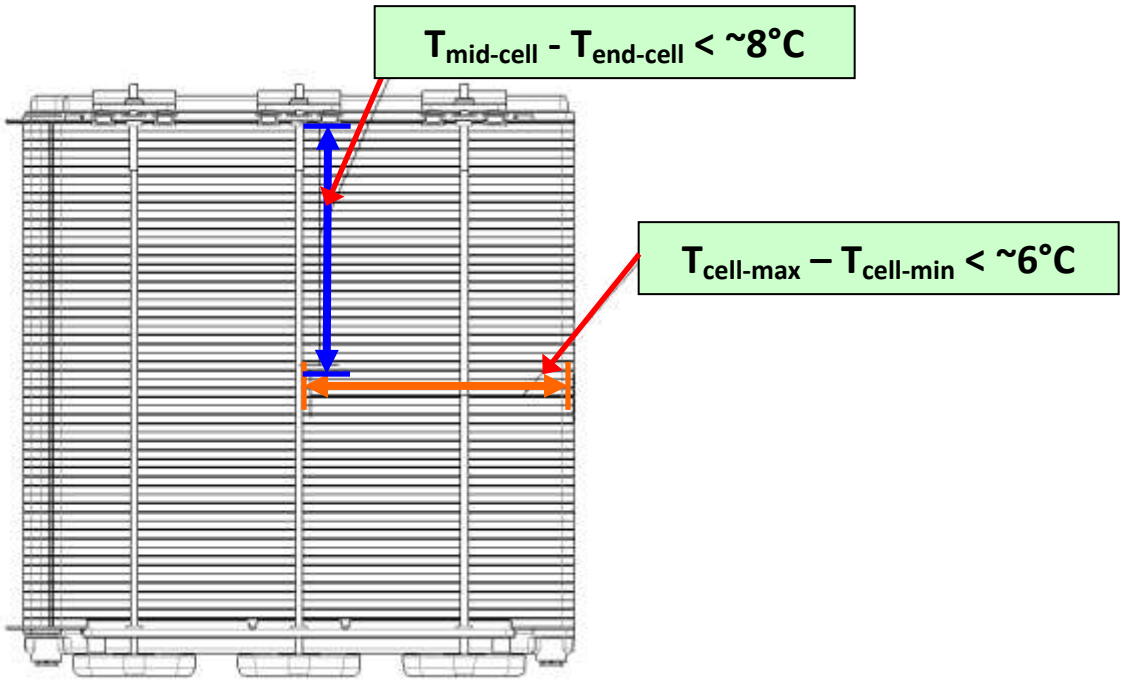


Figure 37: Stack Cathode Outlet Temperature Delta Requirements

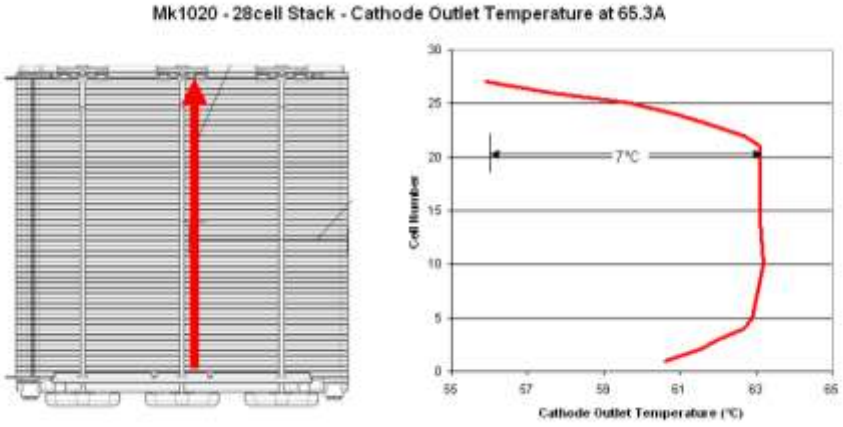


Figure 38: Stack Cathode Outlet Temperature

4.3.4 Coolant Flow – Stack Exhaust

The 1020ACS stack coolant/oxidant exhaust is a stream of slightly oxygen-depleted, humidified air. Except under unusual conditions, the exhaust should not contain liquid water.

WARNING

The fuel cell stack consumes O_2 while operating. If operating the stack in poorly ventilated, small enclosures, care must be taken that O_2 concentrations do not drop below safe levels.

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4.4 Air Stream Component Selection

4.4.1 Cooling Fan

The cooling airflow requirement far exceeds the oxidant airflow requirement. Therefore, the fan needs to deliver enough flow to adequately cool the stack at the maximum stack operating current and at the maximum inlet air temperature. All equations to properly size the fan are presented below and can also be found in *11.0, Page 134*.

4.4.1.1 Fan Selection Equations – Stack Heat Generation

Stack heat generation is a result of inefficiencies inherent to the stack. Maximum heat is produced at maximum current and minimum voltage. As the stack degrades, it becomes less efficient, and more heat and less electrical power are produced. Therefore, EOL stack parameters should be used in stack heat production calculations.

$$q_{created} = n \cdot (1.253 - V_{cell}) \cdot I$$

$q_{created}$ = heat generated (W)
n = number of cells
V_{cell} = cell voltage (V)
I = stack current (A)

Example: If an EOL stack at maximum current is assumed, (*See Section 6.2.4, Page 79*)

If $V_{cell} = 0.5V$ and $I = 75A$, then $q_{created} = 56.48\,{}^{W}/_{cell}$

4.4.1.2 Fan Selection Equations – Required Coolant Mass-Flow

To remove the heat generated by the stack, a specific mass-flow of cooling air is required.

$$\dot{m} = \frac{q_{removed}}{C_p \cdot (T_{stack} - T_{amb} - 0.403 \cdot q_{removed})}$$

$q_{removed}$ = heat removed (W)
$\dot{m}$ = air mass flow $\left({}^{kg}/_{s}\right)$
C_p = air heat capacity $\left({}^{J}/_{kg \cdot K}\right)$
T_{stack} = stack temp $(°C)$
T_{amb} = ambient temp $(°C)$

Example: If the stack is running at T_{opt} for 75A, (*See Section 4.3.2, Page 55*) and the T_{amb} is 40°C,

If $q_{removed} = 56.48\,{}^{W}/_{cell}$ and $T_{stack} = 72°C$ and $T_{amb} = 40°C$ and $C_p = 1006\,{}^{J}/_{kg \cdot K}$

Then $\dot{m} = 6.08 \times 10^{-3}\,{}^{kg}/_{s}\,{}^{1}/_{cell}$

If $\rho_{air} = 1.29\,{}^{kg}/_{m^3}$ at 0°C

Then $\dot{V} = 283\,{}^{slpm}/_{cell}$ where $\dot{V}$ = air volume flow $\left({}^{slpm}/_{cell}\right)$

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4.4.1.3 Fan Selection Equations – Required Oxidant Mass-Flow

The required oxidant airflow needs to provide the exact number of oxygen molecules to complete the fuel cell reaction for a given current. *See Section A.4.2, Page 146 for the equation derivation.*

Example: From *equation A.4.2*, $Air\ Required = \left(0.0166\, slpm\, {}^{1}\!/_{A}\, {}^{1}\!/_{cell}\right) \cdot I \cdot n$

If $I = 75A$,

Then $\dot{V}_{oxidant} = 1.245\, {}^{slpm}\!/_{cell}$

The required coolant airflow from the above example is $\dot{V}_{coolant} = 283\, {}^{slpm}\!/_{cell}$

$Stoich = \dot{V}_{coolant} / \dot{V}_{oxidant} = 227$

4.4.1.4 Fan Selection Equations – Required Coolant Volume-Flow

Since the fan is pulling air through the stack, the stack volume flow $\dot{V}_{stack}$ should be calculated at the stack temperature. A 28 cell stack is assumed in the following example.

Example: If $\dot{m} = 6.08 \times 10^{-3}\, {}^{kg}\!/_{s}\, {}^{1}\!/_{cell}$ and $\rho_{air} = 1.02\, {}^{kg}\!/_{m^3}$ at 72°C and $n = 28$

Then $\dot{V}_{stack} = 599\, {}^{m^3}\!/_{hr}$ or $\dot{V}_{stack} = 9980\, {}^{L}\!/_{\min}$

4.4.1.5 Fan Selection Equations – Stack Pressure Drop

The air flowing through the stack experiences a pressure drop. It should be noted that in a n cell stack, there are $n+1$ plates.

$$\Delta P = \frac{0.0592 \cdot \left(T_{stack} + 273\right)^{1.5}}{T_{stack} + 393} \cdot \frac{\dot{V}_{stack}}{\left(n+1\right)}$$

ΔP = pressure drop $\left(Pa\right)$

T_{stack} = stack temp $\left(°C\right)$

$\dot{V}_{stack}$ = stack volume flow $\left({}^{L}\!/_{\min}\right)$

n = number of cells

Example: If $T_{stack} = 72°C$ and $\dot{V}_{stack} = 9980\, {}^{L}\!/_{\min}$ and $n = 28$

Then $\Delta P = 281\ Pa$

4.4.1.6 Fan Selection Equations – Fan Sizing

1. When sizing fans, the pressure and flow requirements over the entire range of currents, voltage, ambient temperatures, altitudes, etc. should be determined.
2. In addition to the stack pressure drop, other air-stream components such as the ducting, will add additional pressure drop.
3. The fan selected should be oversized to account for unknowns. Use sound engineering practice to determine the amount that the fan should be oversized.
4. The fan selected should have sufficient turndown to not overcool the stack at low current / low ambient temperature operation. *See Section 8.4.4, Page 113 for more information on fan operation in low current / low ambient temperature situations.*
5. In some applications, it may be acceptable to allow the stack to occasionally run above T_{opt} if the fan cannot deliver enough flow to cool the stack. However, running the stack over-temperature will reduce stack lifetime. **Never run the stack above T_{max}.**

Example: $\dot{V}_{stack} = 599\ ^{m^3}/_{hr}$ and $\Delta P_{total} = \Delta P_{stack} + \Delta P_{ducting} = 305\ Pa$

Fan: ebm-papst W1G180-AB31-01 (Used in Ballard lab)

$\dot{V}_{stack} = 599\ ^{m^3}/_{hr}$

$\Delta P_{total} = 305\ Pa$

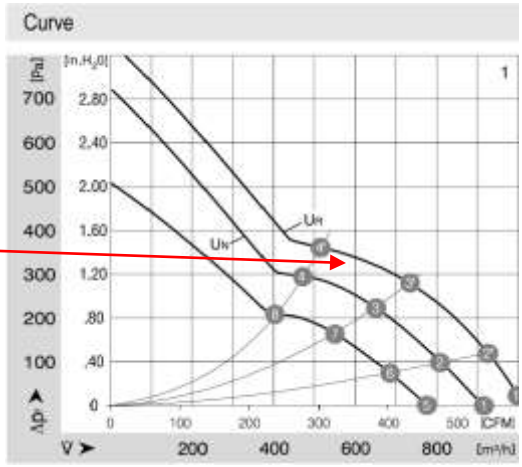


Figure 39: Fan Curve for ebm-papst W1G180-AB31-01

- Since the operating point is below the fan curve, the fan is acceptable for this application.

Ballard strongly recommends that the stack integrator test the cooling system performance, including the cooling fan, over the full range of stack power output and ambient temperatures for the intended application.

5.0 Fuel Stream

The two major functions of the 1020ACS's fuel stream are:

1. **Deliver hydrogen fuel to the stack at the appropriate pressure and flow**.
2. **Periodically remove impurities that have built up on the anode with a purge with the correct flow rate.**

See Section 3.3, Page 39 for information on fuel stream system considerations.

Equations used to calculate fuel requirements are found in Section A.3, 143.

5.1 Fuel Stream Configuration

The fuel stream is configured as shown in the figure below.

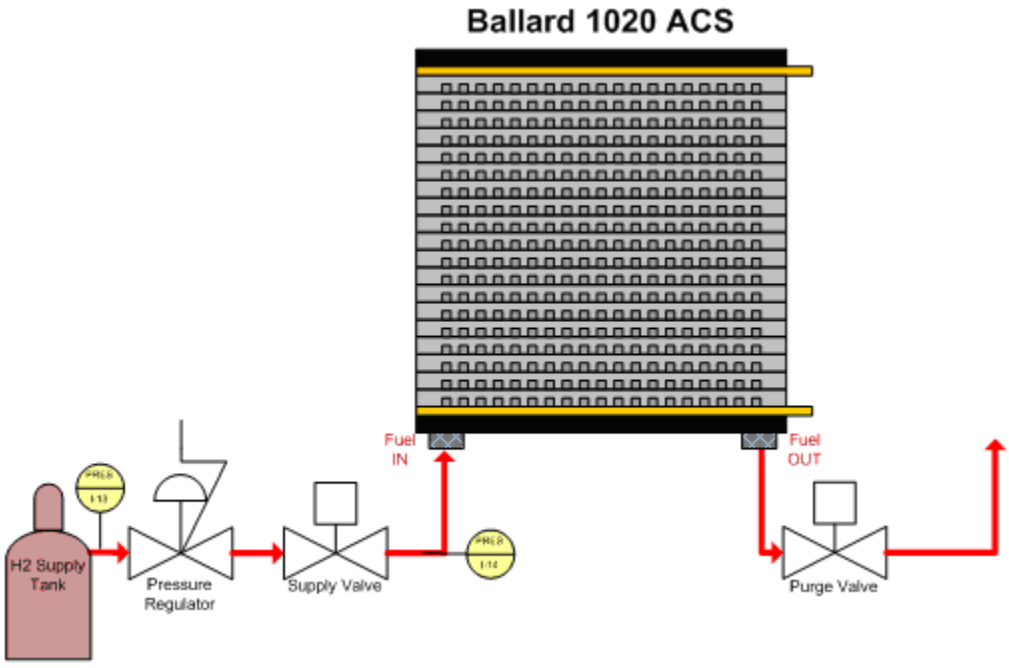


Figure 40: Fuel Stream Configuration

The stack has been designed to operate using a dead-ended configuration to reduce system complexity. The key components in the fuel stream are the pressure regulator, the supply valve, the anode pressure sensor, and the purge valve.

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5.2 Fuel Specification

5.2.1 Fuel Inlet Pressure

Specification	**Required:** 0.16 barg to 0.56 barg **Optimal:** 0.36 barg

1. During a purge, the anode pressure will drop, but the anode pressure should still remain above the minimum acceptable pressure.

See Section 8.7.1, Page 126 for more information on max and min anode H_2 gas pressure alarm values.

See Section 5.3, Page 65 for more information on the anode purge

WARNING **The fuel cell stack and associated system uses pressurized gases, which can be hazardous. Use caution to ensure hydrogen circuits are depressurized before opening any lines or fittings.**

5.2.2 Fuel Stoichiometry

Specification	**Required:** 1.0 **Actual:** 1.07 (Time averaged value including purging)

1. When applied to the hydrogen flow through a fuel cell, the *stoichiometric rate* is defined as the required H_2 gas flow to provide the exact number of H^+ molecules to complete the fuel cell reaction for a given current. The *stoichiometry* (or *stoich*) is defined as the total H_2 gas flow divided by the stoichiometric rate flow.

2. As described in *Section 3.3, Page 39*, the stack is operated with a dead-ended anode configuration. In this configuration, H_2 gas is supplied at the same rate as it is reacted in the stack, and theoretically all the H_2 gas should be consumed.

3. However, inert gases, consisting of fuel impurities, water vapour, and nitrogen diffused from the cathode, build up in the anode and need to be periodically removed. The H_2 gas in the anode is therefore periodically vented to atmosphere, and replaced with fresh H_2 from upstream of the stack. Using the run-time purge parameters outlined in *Section 5.3.4, Page 67*, about 7% of the H_2 consumed is vented to atmosphere.

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5.2.3 Fuel Flow - Continuous

Specification	**Required:** 0.559 slpm/cell

1. The above value is calculated using the equation for H_2 reacted by the stack (*Equation A.3.2, Page 144*), the anode stoich, and the max recommended stack current.

$$Fuel\ Flow = \left(0.00696\, slpm\, {}^{1}\!/_{A}\, {}^{1}\!/_{cell}\right) \cdot 1.07 \cdot 75A = 0.559\, slpm/cell$$

2. During a purge, the flow requirements will increase. The specification for the increase in flow is covered in *Section 5.3, Page 65*, below.
3. Since the purge flow requirement is much larger than the fuel flow requirement, anode components should be sized to meet the purge requirements.

5.2.4 Fuel Temperature

Specification	**Required:** -15°C to 65°C

5.2.5 Fuel Relative Humidity

Specification	**Required:** 0%

1. The stack is designed for use with dry hydrogen, with no external humidification required.
2. The anode is not designed to tolerate large amounts of liquid water. Water accumulation on the anode can block the reactants from reaching the catalyst layer. This is called flooding. *See Section 6.5.2.1, Page 86 for more information on anode flooding.*

5.2.6 Fuel Connections to Stack

The fuel-in and fuel-out connections must be connected to corresponding ports on the stack, as shown in the figure below. The Fuel-In port is on the same side of the stack as the bus plates.

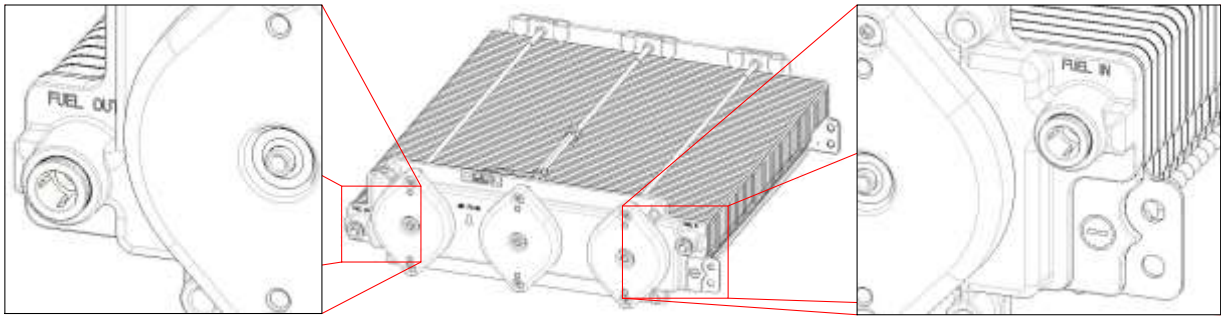


Figure 41: Fuel Inlet and Outlet Ports

5.2.7 Fuel Quality - Considerations

1. The stack is capable of meeting its performance and durability specification if the H_2 fuel is commercial Grade 3.5 hydrogen (99.95% H_2 concentration).
2. Operation using Grade 2.6 hydrogen (99.6% H_2), using the purge parameters detailed in *Section 5.3, Page 65*, will result in a significant irreversible performance loss. Operating the stack using lower grade fuel requires optimization of the purge parameters to improve stack performance.
3. All stack durability testing at Ballard was performed using 99.999% pure (Industrial Grade 5) hydrogen. The impact on stack durability if using Grade 3.5 hydrogen has not been quantified.
4. The impact of fuel quality on performance is highly dependent on the nature and concentration of fuel contaminants. Sulphur-containing compounds can cause significant performance loss and their presence should be avoided as much as possible. Carbon monoxide can also impact stack performance.
5. The below table lists *nominal* and *peak* fuel contaminant values.
6. The stack is capable of meeting its performance and durability specification if the fuel stream meets the *nominal* fuel quality specifications in the table below.
7. The stack can function with higher levels of fuel chemical contamination than *nominal*, but performance and durability may be degraded. If operating above nominal contamination levels, performance loss is expected. Depending on the type of chemical contamination, the performance loss may be recoverable.
8. The composition and concentration of anode impurities in hydrogen can vary significantly depending on how the hydrogen was produced. Ballard recommends that customers look into the composition of their fuel supply. Hydrogen that has gone through a cryogenic liquefaction step during production can generally be considered to be quite pure.

5.2.8 Fuel Quality - Specification

Chemical	Nominal	Peak
Total inert gases(99.95% H_2 min)	< 500 ppm	<19000 ppm
Water	< 5 ppm	<19000 ppm
Total hydrocarbons	< 2 ppm	< 2 ppm
Oxygen	< 5 ppm	< 5 ppm
Helium	< 300 ppm	< 300 ppm
Nitrogen, Argon	< 200 ppm	< 4000 ppm
Carbon dioxide	< 2 ppm	< 2 ppm
Carbon monoxide	< 0.2 ppm	< 10 ppm
Total sulfur compounds	< 0.004 ppm	< 0.004 ppm
Formaldehyde	< 0.01 ppm	< 0.01 ppm
Formic acid	< 0.2 ppm	< 0.2 ppm
Ammonia	< 0.1 ppm	< 0.1 ppm
Total halogenated compounds	< 0.05 ppm	< 0.05 ppm

Figure 42: Fuel Quality Specification

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5.3 Anode Fuel Purge

Periodically, the anode gas needs to be exchanged with fresh hydrogen from upstream of the stack. This is accomplished by opening the purge valve for a specified time to allow fresh hydrogen from upstream of the stack to displace the anode gas. Four major parameters control the stack purge:

5.3.1 Purge Parameters

Purge Rate	**1.** The purge rate is the amount of H_2 gas per time flowing out of the purge valve. **2.** The purge rate can be adjusted by changing anode valve orifice sizes, pressure regulators, tubing sizes, etc. In the Ballard lab, to allow a test station to test stacks with different cell counts, the purge rate is adjusted using a needle valve downstream of the purge valve. **3.** Purge rate is usually not constant over time. The flow rate is highest when the valve is first opened. After the purge valve is opened, the anode pressure droops, and the purge rate reduces.
Purge Duration	**1.** The purge duration is the length of time that the purge valve is opened for a single purge.
Purge Volume	**1.** The purge volume is the volume of gas exchanged in a single purge. **2.** The purge volume is equivalent to the integral of the purge rate over the purge duration. **3.** The purge volume is adjusted by either changing the purge rate or the purge time.
Purge Interval	**1.** The purge interval is the time between purges. **2.** To make the purge interval independent of stack current, the purge interval is measured in Amp·Seconds. **3.** To calculate the Amp·Seconds-since-last-purge, the stack current is integrated with respect to time. Once the Amp·Seconds-since-last-purge is greater than the purge interval, the purge valve opens, and amp-seconds-since-last-purge is reset to zero. **4.** *See Section 5.3.5 and 5.3.6, Page 68 for an example of the purge interval at high and low stack current.*

WARNING **Failure to provide an adequate purge may lead to fuel starvation, cell reversals, and possibly stack fire.**

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5.3.2 Start-Up Purge

During start-up, the gas in the anode also must be exchanged with fresh hydrogen before current can be drawn from the stack. This purge is called the "start-up purge". *For more information on start-ups, see Section 8.2, Page 101.*

Specification			
	Start-Up Purge Rate:	6.0 slpm/cell[1]	**NOTE 1**: During the purge
	Start-Up Purge Duration:	200 ms	**NOTE 2**: The purge volume should also include the tubing volume between the supply valve and stack and purge valve
	Start-Up Purge Volume:	20 mL/cell[2]	
	Start-Up Purge Interval:	N/A	

1. 20mL/cell in 200ms is equivalent to an average flow rate of approximately 6 slpm/cell.
2. The start-up purge flow rate should be measured with a hydrogen filled anode. A hydrogen filled anode will have a higher flow rate than an air-filled anode.
3. The speed at which the H_2 / air front passes through the stack determines the amount of performance degradation per start-up. Purging the specified purge volume in the specified purge duration is important to minimize the irreversible performance loss caused by each start-up. *For more information on performance degradation at start-up, see Section 7.3, Page 94.*
4. If the volume of the anode plumbing is significant when compared to the stack anode volume, the purge rate, volume, and duration will need to be increased to ensure that the start-up purge is adequate. *See Section 3.3.2, Page 40 for more information on the anode plumbing.*
5. The start-up purge must displace all the air in the anode and the anode plumbing with hydrogen. Partially exchanging the air can lead to damaging corrosion currents. *See Section 7.2, Page 92.*
6. A load may be drawn from the stack during the start-up purge to reduce the severity of the start-up corrosion. See *Section 8.2.3, Page 103.*

The system commissioning procedure should include filling all normally-H_2-filled components with H_2 gas. (i.e. hoses from the hydrogen bottles to the system) Drawing current from a stack that has a mixture of H_2 gas and air on the anode will cause significant stack damage.

5.3.3 Measuring Purge Volume

1. A useful method of measuring the purge volume is to collect the anode purge gas over a series of purges and measure the gas volume. This can be performed by purging into an inverted water-filled graduated cylinder, or some other variable-volume container, as shown in the figure. This method is used in the Ballard lab.

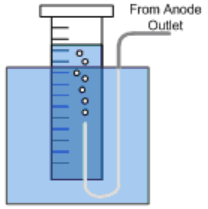


Figure 43: Measuring Purge Volume

2. Another method is to collect the purge gas in a plastic bag. The volume of the gas in the bag is then measured by submerging the bag in a water-filled graduated cylinder. This method does not require the purged gas to flow through water, which should increase the accuracy of the result.

5.3.4 Run-Time Purge

1. During operation, impurities build up in the anode. The inert gases build-up is caused by inert gases in the fuel, nitrogen diffusing in from the cathode, and product water.
2. The inert gases cause a reduction in performance. The magnitude of the performance reduction increases as the inert gases concentration level increases.
3. To remove the inert gases, the anode gas needs to be periodically exchanged. This is accomplished by opening the purge valve for a specified time to allow fresh hydrogen from upstream of the stack to displace the anode gas. This purge is called the "run-time" purge.

Specification			
	Run-Time Purge Rate:	2.4 slpm/cell or greater[1]	**NOTE 1**: During the purge
	Run-Time Purge Duration:	Less than 500 ms	
	Run-Time Purge Volume:	20 mL/cell	
	Run-Time Purge Interval:	2300 Amp·Seconds	

1. 20 mL/cell in 500 ms is equivalent to an average flow rate of approximately 2.4 slpm /cell.
2. The run-time purge parameters result in a H_2 utilization of 93% or 7% of the total H_2 used is vented to atmosphere.
3. Since the run-time purge rate is much smaller than the start-up purge rate, purge valves are usually sized to meet the start-up purge rate.
4. Purge volumes smaller than 20 mL/cell may not remove all the anode inert gases. With a sufficient purge, stack voltage between purges will remain relatively constant (expect <1-2% change in stack voltage).
5. An indicator of a non-optimal purge strategy is a "sawtooth" stack voltage, as shown in *Section 5.3.5 and Section 5.3.6*, below.
6. Another indication of a non-optimal purge strategy is cell-to-cell voltage variation, as shown in the figure to the right. Cell-to-cell voltage variation indicates fuel starvation in regions of the cell. Fuel starvation can accelerate performance degradation.

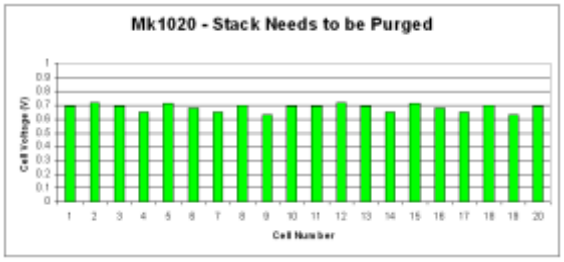
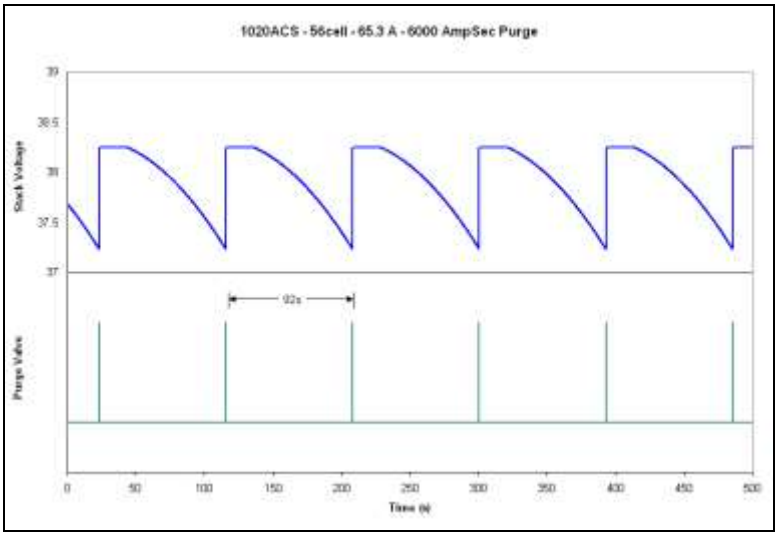


Figure 44: Cell to Cell Voltage Variability Indicates a Purge is Required

7. When tuning purge parameters in a lab environment, a CVM is very useful to measure cell-to-cell variability. *See Section 3.4.2.4, Page 43 for more information on CVMs.*
8. Purging less frequently decreases fuel consumption. However, purging less frequently also results in a reduced performance, which may not translate into increased fuel efficiency. If the application demands high fuel efficiency, contact Ballard for strategies to reduce the purge requirements.
9. All testing in the Ballard lab uses a 20 mL/cell, 200 ms, 2300 Amp·Sec run-time purge.

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5.3.5 Purge – 6000 Amp·Sec Example



In the figure to the left, the stack is run at a constant 65.3 A. With a purge interval of 6000 Amp·Sec, the purge valve opens every 93 s. The 6000 Amp·Sec purge interval results in a "sawtooth" pattern.

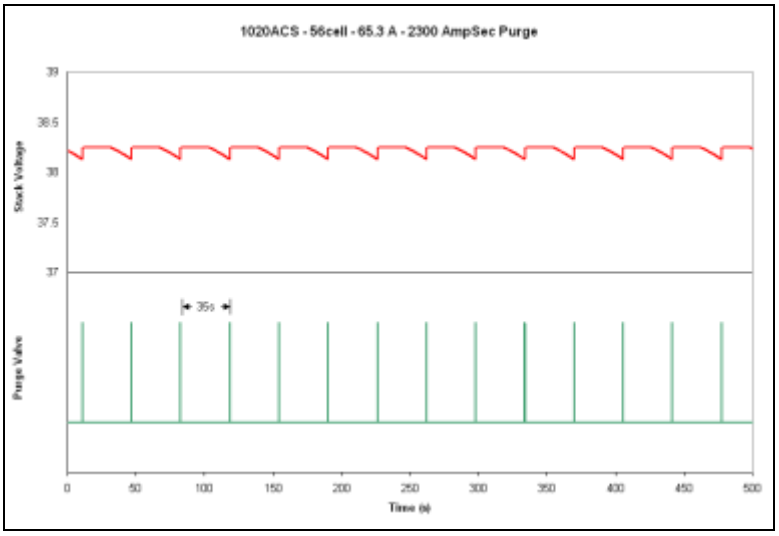
Purge Interval: $6000A \cdot s$

Stack Current: $65.3A$

Time: $92s$

Figure 45: 6000 Amp·Sec Purge Interval

5.3.6 Purge – 2300 Amp·Sec Example



In the figure to the left, the stack is run at a constant 65.3 A. With a purge interval of 2300 Amp·Sec, the purge valve opens every 35 s. The 2300 Amp·Sec purge interval results in a small voltage drop between purges.

Purge Interval: $2300A \cdot s$

Stack Current: $65.3A$

Time: $35s$

Figure 46: 2300 Amp·Sec Purge Interval

WARNING

Hydrogen is a colourless, odourless, highly flammable gas. Hydrogen purged from the fuel cell must be managed in a safe manner. It is recommended to direct the purged hydrogen into the inlet of the ventilation fan. Hydrogen is non-toxic, but can cause asphyxiation by displacing the oxygen in the air. There are no warning symptoms before unconsciousness results.

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5.4 Anode Pressure Drop

1. Since the stack is operated in a dead-ended configuration, the stack anode pressure drop only affects the purge flow through the stack.
2. In most systems, the purge valve orifice is used to control the purge flow, and is the largest restriction in the fuel stream.
3. An estimate of the stack anode pressure drop is given by:

$$\Delta P = 21.31 \cdot \dot{m}_{cell}$$ where ΔP = pressure drop $(mbar)$

$\dot{m}_{cell}$ = H_2 flow rate per cell $(slpm)$

5.5 Purge Exhaust

1. The purge exhaust is a mixture of impurities and H_2 gas, but the majority is H_2 gas.
2. Hydrogen is flammable in air and will burn over a wide range of concentrations: 4% to 75% by volume.
3. The purge exhaust must be properly positioned to prevent situations where the purge exhaust could potentially ignite. Two strategies for the purge exhaust are:
   - **A.** Dilute the purge exhaust in the air stream. The exhaust should mix with the air stream *downstream* of the stack, as shown in the figure below.
   - **B.** Vent the exhaust to atmosphere, outside of the system enclosure.
4. The placement of the purge exhaust must adhere to all applicable codes and standards.

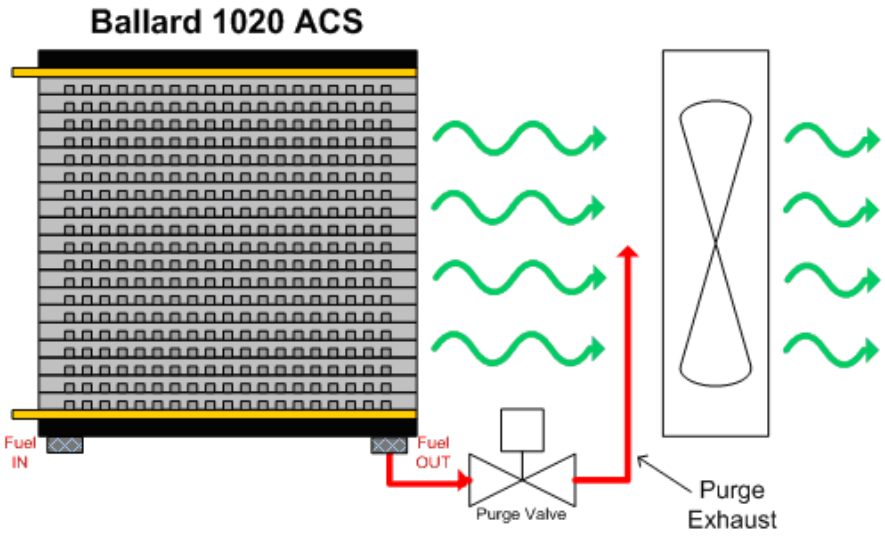


Figure 47: Purge Exhaust – Into Air Stream

5.6 Anode Component Selection

5.6.1 Pressure Regulator

1. The pressure regulator regulates the pressure from the H_2 storage pressure to the stack fuel pressure, 0.36 barg. *See Section 5.2.1, Page 62.*
2. A *dual-stage* regulator is recommended. A dual-stage regulator provides a consistent outlet pressure over a wide range in inlet pressures.
3. The pressure regulator must supply H_2 gas at the start-up purge flow-rate. *See Section 5.3.2, Page 66.*
4. If the pressure regulator cannot supply sufficient flow at the start-up flow rate, an accumulator should be used. *See Section 5.6.6, Page 71.*

5.6.2 Pressure Sensor

1. The anode pressure sensor should be capable of measuring the anode pressure above and below the optimum anode pressure, as described in *See Section 5.2.1, Page 62.* A range of 0 barg to 0.7 barg should be sufficient for most application.
2. In a laboratory system, it is useful to measure the anode pressure after shutdown. *See Section 8.5.2, Page 114 for more information on the anode after shutdown*. A range of –1.0 barg to 1.0 barg is sufficient.

5.6.3 Supply Valve

1. The supply valve controls the flow of hydrogen to the stack. It is positioned upstream of the stack.
2. The supply valve must be able to open and seal against an upstream pressure of 0.36 barg.
3. The supply valve must be able to open and seal against a downstream pressure of -0.7 barg to 0.36 barg. *See Section 8.5.2, Page 114 for more information on the anode pressure after shutdown*.
4. It is very important that the supply valve does not leak when in the closed position and allow hydrogen into the anode when the stack is off. *See Section 7.6, Page 96 for more information on stack damage caused by a leaky supply valve.*
5. The supply valve should not restrict the flow of H_2 gas. In the Ballard lab, the supply valve used has a 1/2” orifice, but using a smaller orifice is possible.

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5.6.4 Purge Valve

1. The purge valve is used to control the flow of hydrogen from the stack. It is positioned downstream of the stack.
2. The purge valve must be able to open and seal against an upstream pressure of -0.7 barg to 0.36 barg. *See Section 8.5.2, Page 114 for more information on the anode pressure after shutdown*.
3. The purge valve must be able to open and seal against a downstream pressure of 0 barg (atmospheric pressure).
4. It is very important that the purge valve does not leak when in the closed position and allow air into the anode when the stack is bleeding down during shutting down. *See Section 7.5, Page 96 for more information on the stack damage caused by a leaky purge valve.*
5. The purge valve orifice should be sized to provide the proper start-up purge rate. *See Section 5.3.2, Page 66.* In the Ballard lab, a solenoid valve with a 1/2" orifice is used in conjunction with a needle valve downstream to adjust the purge rate.
6. In most applications, the purge valve will be made from stainless steel, as described in *Section 3.3.3, Page 40.*

5.6.5 Pressure Relief Valve (*optional)*

1. The pressure relive valve protects the anode if there is an anode overpressure event.
2. The pressure relief valve is positioned downstream of the pressure regulator and upstream of the supply valve.
3. The pressure relief valve should be set to open if the anode pressure goes above 1.0 barg.

5.6.6 Accumulator (*optional*)

1. The accumulator is a hydrogen reservoir positioned downstream of the pressure regulator and upstream of the supply valve. The accumulator should be at least 2X the internal volume of the stack.
2. The accumulator is at stack pressure, and during a purge, the accumulator provides additional H_2 gas to maintain a higher anode pressure, increasing the effectiveness of a purge.

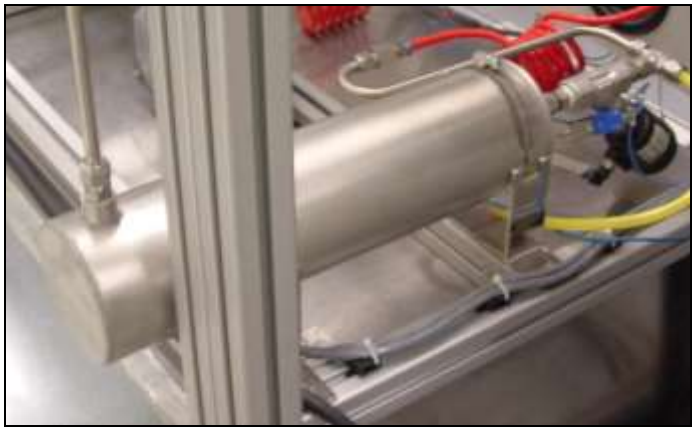


Figure 48: Accumulator in the Ballard Lab

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5.7 Fuel Leaks

1. A fuel leak is defined as H_2 gas at above atmospheric pressure escaping from the stack anode. Internally at Ballard, fuel leaks are often referred to as "transfer leaks."
2. As the stack degrades, the proton-exchange-membrane (PEM) will slowly develop holes in the membrane, allowing H_2 gas to leak out of the anode.
3. Although much less likely than membrane leaks, the anode seals can also leak, allowing H_2 gas to escape through the seals between the MEA and the bipolar plate.
4. An anode plumbing leak is defined as H_2 gas escaping from the anode plumbing. Anode plumbing leaks cause the stack to behave in a similar fashion to an anode fuel leak.

5.7.1 Fuel Leak – BOL Specification

Specification	**Specification:** Less than 5 sccm/cell (**H_2 filled anode at 0.36 barg**)

1. In the Ballard lab, the BOL leak test is performed using N_2 gas at 0.5 barg.

   Ballard Lab Specification: Less than 0.2 sccm/cell (**N_2 filled anode at 0.5 barg**)

See Section 11.1.10, Page 139 for a schematic and description of the Ballard lab leak test set-up

5.7.2 Fuel Leak – Ballard Lab EOL Specification

Specification	**Specification:** Greater than ~6.5 sccm/cell (**N_2 filled anode at 0.5 barg**)

1. In the Ballard lab, a stack will almost always reach EOL based on performance, not fuel leak. *See Section 6.2.4, Page 79 for more information on the EOL stack performance threshold.*
2. Depending on the application, the system might be able to tolerate a leak larger than the Ballard EOL fuel leak threshold.
3. A stack with 1500 hrs and 1000 on-off cycles is expected to have a leak of 12 sccm/cell (**H_2 filled anode at 0.36 barg**).

5.7.3 Fuel Leaks – Effect on Stack Fuel Efficiency

1. As described in *Section 6.3.3, Page 81*, the amount of fuel leaked is negligible compared to the amount of fuel reacted or purged.
2. In most applications, the fuel efficiency reduction caused by a fuel leak will not limit the life of the stack.

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5.7.4 Fuel Leaks – Effect on Stack Performance

1. If a cell in a stack has a fuel leak, the cell will go from OCV to 0 V before the other cells in the stack, as shown in the figure below.
2. The presence of a fuel leak (or a plumbing leak or purge valve leak) dramatically increases performance degradation through carbon corrosion currents after shutdown, as described in *Section 7.4, Page 95*.
3. The overall stack performance degradation rate is broken into two parts: Before fuel leak initiation and after fuel leak initiation, as described in *Section 6.7.2, Page 90*.

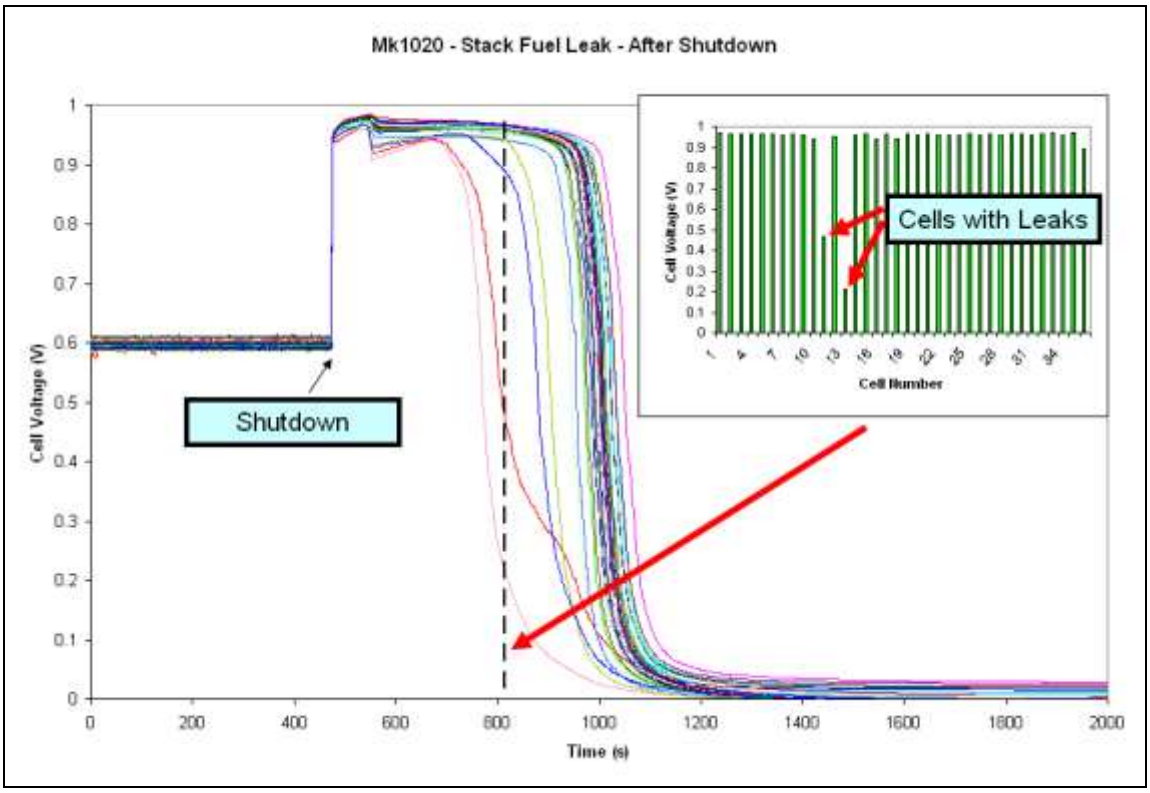


Figure 49: Stack Fuel Leaks – After Shutdown

5.7.5 Fuel Leaks – Prevention

1. If the stack is operated for long enough, fuel leaks will initiate. To increase the time to fuel leak initiation:
   - **A. Reduce the time cell voltages are above 0.80 V –** Membrane chemical degradation is more severe at high cell voltages. Reducing the time at high cell voltages will delay fuel leak initiation. However, time at high cell voltages, such as after a shutdown, is unavoidable.
   - **B. Operate the stack at lower temperatures** – Membrane chemical degradation is more severe at high temperatures. Since the stack should always be operated at T_{opt}, as described in *Section 4.3.2, Page 55*, reducing the operating temperature requires a reduction in operating current.

5.7.6 Fuel Leaks – Detection

1. Fuel leaks, H_2 gas at above atmospheric pressure leaking from the anode into the air stream, can create a potentially explosive mixture of H_2 gas in air. Hydrogen is flammable in air and will burn over a wide range of concentrations: 4% to 75% by volume.
2. Two methods to detect a fuel leak are:
   - **A. Use a hydrogen gas sensor in the system –** A hydrogen sensor can measure the concentration of H_2 gas in air. If the sensor is placed in the correct position in the system, it should be able to detect fuel leaks and plumbing leaks.
   - **B. Perform an anode pressure decay test –** To perform an anode pressure decay test, the anode is filled with hydrogen, the load is removed, and the both the supply valve and purge valves are closed. The anode pressure will then reduce. Measuring the change of the anode pressure in time will give an indication of how much H_2 gas is leaking out of the stack anode and the anode plumbing.

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6.0 Stack Performance

6.1 Voltage, Current, and Power Range

6.1.1 Current Range

Specification	**Current Range:** 0 A to 75 A.

1. 75A is the maximum current where the stack can provide continuous, stable output power.
2. The stack performance end-of-life (EOL) point is reached when the stack is unable to produce the power required for the application at 75 A.
3. A negative current should never be applied to the stack. Negative stack currents will cause irreversible stack damage.
4. Current should never be drawn from the stack without H_2 gas on the anode or if the anode is a mixture of H_2 gas and air. Drawing current without hydrogen on the anode will cause irreversible performance loss. *See Section 7.7, Page 97 for more information on fuel starvation.*
5. Operating the stack at a lower current improves stack durability, but the current should be high enough to keep the average cell voltage below ~0.80 V. *See Section 6.6, Page 88 for more information on how to improve stack durability.*
6. All Ballard 1020ACS life testing performed at 65.3 A. This is considered the design full power operating point. *See Section 6.7.1, Page 89 for the Ballard Lab Standard Test Cycle.*

6.1.2 Voltage Range

Specification	**Cell Voltage Range:** 0.5 V/cell to 1 V/cell

1. **Max Cell Voltage:** At 0 A stack current, the stack voltage is ~1 V/cell. However, if possible, the voltage should be kept below 0.80 V to reduce fuel leak initiation. *See Section 5.7.5, Page 73.*
2. **Min Cell Voltage:** A stack provides peak power at 0.5 V/cell. *See Section 6.2.5, Page 80 for more information on peak power.* Cell voltages lower than 0.5 V/cell are acceptable in specific situations, such as start-up. *See Section 8.3.3, Page 108.*
3. Cells should always be greater than 0V to avoid irreversible damage. *See Section 6.5.2.1, Page 86 for more information on cell reversals.*
4. **Stack Voltage:** $V_{stack} = V_{cell} \times n$ where $n =$ number of cells

6.1.3 Power Range

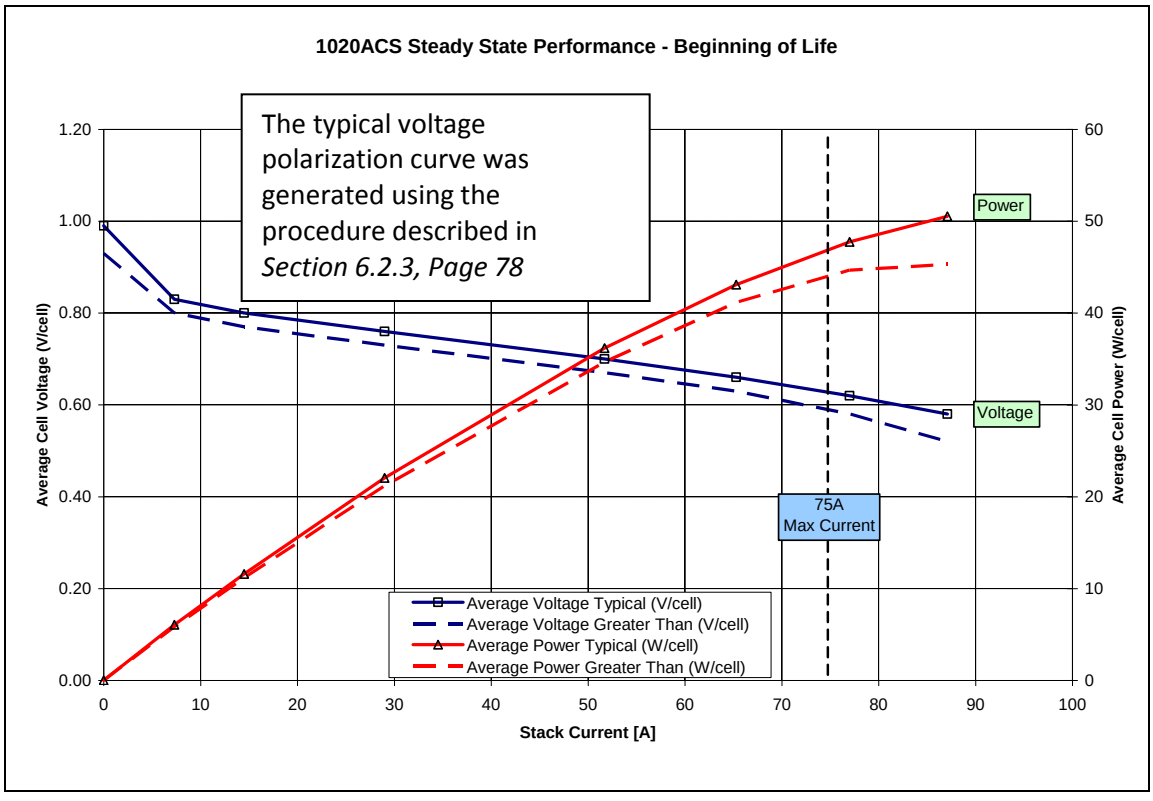
1. **Stack Power:** $P_{stack} = V_{cell} \times I_{stack} \times n$ where $n =$ number of cells

6.2 Stack Performance

6.2.1 BOL Stack Performance

1. The figure below is a polarization curve for a fully-conditioned beginning-of-life (BOL) 1020ACS stack. The procedure used to generate the polarization curve is outlined in *Section 6.2.3, Page 78*. When generating a polarization curve to compare with Ballard data, the same procedure should be used.
2. The stack does not deliver rated power instantaneously. *See Section 8.3, Page 105 for more information on how to take the stack from an off state to where it can deliver rated power.*
3. Other performance factors that may affect the 1020ACS stack that are not taken in to account in this performance summary include:

   - NOPL/Transport. *See Section 6.5.1.2, Page 85 for more information on NOPL.*
   - Air quality. *See Section 4.2.6 , Page 52 for more information on air quality*.
   - Ambient conditions/altitude. *See Section 4.2.2, Page 50 for more information on altitude*.
   - Operating time & on/off cycles. *See Section 6.7.4, Page 91 for specific information on run-time performance degradation.*



Stack Current (A)	0	7.3	14.5	29.0	51.7	65.3	77.0	87.1
Avg. Cell Voltage Greater Than (V/cell)	0.93	0.80	0.77	0.73	0.67	0.63	0.58	0.52
Avg. Power Greater Than (W/cell)	0	5.8	11.2	21.2	34.6	41.1	44.7	45.3
Avg. Cell Voltage, Typical (V/cell)	0.99	0.83	0.80	0.76	0.70	0.66	0.62	0.58
Avg. Power, Typical (W/cell)	0	6.1	11.6	22.0	36.2	43.1	47.7	50.5

Note 1: The 77.0 A and the 87.1 A points are given for reference only. The stack should never be run over the 75 A max stack current during normal operation.

Figure 50: Beginning-of-Life Stack Performance

6.2.2 Bus Plate Voltage Loss

1. All testing at Ballard measures voltage at the cell level and not at the stack level. Measuring voltage at the cell level generates data that is independent of stack size. However, the stack power connections are made at the bus plates, and to accurately predict the stack output voltage, the voltage loss associated with bus plates must be taken into consideration.

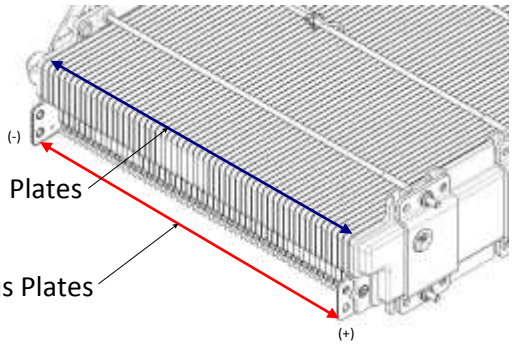


Figure 51: Measuring Stack Voltage across the Bus Plates

(Single thermistor assembly shown)

2. Testing at Ballard shows that the total resistance of the bus plates is ~2.2 mΩ, as shown in Figure 52, right.

3. For example, at 65.3 A, the bus-plates are responsible for a ~0.14 V loss.

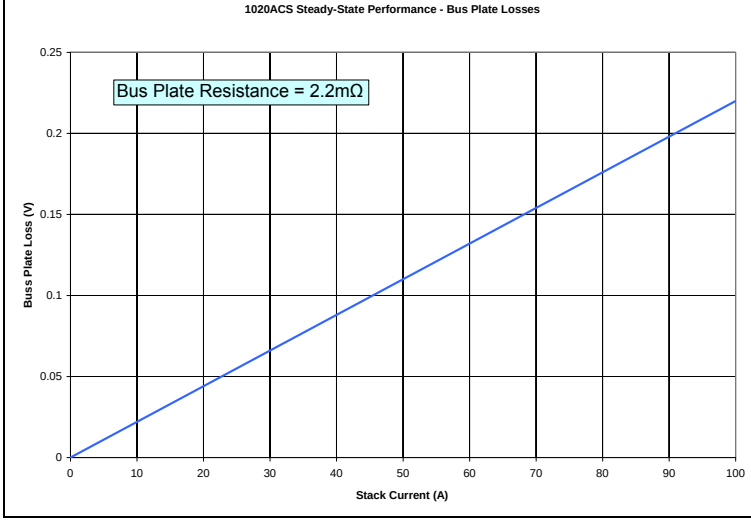


Figure 52: Bus Plate Voltage Loss

6.2.3 Ballard Polarization Curve Generation Procedure

1. Ballard uses a specific procedure to generate polarization curves. If customer polarization data is compared to the performance data presented in this manual, the following procedure should be followed to generate the polarization curve.

2. Note that the T_{opt} values listed in the "Procedure" column are different than the T_{opt} values used in normal operation. This procedure uses legacy T_{opt} values.

Air Stream (Ballard Lab Conditions)	Fuel Stream (Industrial Grade 5 H_2)	Procedure
▪ **Temp:** 18°C-24°C, ▪ **RH:** 25%-30% ▪ **Pressure:** Sea Level ▪ **Flow:** Controlled so that the stack temp is equal to temperatures listed in "Procedure" ▪ **Quality**: Low levels of common urban pollutants such as NO_x and SO_x ▪ **Stoich:** greater than 80	▪ **Temp:** 18°C-25°C, ▪ **RH:** 0% ▪ **Pressure:** 0.36 barg ▪ **Purge:** 20 mL/cell in 200 ms, every 2300 Amp·Sec (*See Section 5.3, Page 65 for more info on purge*)	**1.** 60 min at 65.3 A, **63°C**, avg. last 5 min **2.** 20 min at 87.1 A, **75°C**, avg. last 5 min **3.** 20 min at 77.0 A, **69°C**, avg. last 5 min **4.** 20 min at 65.3 A, **63°C**, avg. last 5 min **5.** 20 min at 51.7 A, **59°C**, avg. last 5 min **6.** 20 min at 29.0 A, **49°C**, avg. last 5 min **7.** 20 min at 14.5 A, **36°C**, avg. last 5 min **8.** 20 min at 7.3 A, **30°C**, avg. last 5 min **9.** 10 min at 0 A, **30°C**, avg. last 5 min

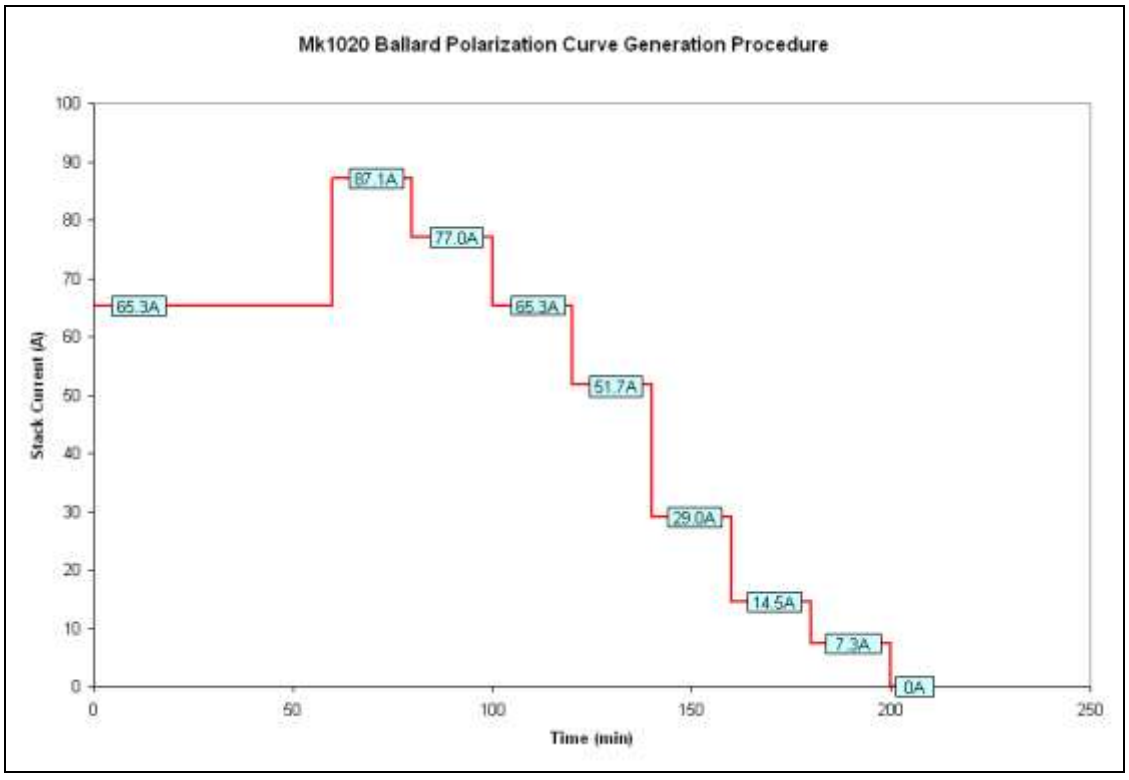


Figure 53: Ballard Polarization Curve Generation

6.2.4 EOL Stack Performance

A stack is determined to be at end-of-life (EOL) if either:

1. **The stack performance no longer meets requirements -** The stack cannot deliver the required power, current, or voltage. *See Section 9.0, Page 131 for more information on stack sizing.*
2. **The stack fuel leak rate exceeds allowable limits -** *See Section 5.7, Page 72 for more information on stack fuel leaks.*

Ballard defines the end-of-life point to be when the *peak power point* has been reached at the maximum stack current, 0.5 V/cell at 75 A*. See Section 6.2.5, Page 80 for more information on Peak Power.* EOL criteria will be different for different applications. The figure below is a polarization curve for an end-of-life (EOL) 1020ACS stack. The procedure used to generate the polarization curve is outlined in *Section 6.2.3, Page 78.*

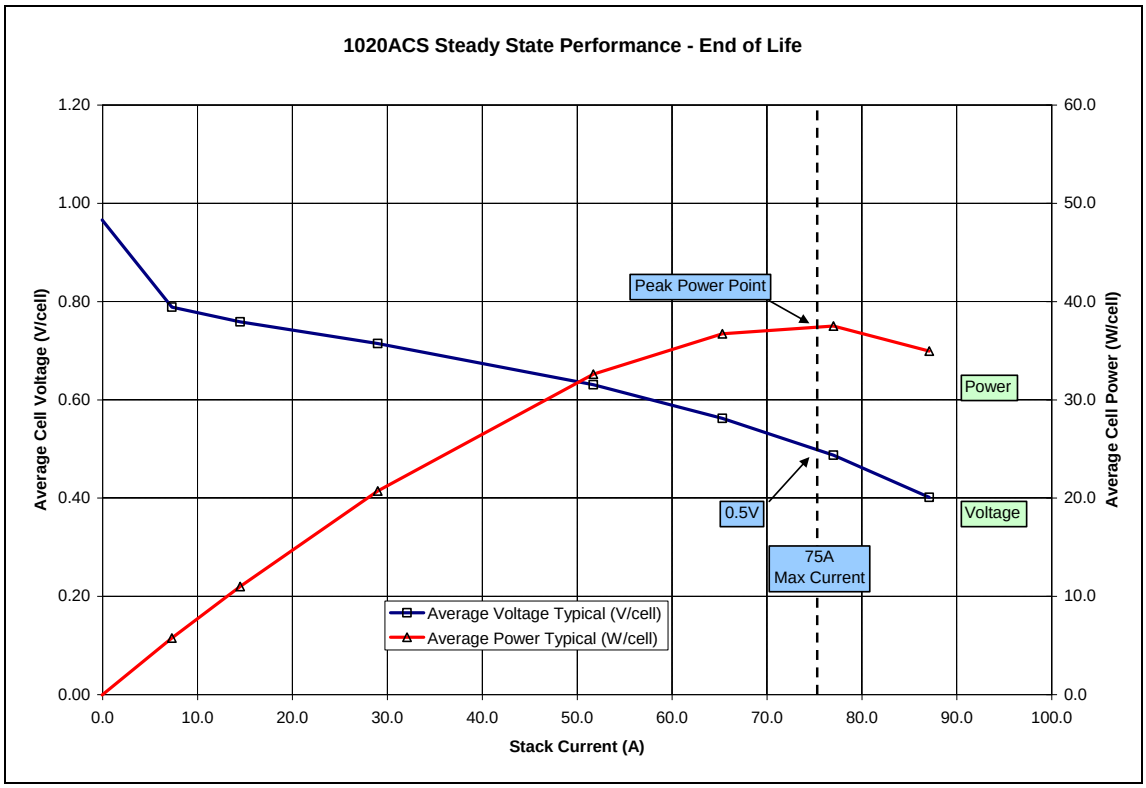


Figure 54: End-of-Life (EOL) Stack Performance

6.2.5 Peak Power

1. Increasing the stack current will cause a reduction in the stack voltage, as shown in Figure 54, below. The stack power, which is the product of the stack current and the stack voltage, will reach a maximum if the stack current is increased to a given point. The maximum is referred to as *Peak Power*. The stack should never be operated above *peak power* during steady-state operation.
2. *Peak Power* is the maximum power that a stack is capable of producing. The stack current at *Peak Power* is the *Peak Power Current*, and the stack voltage is the *Peak Power Voltage*.
3. At BOL, the *Peak Power Current* is ~100 A, and since the maximum recommended stack current is 75 A, the *peak power current* of a BOL stack is outside stack operating current range.
4. As the stack degrades, or if it is operated non-optimally, or after start-up, the *Peak Power Current* reduces.
5. The *Peak Power Voltage* is ~0.5 V/cell, and is constant over the life of the stack. If the stack is operated at voltages lower than 0.5 V/cell, the stack is operating beyond the peak power point.
6. There is no benefit to operating at currents higher than *Peak Power Current* or voltages below the *Peak Power Voltage* during steady-state operation, because the efficiency and power are less than at *Peak* Power. However, during start-up, operating at currents above and voltages below *Peak Power* is acceptable. *See Section 8.3.3, Page 108 for more information on start-ups.*
7. In some systems, the power electronics are designed to increase the stack current until the desired output power level is reached. However, if the current is increased beyond the peak power point, the stack is delivering less power than at lower currents, but the power electronics will increase the current to the maximum allowable. This is known as “Power Latching” and should be avoided. If the load is limited by a minimum stack voltage threshold (0.5 V/cell), power latching is avoided.

6.3 Stack Fuel Consumption and Fuel Efficiency

The stack consumes hydrogen fuel through three possible mechanisms: reacting, purging, and leaking.

1. **Reacting –** The stack reacts H_2 gas with oxygen to produce electricity, water, and waste heat. Of the three fuel consumption mechanisms, reacting is the only one to produce useful electrical energy.
2. **Purging –** Periodically, the anode gas must be exchanged to remove any built-up impurities. The purge gas is vented to atmosphere.
3. **Leaking –** Hydrogen leaking through holes in the MEA or seals to atmosphere.

6.3.1 Fuel Consumption – Reacted H_2

1. The amount of H_2 reacted by the stack is a function of stack current and number of cells.

   $$H_2\ Reacted = \left(0.00696\, slpm\,{}^{1}\!/_{A}\,{}^{1}\!/_{cell}\right)\cdot I \cdot n, \quad \text{where } n = \text{ number of cells}$$

2. Example: If $I = 65.3A$, $H_2\ Reacted = 0.454\ {}^{slpm}\!/_{cell}$
3. The derivation of the above equation can be found in *Section A.3.2, Page 144*.

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6.3.2 Fuel Consumption – Purged H_2

1. Periodically exchanging the anode gas with fresh H_2 anode gas from upstream of the stack removes anode impurities, which are mostly N_2 gas, water, and other inerts.
2. This anode gas is exchanged by opening the purge valve for a specified time.
3. The gas volume displaced in a single purge should be $20\,{}^{mL}/_{cell}$, which is equivalent to 2 anode volumes. The stack should be purged every $2300 Amp \cdot Sec$.
4. Using $20\,{}^{mL}/_{cell}$ per purge every $2300 Amp \cdot Sec$, for a given amount of hydrogen reacted, an additional 7% is required for purging.
5. Equations that describe the purging are found in *Section A.3.3, Page 144*.

See Section 5.3, Page 65 for more information on purging.

6.3.3 Fuel Consumption – Leaked H_2

1. Holes in the membrane or seals cause hydrogen to leak from the above-atmospheric pressure anode to the atmosphere.
2. The stack fuel leak magnitude is expected to increase over the life of the stack.
3. In the Ballard lab, with a 0.36 barg H_2 anode pressure, the expected fuel leak at EOL is ~$12\,{}^{sccm}/_{cell}$, and the BOL fuel leak specification is less than $5\,{}^{sccm}/_{cell}$.
4. H_2 consumption through fuel leak is usually ignored in calculations because it is much smaller than the H_2 consumed through from reacting and purging. Leaked H_2 gas is approximately 2 orders of magnitude less than reacted H_2 gas, as calculated in *Section 6.3.1*, above.

See Section 5.7, Page 72 for more information on the fuel leaks.

6.3.4 Stack Fuel Efficiency

1. Hydrogen thermodynamic efficiency is defined as

$$H_2\ Efficiency = \frac{Energy\ Delivered}{H_2\ Consumed} \qquad \left(kW \cdot hrs/kg\right)$$

2. The H_2 consumed should include the sum of H_2 reacted and H_2 purged.
3. The energy delivered can be the energy delivered at any point in the system. For example, the stack efficiency would use the energy delivered by the stack. The system efficiency would use the energy delivered by the system.
4. More equations that describe hydrogen efficiency can be found in *Section A.3.5, Page 145.*

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6.4 Stack Efficiency

Unlike an internal-combustion engine, a fuel cell does not have a maximum-efficiency point in the middle of the operating range. The efficiency of a fuel cell decreases as the output increases.

6.4.1 Stack Efficiency

1. The chemical energy stored in the H_2 fuel is converted into a combination of electrical and heat energy by the fuel cell stack. The electrical energy is useful, and the heat energy is wasted energy and must be removed from the system. The below equations describe the balance between electrical energy production and heat energy production in a fuel cell.

$$Total\ Energy = Electrical\ Energy + Heat\ Energy \quad (W)$$

$$Total\ Energy = V_{Stack} I_{Stack} + (1.23V - V_{Stack}) I_{Stack} \quad (W)$$

2. As the stack current increases, the stack voltage decreases along the polarization curve, as shown in the below figure.

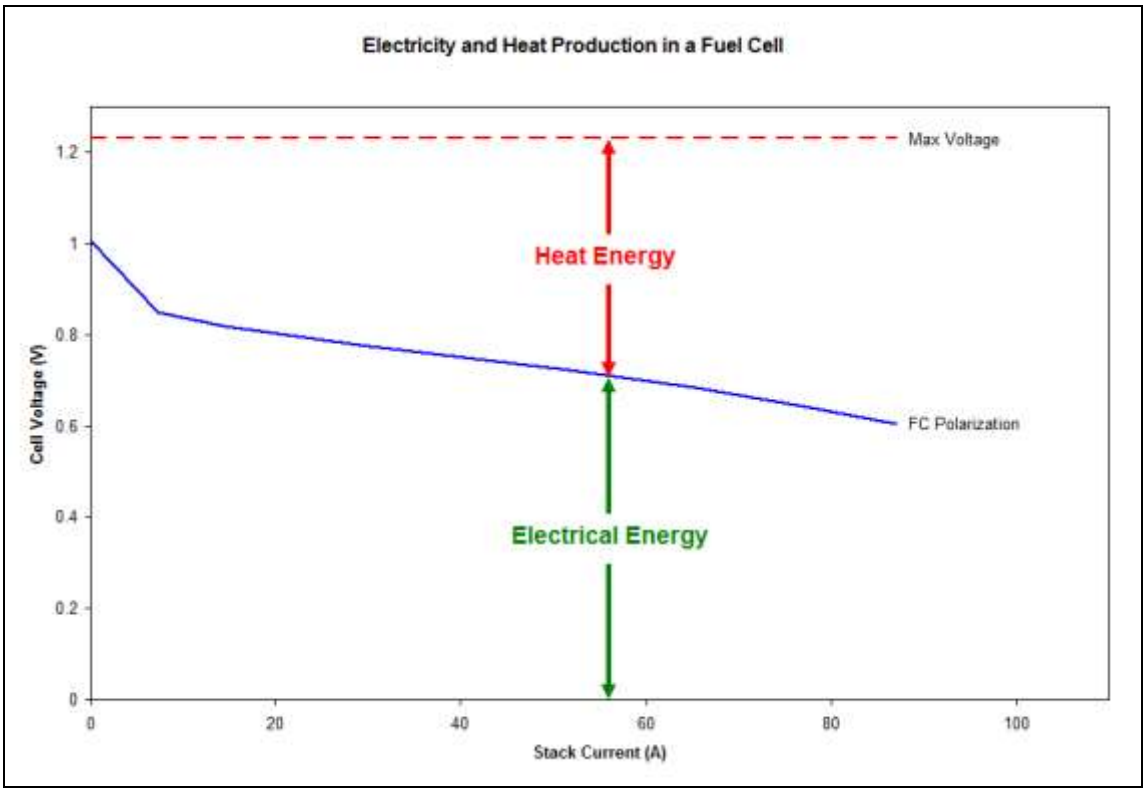


Figure 55: Electricity and Heat Production in a Fuel Cell

3. To maximize efficiency, the amount of electrical energy must be maximized and the amount of heat energy must be minimized. The most efficient operating point is at 0 A, but at 0 A, the stack is not producing any power.

4. In an application where efficiency is important, the trade-off between efficiency, stack output power, and stack cost must be analyzed.

5. As the stack performance decreases over life, the stack efficiency will decrease.

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6.5 Stack Performance Degradation

Stack performance degradation can be split into two broad categories:

1. **Reversible Performance Degradation** – Reversible performance degradation is generally caused by running the stack outside specification, and the performance loss can be recovered.
2. **Irreversible Performance Degradation** – Irreversible performance degradation cannot be recovered.

6.5.1 Reversible Performance Degradation

6.5.1.1 Sources of Reversible Performance Degradation

Running at Non-Optimum Stack Temperature	▪ Running the stack at temperatures above the Optimum Temperature (T_{opt}) causes the membrane to dry out and become more electrically resistive, reducing performance. Running the stack at temperatures below T_{opt} results in water accumulation in the cells, resulting in performance loss due to increased mass transport effects. ▪ *See Section 4.3.2, Page 55 for more information on T_{opt}.*
Running at Altitude	▪ Running at altitude reduces the cathode pressure, reducing performance. ▪ *See Section 4.2.2, Page 50 for more information on altitude*.
Running in Cold Ambient Temperatures	▪ Running in cold ambient temperatures causes large temperature gradients between the stack air inlet (T_{amb}) and the stack air outlet (T_{opt}). Running regions of the stack at temperatures significantly cooler than T_{opt} reduces performance. ▪ *See Section 4.2.1, Page 50 for more information on cold ambient temperature operation.*
Oxidant Contaminants	▪ Specific airborne contaminants, such as NO_x, can cause significant performance loss, but this loss is recoverable through current pulses, air-starves, or running the stack at high current densities. For example, in the Ballard lab, stack performance rises and falls with atmospheric NO_x levels. It should be noted that other oxidant contaminants can cause irreversible performance loss. ▪ *See Section 4.2.6, Page 52 for more information on oxidant contamination.* ▪ *See Section 8.6, Page 118 for more information on current pulses and air starves.*

Sources of Reversible Performance Degradation (*continued*)

Low Cathode Stoich

- Running at low stoich reduces the amount of O_2 gas available for the fuel cell reaction, reducing performance.
- *See Section 4.2.4, Page 51 for more information on air stoich.*

Running for Extended Periods at High Cell Voltages (> 0.7 V)

- Running at cell voltages greater than 0.70 V for extended periods of time results in the growth of a platinum oxide (PtO_x) layer on the catalyst that lowers cell performance.
- Conversely, running at cell voltages lower than 0.70V will remove the layer and improve performance.
- As shown in the figure below, the further away the cell voltage moves above or below ~0.70 V, the faster the PtO_x layers grows or reduces.

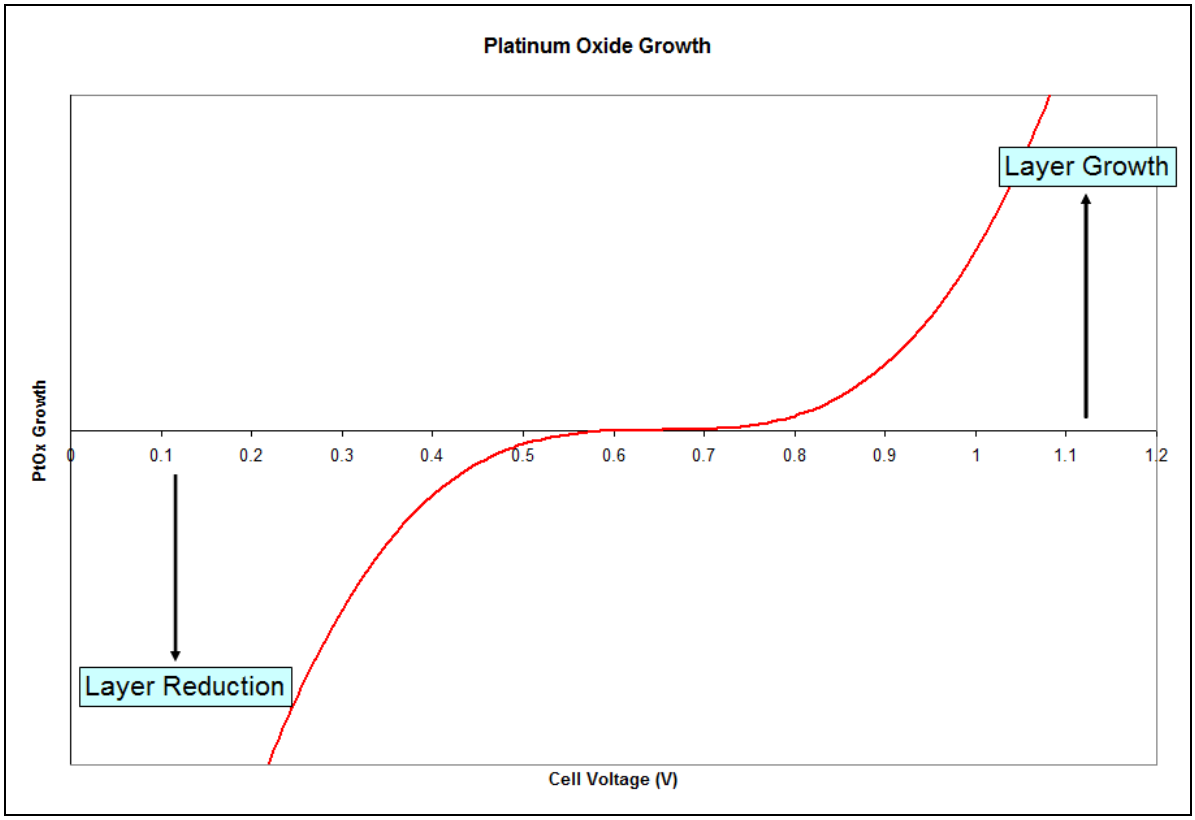


Figure 56: PtO_x Layer Growth as a Function of Cell Voltage

- High-current / low-voltage operation, air-starves, or current pulses can be used to reduce the oxide layer.
- *See Section 8.6, Page 118 for more information on current pulses and air starves.*

6.5.1.2 Non-Operational Performance Loss (NOPL)

1. Non-Operational-Performance-Loss (NOPL, pronounced “nopple”) is the term used to define reversible performance loss that occurs if the stack is not used for a period of time.
2. NOPL increases with increased storage time and increased storage temperature, but should reach a maximum performance loss of no more than 70 mV. Testing over 13 weeks at worst-case storage conditions resulted in a voltage loss of about 1.5 mV/cell/week at 65 A.
3. NOPL results from the growth of an oxide layer on the cathode catalyst that decreases performance. The magnitude of NOPL can be limited by minimizing the storage time and avoiding high storage temperatures. *See Section 10.2, Page 132 for more information on stack storage*.
4. NOPL is recovered by chemically reducing the oxide layer. This is accomplished by drawing current from the stack at low cell voltage. The oxide layer can be removed by:
   A. **Operating the stack at a high current density –** At high current densities, the cell voltages are low while current is drawn from the stack.
   B. **Air-starves -** *See Section 8.6, Page 118*
   C. **Current Pulses –** *See Section 8.6, Page 118*

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6.5.2 Irreversible Performance Degradation

6.5.2.1 Sources of Irreversible Performance Degradation

Running The Stack (unavoidable)	▪ Stack performance degrades during normal operation; running the stack wears it out. However, the magnitude of the degradation is predictable, if the run-time procedure is followed. ▪ *See Section 6.7.4, Page 91 for specific information on run-time performance degradation.*
Air-Air Starts (unavoidable)	▪ If the stack is an air-air state (air on cathode, air on anode), filling the anode with H_2 can induce internal corrosion currents in the stack, causing irreversible performance degradation through cathode catalyst support corrosion. The magnitude of the degradation is predictable, if the start-up procedure is followed. ▪ *See Section 6.7.4, Page 91 for specific information on start-up performance degradation.* ▪ *See Section 8.2, Page 101 for more information on start-ups.* ▪ *See Section 7.3, page 94 for more information start-up degradation mechanisms.*
Fuel Starves / Cell Reversals	▪ **Fuel Starve –** Not providing enough protons/electrons to complete the fuel cell reaction for the amount of current being drawn from the stack. When H_2 gas comes into contact with the catalyst layer, it splits into H^+ and e^-. If a cell cannot produce enough electrons/protons for the current being drawn, it is fuel starved. ▪ **Cell Reversal –** A cell dropping below 0 V, when current is being pulled from the cell. A cell reversal is most commonly caused by a fuel starve. If the cell can produce enough electrons/protons, the cell voltage is above 0 V. If it cannot, the cell voltage will drop below 0 V. A cell reversal can also be caused by cell dehydration. ▪ A fuel starve / cell reversal is caused by: ▪ **Flooding –** Water blocking the anode flow field channels and preventing H_2 gas from reaching the catalyst layer. This prevents the H_2 gas from splitting into protons and electrons. ▪ **Inadequate Purge –** Build up of anode impurities because of an inadequate purge. ▪ **Stack fuel leak / plumbing leak –** If there is large enough anode leak, enough H_2 gas can leak out to cause a fuel starve. ▪ **Too much current draw –** Drawing too much current from the cell. The rate that a cell can split H_2 into protons and electrons is dependent on the state of the cell. For example, more current can be drawn from a cell if it is at BOL and T_{opt}, than a cell at EOL and 20°C less than T_{opt}.

Sources of Irreversible Performance Degradation (*continued*)

Fuel Starves / Cell Reversals (*continued*)	▪ If the cell cannot produce enough protons/electrons from the H_2 fuel to complete the fuel cell reaction, portions of the carbon catalyst support layer are reacted to make up the difference. This is known as carbon corrosion, and is very damaging to the cell. If carbon corrosion is occurring, the cell voltage will be below 0 V. ▪ It should be noted that air starving a cell does not have the same effect as fuel starving. *See Section 8.6.1, Page 118 for more information on air starves.* ▪ *See Section 7.7, Page 97 for more information on fuel starvation.*
Cathode Contamination	▪ Certain airborne contaminants, such as H_2S and SO_x, can cause significant irreversible performance loss. ▪ *See Section 4.2.6, Page 52 for specific information on oxidant contamination.*
Anode Contamination	▪ Fuel contamination can cause significant irreversible performance loss. ▪ *See Section 5.2.7, Page 64 for specific information on fuel contamination.*
Plumbing Leaks, Fuel Leaks After Shutdown	▪ If both H_2 gas and oxygen/air are present on the anode of a single cell, corrosion currents will be present. Corrosion currents cause irreversible performance degradation. ▪ After shutdown, air leaking into the anode will cause air and H_2 to be present in different regions of a cell, causing corrosion currents. ▪ *See Section 7.0, Page 92 for more information on carbon corrosion.*
Running Over-Temperature	▪ Running the stack at high temperatures increases stack degradation rates. Running at extreme temperatures will permanently damage the stack. ▪ *See Section 4.3.2, Page 55 for more information on stack temperatures.*
High Cell Voltages	▪ The membrane chemically degrades faster at high cell voltages than at low cell voltages. Keeping the cell voltage below 0.80 V/cell will help to minimize membrane degradation. ▪ At shutdown, when the load has been removed from the stack and there is still H_2 on the anode, high cell voltages are unavoidable. *See Section 5.7.5, Page 73 for information on preventing fuel leaks.*
Ripple currents	▪ Ripple current applied by the power electronics should be avoided because it accelerates performance degradation and increases the risk of fuel starvation. ▪ *See Section 3.4.2.3, Page 42 for information on ripple currents.*

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6.6 Tips to Maximize Stack Life

Tips to Maximize Stack Life	
Start-Up Purge	▪ To prevent carbon corrosion on start-up, the start-up purge should be executed as described in *Section 5.3.2, Page 66*. ▪ *See Section 7.3, Page 94 for more information on corrosion currents.*
Load-On-Start	▪ To reduce the effect of carbon corrosion, a load should be drawn from the stack during the start-up purge as described in *Section 8.2.3, Page 103.*
Running at T_{opt}	▪ Running at temperatures below T_{opt} can lead to anode flooding and fuel starvation. Running above T_{opt} will increase stack performance degradation. ▪ *See Section 4.3.2, Page 55 for more information on T_{opt}.*
Running at Low Stack Currents/Low Stack Temp	▪ Stack performance degradation is reduced at lower currents because at lower currents, the associated T_{opt} is lower. Also, at lower currents, the stack has more capacity to degrade before EOL triggers such as peak power and cell reversals become an issue.
Avoid Start-Up / Shutdown Cycles (Air-Air Starts)	▪ Filling the anode with hydrogen before starting the stack and air diffusing back into the anode after a shutdown results in corrosion currents that contribute to stack wear-out. The number of stack start-ups should be minimized. ▪ *See Section 7.3, Page 94 for more information on start-up corrosion.*
Reduce Time at OCV / High Cell Voltages	▪ Time at cell voltages above 0.80 V should be minimized because membrane degradation occurs at high cell voltages. ▪ *See Section 5.7.5, Page 73 for more information on preventing fuel leaks.*
Run-Time Purge	▪ The run-time purge should be executed as described in *Section 5.3.4, Page 67* to prevent fuel starves.
Reduce Operating Current	▪ Add cells to the stack to reduce the stack current. ▪ Adding cells also provides more margin for degradation over life. ▪ *See Section 9.0, Page 131 for more information on stack sizing.*

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6.7 Stack Durability Estimates

6.7.1 Ballard Lab Standard Test Cycle

1. In the Ballard lab, durability testing is performed using the standard test cycle. There are ten parts to the standard cycle: the first nine are a 45 minute run at 63°C, each followed by a 30min off period to allow the stack to bleed down. The tenth cycle is a 45 minute run at 73°C, simulating high ambient temperature operation, followed by a 60 minute off period.

2. Unlike the shutdown procedure outlined in *Section 8.5, Page 114*, just after shutdown the Ballard test cycle opens the purge valve for 1sec with the supply valve closed to reduce the amount of H_2 gas in the anode. This decreases the time require for air to diffuse onto the anode, decreasing the time between successive air-air starts.

3. It should be noted that the 63°C temperature is not the T_{opt} for 65.3A that is listed in *Section 4.3.2, Page 55.* The 63°C is a legacy value.

Standard Test Cycle

Step	Time	Current	Temp
#1	45 min 30 min	65.3 A 0 A	63°C
#2	45 min 30 min	65.3 A 0 A	63°C
:	: :	: :	: :
#10	45 min **60 min**	65.3 A 0 A	**73°C**

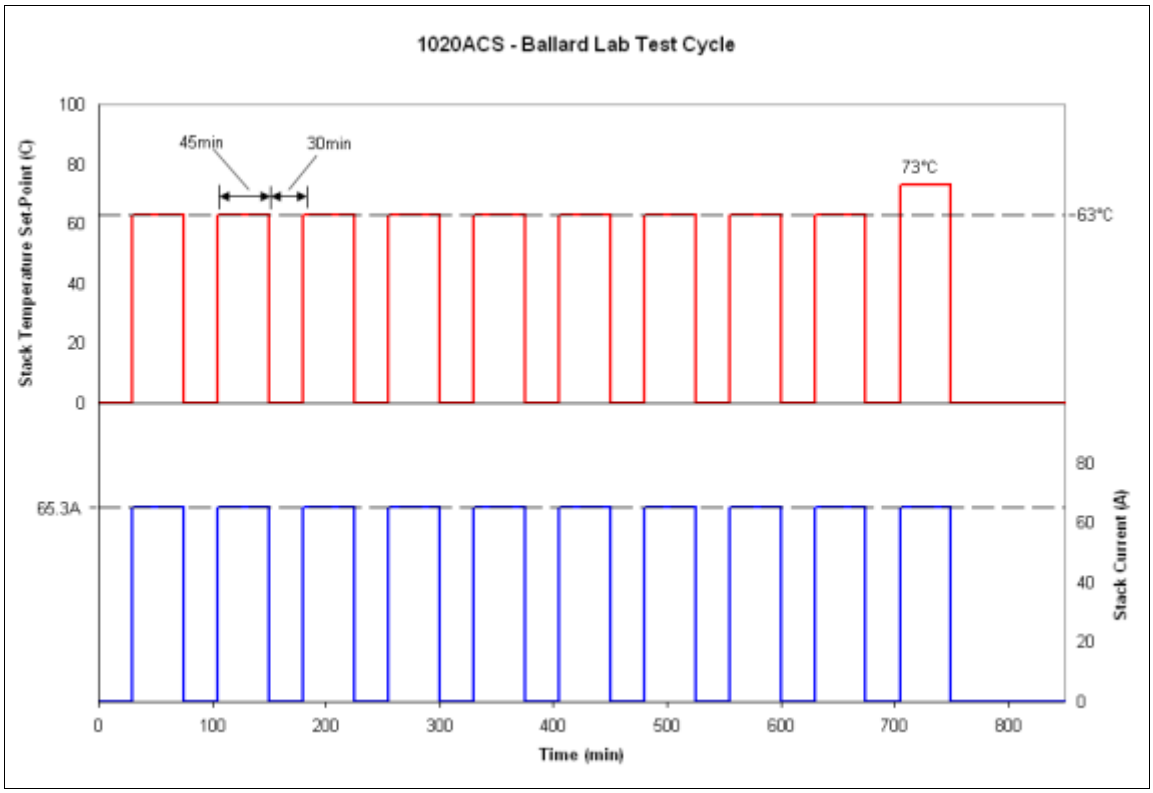


Figure 57: Ballard Lab Test Cycle

6.7.2 Stack Durability Plot

1. Below is a stack durability plot that was generated by running a stack using the Ballard Lab Standard Test Cycle, as described *in Section 6.7.1, Page 89*. In the below plot, the x-axis corresponds to the number of cycles or air-air starts. The primary Y-axis corresponds to the average cell voltage at 65.3 A and 63°C, and the secondary y-axis is the transfer leak rate per cell.
2. The stack performance degradation can be broken up into two distinct regions:
   - **A. Before Fuel Leak Initiation –** Before the fuel leak has initiated, the performance degradation is fairly constant.
   - **B. After Fuel Leak Initiation –** After the fuel leak has initiated, the performance degradation increases. The increase in degradation is caused by air transferring into the anode after shutdown, causing corrosion currents. *See Section 7.4, Page 95 for more information on corrosion currents after shutdown caused by a fuel leak.*
3. Cell-to-cell variability increases with the fuel leak. In most cases, there are a small number of leaky cells. These leaky cells have a higher degradation rate, increasing stack cell-to-cell variability, as well as the overall stack degradation rate.

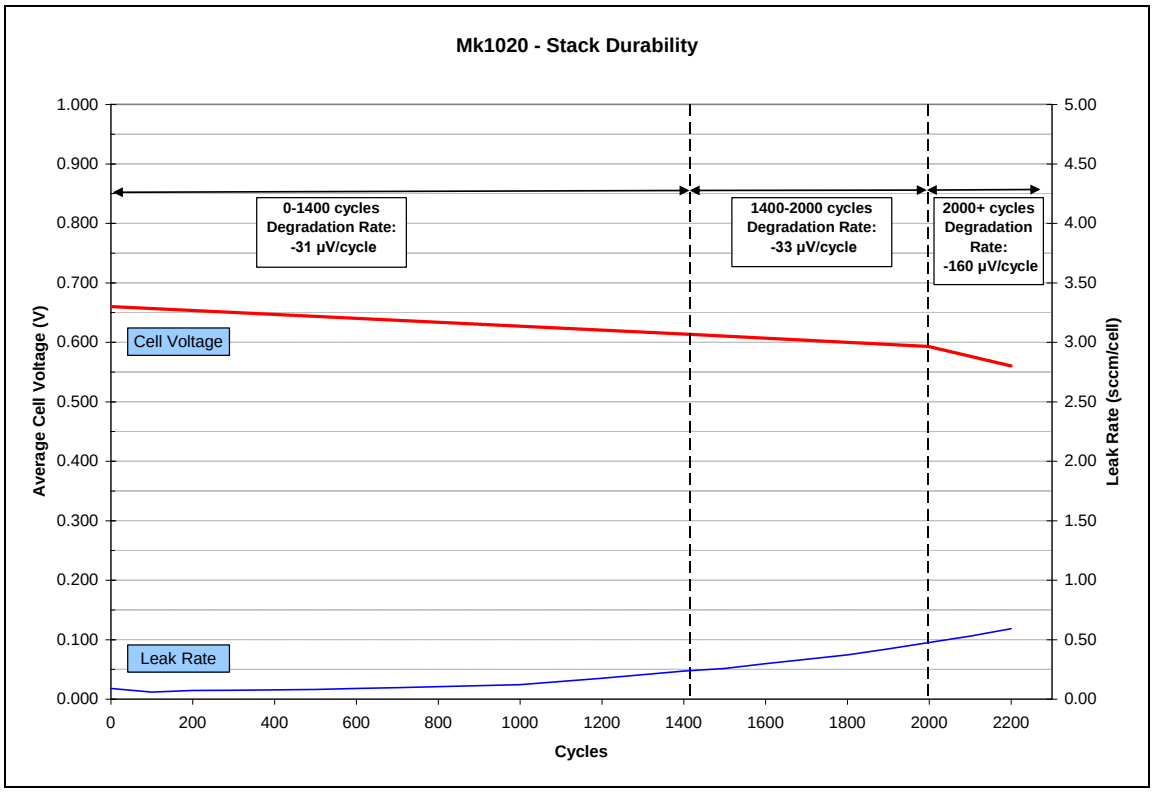


Figure 58: 1020ACS Durability at 65.3A

6.7.3 Stack Life Estimates

1. Testing in the Ballard lab has shown a stack running on the 1020ACS Ballard Standard Lab Test Cycle has a mean lifetime of 1650 cycles (45 minute run time per air-air start) before EOL cell voltage is reached. *See Section 6.2.4, Page 79 for the Ballard EOL definition.*
2. *See Section 6.7.1, Page 89 for a description of the Ballard Lab Standard Test Cycle.*

6.7.4 Start-Up and Run-Time Degradation Estimates

1. If the stack is operated within specification, the two primary sources of performance degradation are run-time degradation (hours on the stack) and air-air starts (number of stack start-ups when there is air on the anode before the start-up).
2. The results of steady-state degradation tests were combined with the results from the Ballard test cycle to produce the following degradation rates.

Run-Time Degradation	Degradation per cell at 65.3 A is $Deg_{run-time} = 12\ ^{\mu V}/_{hour}$
Air-Air Start Degradation	Degradation per cell at 65.3 A is $Deg_{air-air} = 30\ ^{\mu V}/_{start}$
Total Degradation	$Deg_{total} = Deg_{run-time} \times hours\ +\ Deg_{air-air} \times starts$ (at 65.3A)

Example: If $hours = 1000$ and $starts = 1000$ and $NumCells = 56$

Then $Deg_{total} = 12\ ^{\mu V}/_{hour} \times 1000 hrs\ +\ 30\ ^{\mu V}/_{start} \times 500 starts = 15\ ^{mV}/_{cell}$

Then $Deg_{stack} = 15\ ^{mV}/_{cell} \times 56 cells = 0.84 V$

See Section 9.0, Page 131 for information on stack sizing.

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7.0 Carbon Corrosion & Fuel Starvation

7.1 Fuel Cell Reaction During Normal Operation

1. The figure below is a schematic of the fuel cell reaction that occurs in the stack during normal operation.

A. On the anode, H_2 gas splits in two protons and two electrons, and the protons travel through the proton-exchange-membrane (PEM). The electrons travel through the anode bi-polar plate, through the external load, and into the cathode bi-polar plate.

B. On the cathode, the protons, electrons, and O_2 gas recombine to form H_2O.

C. The overall fuel cell reaction results in a voltage across the cell of ~1 V at no load, and the voltage decreases with increasing current draw. *See Section 6.2, Page 76 for more information on stack performance.*

Cathode Reaction

$$4H^+ + 4e^- + O_2 \rightarrow 2H_2O$$

Anode Reaction

$$2H_2 \rightarrow 4H^+ + 4e^-$$

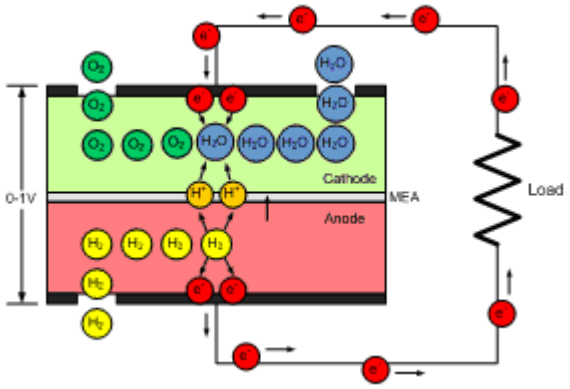


Figure 59: Normal Fuel Cell Operation

7.2 Carbon Corrosion Currents

1. However, if a mixture of H_2 gas and air are on the anode, the resulting corrosion currents cause irreversible damage to the stack. The figure below shows a single cell with a mixture of H_2 gas and air in the anode.

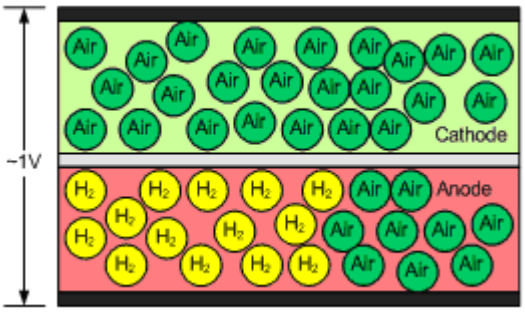


Figure 60: Mixture of Air and H_2 Gas in the Anode

Carbon Corrosion (*continued*)

2. In regions of the anode that contain H_2 gas, there is a voltage across the cell. In regions of the anode that contain air, there is no voltage. Since the bi-polar plates are conductive, the voltage created in the H_2/Air region is applied to the Air/Air region. This effectively creates two electrically separate regions of the cell, connected by the bi-polar plates. The figure below shows the two separate regions.

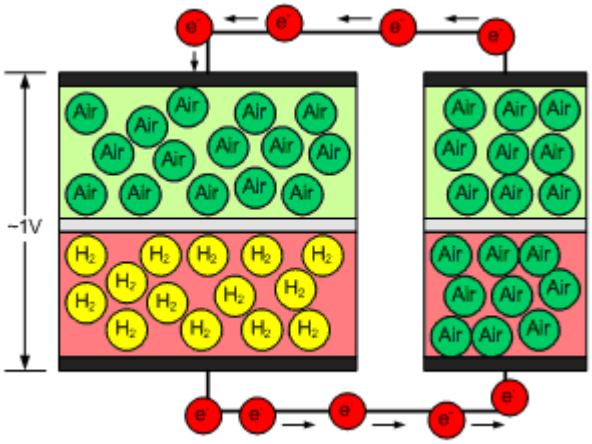


Figure 61: Air and H_2 Gas on the Anode Create Two Cells

3. In the H_2/Air region, the cell behaves like it would under normal operation, and applies ~1 V across the bi-polar plates.

4. In the Air/Air region, the applied voltage causes a current to flow. This current is known as a *corrosion current*. However, since in the Air/Air region there is not any H_2 gas to provide protons and electrons, the carbon catalyst support structure reacts with water to create CO_2, protons, and electrons, consuming the carbon catalyst support.

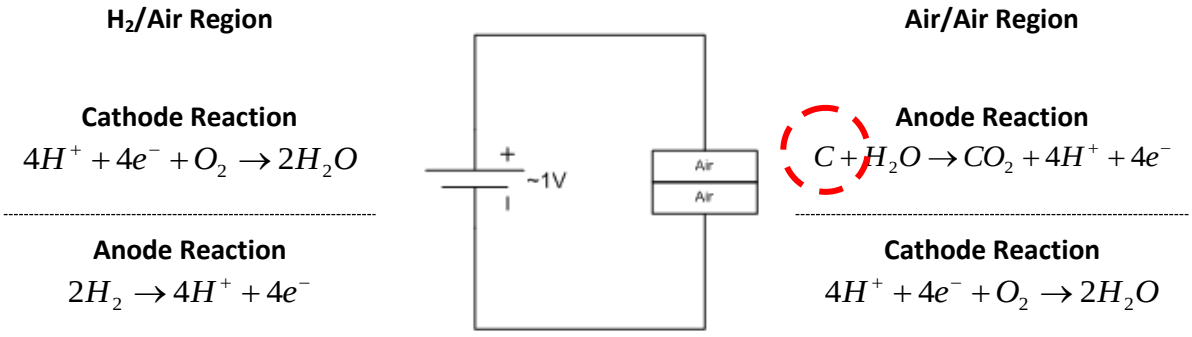


Figure 62: Mixture of Air and H_2 Gas in the Anode

5. Consuming the catalyst support structure permanently reduces stack performance. Therefore, stack exposure to carbon corrosion currents caused by a mixture of air and H_2 on the anode should be minimized.

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7.3 Carbon Corrosion Currents – Start-up

1. If the stack has not been operated for ~30 minutes, the anode is filled with air. On start-up, before the stack can deliver power, the air in the anode needs to be exchanged with fresh hydrogen from upstream of the stack. As the H_2 gas displaces the air, there is a hydrogen/air front that moves through the anode from the anode inlet to the outlet.
2. Behind the front, the cell is H_2/Air, and in front of the front, the cell is Air/Air. The two regions in the cell form corrosion currents, as outlined in *Section 7.2, Page 92*, above.
3. This mechanism is known as start-up corrosion, and start-up corrosion is a significant contributor to the overall performance degradation of the stack.

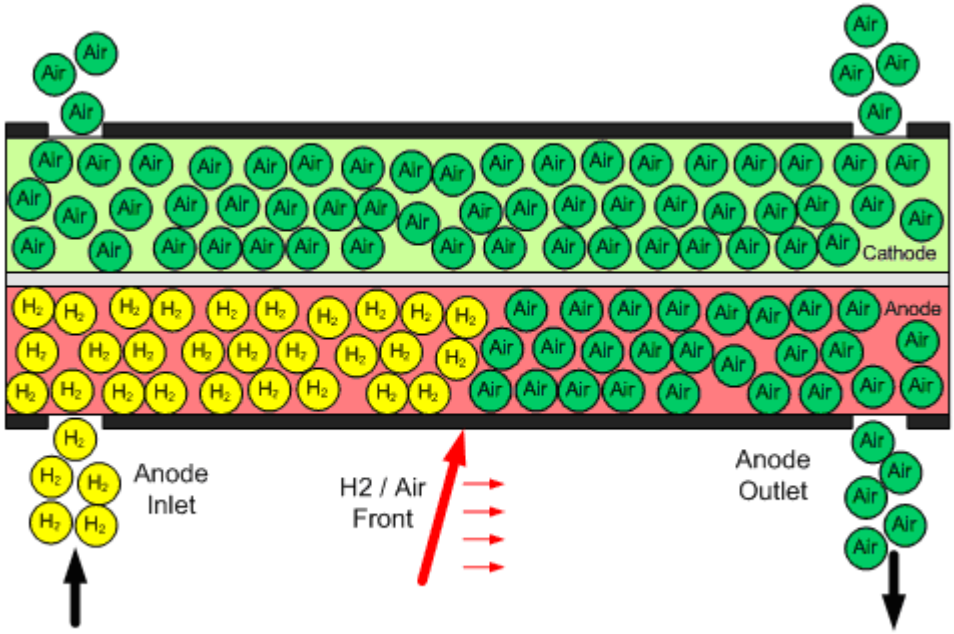


Figure 63: H_2/Air Front During Start-Up Purge

4. To limit start-up corrosion currents and minimize performance degradation on start-up:

A. Perform the start-up purge as described in *Section 5.3.2, Page 66*.

1 Purging the specified start-up purge volume ensures that all the air in the anode is displaced with H_2 gas.

2 Purging the volume within the specified time ensures the H_2/Air front moves quickly through the stack, limiting the time available for corrosion currents to consume the catalyst support structure.

B. Apply a load during the start-up purge as described in *Section 8.2.3, Page 103*.

1 The catalyst support corrosion reaction is greatly reduced at cell voltages less than ~0.80V. Applying a 15 A load during the start-up purge will suppress the cell voltage and prevent carbon corrosion.

2 Care should be taken that the start-up purge load does not pull cells into reversal.

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7.4 Carbon Corrosion Currents – Fuel Leak

1. During normal stack operation, when the anode is pressurised to ~0.36 barg, small fuel leaks do not affect cell performance or durability because pressurised H_2 gas flows out of the anode into the air stream.
2. However, after shutdown, fuel leaks can lead to corrosion currents. As described in *Section 8.5.2, Page 114*, a short time after shutdown, the consumption of H_2 gas in the anode will reduce the anode pressure below atmospheric. If there is a fuel leak, air is sucked into the anode at the leak location. This results in an Air/Air region in the cell when the majority of the cell is at H_2/Air, causing corrosion currents.
3. As described in *Section 6.7.2, Page 90*, the performance degradation rate of the stack increases significantly after a fuel leak has been initiated.
4. Before a fuel leak has been initiated, air diffuses more uniformly into the anode after shutdown, reducing the magnitude of the carbon corrosion currents.

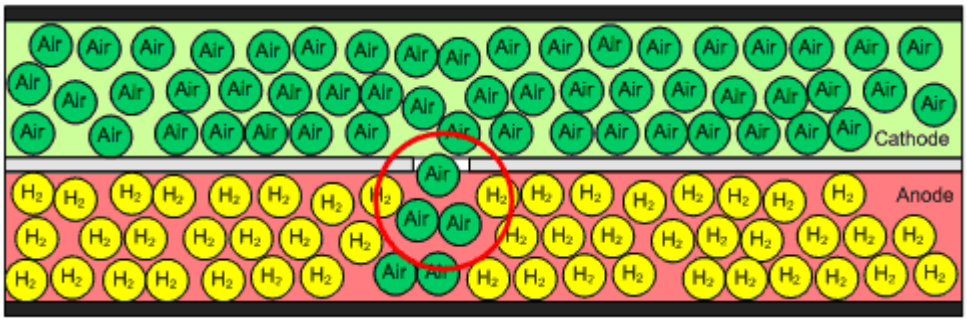


Figure 64: Corrosion Currents Resulting from a Fuel Leak

5. After shutdown, the cell with the fuel leak is the first to reduce from OCV (~1 V), as shown in the figure below. *See Section 5.7.4, Page 73 for a plot of a leaky stack after shutdown*.

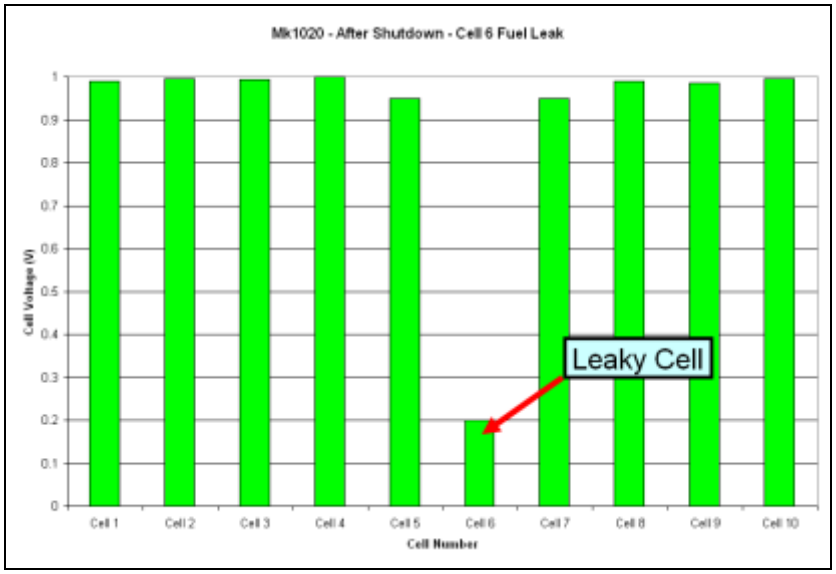


Figure 65: Cell Voltage After Shutdown for a Leaky Stack

7.5 Carbon Corrosion Currents – Leaky Purge Valve

1. A leaky purge valve has the same effect as a fuel leak. Air introduced into a hydrogen-filled anode results in corrosion currents.
2. An indicator of a leaky purge valve is cell-1 being the first cell to reduce from OCV.

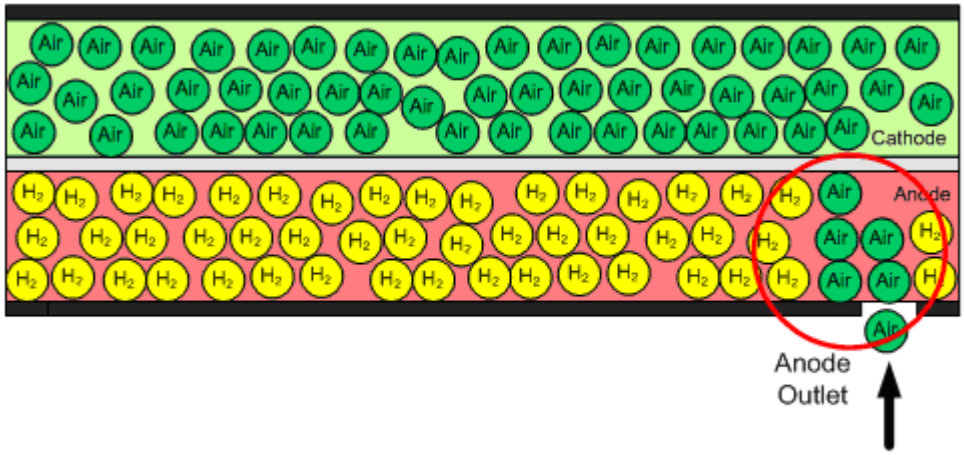


Figure 66: Corrosion Currents Resulting from a Leaky Purge Valve

See Section 5.6.4, Page 71 for more information on purge valves.

7.6 Carbon Corrosion Currents – Leaky Supply Valve

1. A leaky supply valve will continue to let H_2 gas into the anode after shutdown. This results in a region of H_2 gas in an otherwise air-filled anode, resulting in carbon corrosion currents.
2. A leaky supply valve usually results in the stack never reducing to 0 V.
3. A leaky supply valve usually results in severe stack damage because the leak is continuous and the stack never reaches an Air/Air state.

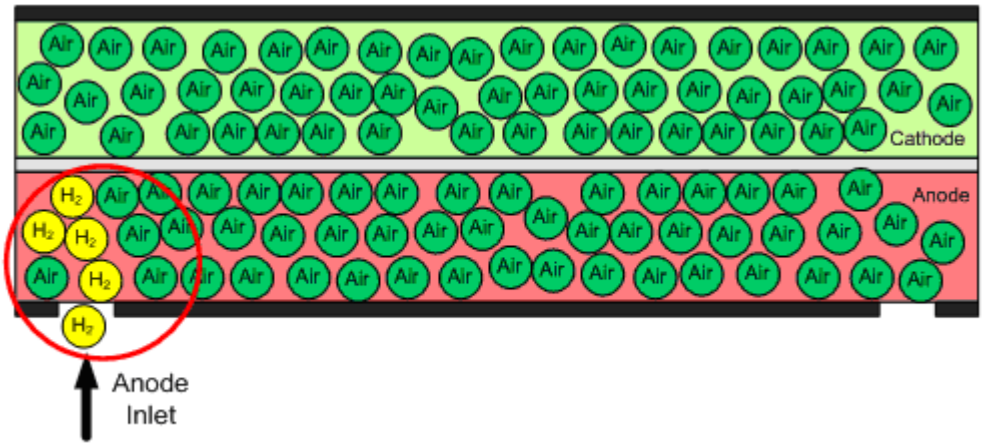


Figure 67: Corrosion Currents Resulting from a Leaky Supply Valve

See Section 5.6.3, Page 70 for more information on supply valves.

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7.7 Fuel Starvation

1. Under normal operation, when current is drawn from a cell, the electrons are supplied by the H_2 gas splitting into electrons and protons. However, if the cell cannot deliver enough electrons, the cell is fuel starved.
2. The cell must provide electrons to meet the current draw. After the cell has reached its capacity to supply electrons through disassociating H_2, the remaining electrons are delivered by reacting the carbon catalyst support with water to form CO_2, protons, and electrons. This irreversibly damages the cell and permanently reduces performance.

Normal Operation

Cathode Reaction

$$4H^+ + 4e^- + O_2 \rightarrow 2H_2O$$

Anode Reaction

$$2H_2 \rightarrow 4H^+ + 4e^-$$

Fuel Starve

Cathode Reaction

$$4H^+ + 4e^- + O_2 \rightarrow 2H_2O$$

Anode Reaction

$$C + H_2O \rightarrow CO_2 + 4H^+ + 4e^-$$

3. During a fuel starve, the cell voltage drops below 0V and goes into reversal, permanently damaging the stack. If the reversal is deep enough, stack fires can result. *See Section 8.7.2, Page 127 for more information on how to detect a reversal.*
4. Air starves (limiting the amount of O_2 gas) are different than fuel starves. *See Section 8.6, Page 118 for more information on air starves.*

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8.0 Stack Operation

To successfully operate the 1020ACS, the following procedures must be followed to ensure that performance, durability, and safety specifications can be met.

There stack can be in five major states:

1. **Standby –** The stack is not consuming reactants or delivering power.
2. **Start-Up –** *Start-up* takes the stack from S*tandby* and puts it into a state where current can be safely drawn from the stack.
3. **Warm-Up –** *Warm-Up* takes the stack from *Start-Up* and puts it into a state where rated power can be safely delivered by the stack in *Run*.
4. **Run –** The stack is delivering rated power.
5. **Shutdown –** *Shutdown* takes the stack from *Run* and puts it into *Standby*.

8.1 Standby

In *Standby*, the stack is not consuming reactants or delivering power.

8.1.1 Standby – Procedure

1. When in *Standby*, the stack is not consuming reactants, and it should also be in a state where after start-up, it can deliver power.
2. If the stack is not frequently exercised, the stack can experience Non-Operational-Performance-Loss (NOPL). *See Section 6.5.1.2, Page 85 for more information on NOPL.*

Actuator	Cmd	Notes
Fan:	0	▪ In *Standby*, all actuators are in their safe state.
Supply Valve:	Closed	
Purge Valve:	Closed	
Stack Contactor:	Open	
Stack Current:	0	

CAUTION

Irreversible performance loss will occur if current is drawn from the stack without reactants present. If the stack contactor is open, current cannot be drawn from the stack.

8.1.2 Standby – Stack Temperature

Specification	**Required:** -10°C or greater

1. Testing in the Ballard environmental chamber has shown that the stack can be started repeatedly, consistently, and without performance loss if the stack and oxidant inlet temperature are -5°C or greater.
2. Start-ups at -10°C are possible, but a series of repeated start-ups at -10°C will often cause cells to go into reversal, causing irreversible performance loss.
3. The stack should be maintained at -5°C or above when in standby to ensure it can be successfully started.
4. Heaters can be used to keep the stack warm in *Standby*. Use of heaters will result in a decrease of relative humidity in the surrounding environment, unless water is added while the heaters are active. Air humidity of <5% RH may impact the ability of the stack to start. *See Section 8.1.3, Page 100 for more information on ambient RH during Standby.*
5. The stack cannot deliver full rated power immediately after it has been started. A warm-up period is required after start-up to allow the stack time to reach its operating temperature, to hydrate the membrane, and to increase catalyst performance through air starves or current pulse. *See Section 8.6, Page 118 for more information on air starves and current pulses and Section 6.5.1.2, Page 85 for more information on non-operational-performance-loss.*
6. The time required for the stack to go from standby to a state where it can deliver rated power is influenced by standby stack temperature. A lower standby stack temperature results in an increase in the time required to reach rated power*. See Section 8.3.2, Page 106 for more information on warm-up time.*

See Section 4.2.1, Page 50 for more information on ambient air temperature.

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8.1.3 Standby – Ambient Air RH

Specification	**Required:** 2% or greater

1. Testing at Ballard shows that the stack can be started and reach full power output in ~100 s if the ambient temperature is lab temperature (~20°C) and the ambient air RH is 5% or above.
2. If the stack is started when the ambient air RH is less than 2%, the stack requires >200 s to go from start-up to full power. At <5% ambient RH, the membrane is sufficiently dehydrated to cause performance loss, limiting the amount of current that can be pulled from the stack. However, the water required to hydrate the membrane is created when current is drawn from the stack. The combination of these two effects results in an increased time to start-up the stack.
3. Low humidity conditions can be encountered in extreme environments, or may result when the stack is placed in an enclosure subject to heating. This heating may be the result of solar loading of an enclosure; heating from adjacent sources such as other piece of equipment in the same enclosure; or through the use of heaters to maintain the stack within the recommended start-up temperature range in a cold environment. Heating the air surrounding the system without adding moisture results in decreased relative humidity.
4. During low ambient humidity start-up, the amount of current drawn from the stack should be limited such that average cell voltage does not decrease below 0.5 V/cell. *See Section 8.3.3, Page 108 for more information on stack current during warm-up.*
5. The figure below shows stack start-ups from varying ambient humidities at lab ambient temperature. The start-ups were performed after allowing the stacks to equilibrate with the ambient environment. Load ramp rate during these start-ups was regulated by drawing as much current from the stack as possible while maintaining a minimum stack voltage equivalent to 0.5 V/cell.

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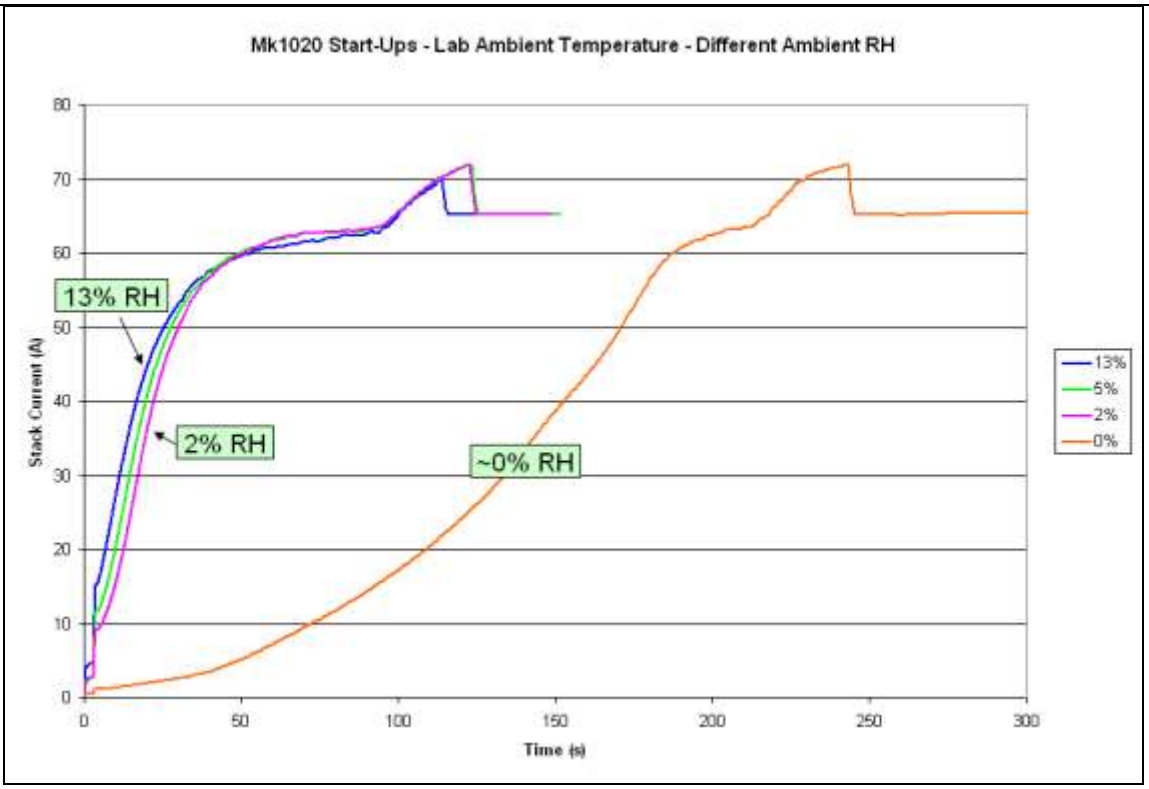


Figure 68: Start-Ups at Different Ambient RHs

See Section 4.2.3, Page 51 for more information on ambient air RH.

8.2 Start-Up

Start-up takes the stack from S*tandby* and puts it into a state where current can be safely drawn from the stack.

8.2.1 Start-Up - Procedure

1. The fan is started before the start-up purge to give the fan time to spool-up as well as to provide airflow to dilute the start-up purge.
2. The anode gas needs to be exchanged with fresh hydrogen from upstream of the stack before the stack can deliver power. The purge must meet certain specifications to minimize the irreversible performance loss caused by exchanging air in the anode with H_2 gas. *See Section 7.3, Page 94, for more information on corrosion during start-up. See Section 5.3.2, Page 66 for specific information on the start-up purge.*
3. A load is applied during the initial purge to pull down the average cell voltage, reducing the severity of the start-up corrosion. *See Section 8.2.3, Page 103 for more information on the start-up load.*

Step	Actuator	Cmd	Notes
1	**Fan:** **Supply Valve:**	*See Notes* Closed	▪ Allow the fan time to spool up to provide ventilation and air flow to the anode and cathode for the initial

	Purge Valve: **Stack Contactor:** **Stack Current:**	Closed Open 0	purge. ■ The start-up fan speed should be high enough to get the fan turning and to provide enough flow for the initial purge.
	After 3 seconds… (Fan spool up time)		
2	**Fan:** **Supply Valve:** **Purge Valve:** **Stack Contactor:** **Stack Current:**	See Step 1 Open Open Open *See Below*	■ Both the supply and purge valves are open to exchange the anode gas with fresh H_2 from upstream of the stack. The fan provides airflow to the anode and the cathode. The stack current is the start-up load (*See Below*).
	After 0.2 seconds… (Start-up purge duration)		
3	**Fan:** **Supply Valve:** **Purge Valve:** **Stack Contactor:** **Stack Current:**	Fan Min Open Closed Closed *See Below*	■ Once the start-up purge is finished, *Start-Up* is complete. ■ The fan is set to the minimum set-point to allow the stack to warm-up as quickly as possible.
	Transition to Warm-Up		

8.2.2 Start-Up – Timing Diagram

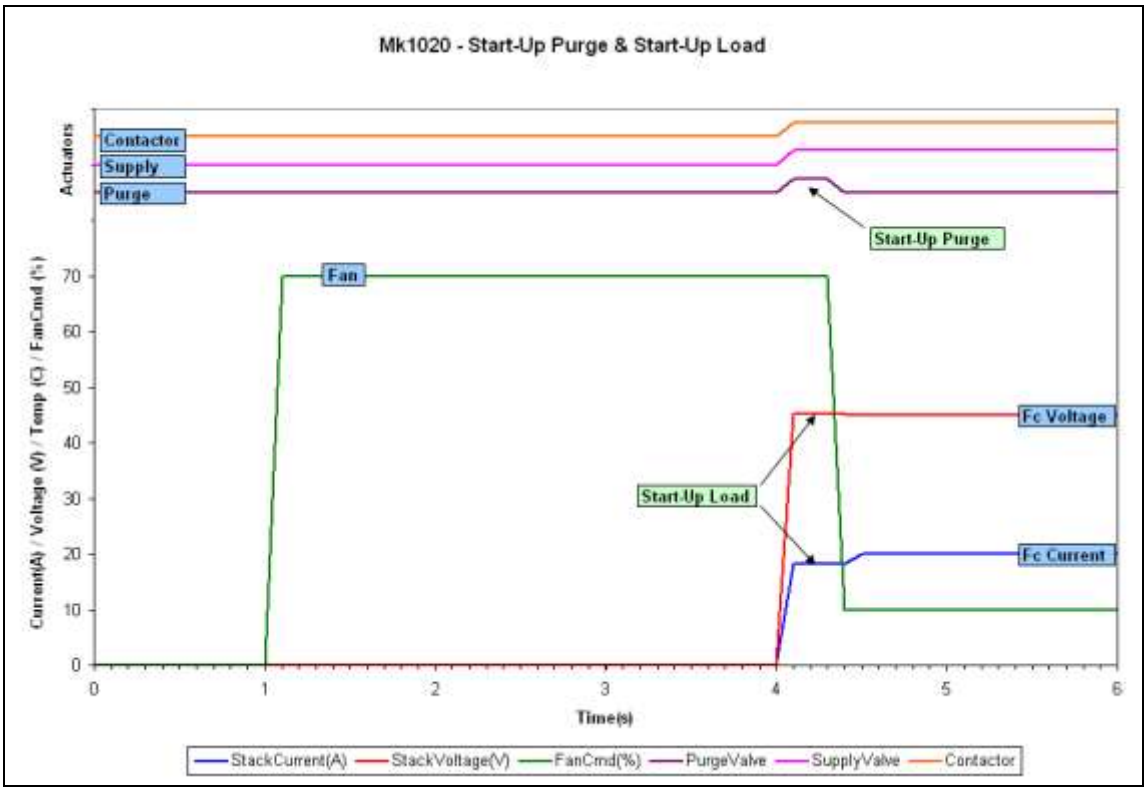


Figure 69: Start-Up Procedure – 56 cell stack

8.2.3 Start-Up - Start-Up Purge Load

During the start-up purge, the process of exchange the air in the anode with H_2 gas causes corrosion currents that irreversibly damage the stack. In addition to meeting the start-up purge requirements described in *Section 5.3.2, Page 66*, applying a load to pull down the stack voltage during the initial purge helps to minimize the performance loss.

Specification	**Cell Voltage:** Average Cell Voltage less than 0.80 V No cells in reversal

1. The start-up load should pull down the average cell voltage to reduce the degree of carbon corrosion. Several methods are available to draw the start-up load.

- **A. Power Electronics –** If power electronics are connected directly to the stack, current can be pulled from the stack during the initial purge. There must be H_2 gas on the anode before the load is drawn or cells can be pulled into reversal, irreversibly damaging the stack.
- **B. Power Resistor –** If a power resistor is connected in parallel with the stack, the resistor can be used to provide the start-up load. The advantage of a power resistor is that it cannot pull cells into deep reversal if reactants are not present because it is a passive electrical element. The size of the resistor is dependent on the number of cells in the stack.

2. The start-up load required is ~15 A at BOL. At EOL, the load requirement reduces.

3. The start-up load should be applied as soon as possible after the supply valve and purge valves are opened, without pulling cells into reversal.

- **A.** There is a delay between when the purge valves and the supply valves are opened and when H_2 gas reaches the cells furthest away from the anode inlet. If the start-up load is applied too quickly, these cells can fuel-starve and go into reversal, as shown in the figure below.
- **B.** In a low cell-count stack (10 cell), the start-up load can be applied at the same time as the valves are opened.
- **C.** In a high cell-count stack (80 cell), a delay is required between when the valves re opened and when the load is applied.
- **D.** During development, an oscilloscope should be used to measure the voltage of the last cell in the stack during start-up to see if it goes into reversal. If the last cell does go into reversal, the delay should be increased.

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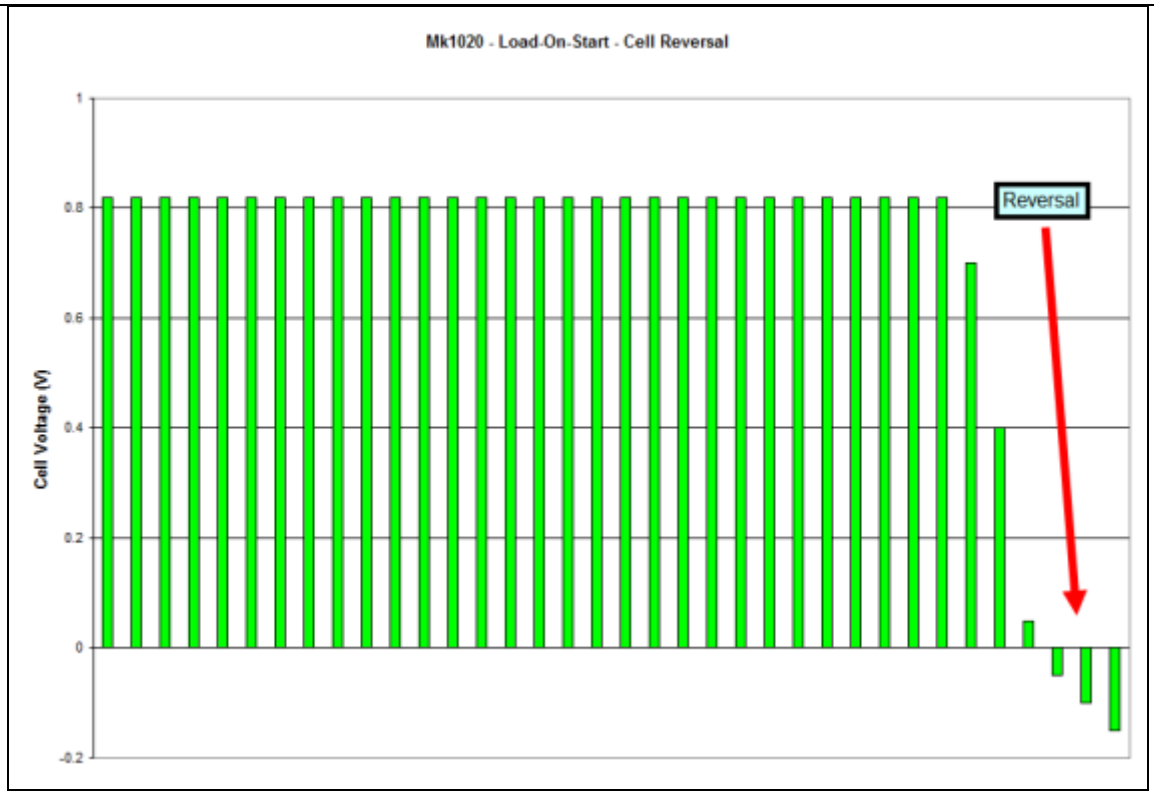


Figure 70: Start-Up Purge Load – Cell Reversal

See Section 7.3, Page 94 for more information on corrosion during start-up

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8.3 Warm-Up

Warm-Up takes the stack from *Start-Up* and puts it into a state where rated power can be safely delivered by the stack in *Run*.

8.3.1 Warm-Up – Procedure

1. In *Warm-Up*, the stack is brought up to temperature, the membrane is hydrated, and the catalyst performance is increased to enable the stack to deliver rated power. The time required for the 1020ACS stack to reach rated power output is a function of the time required for the stack to heat to the optimal operating temperature and the time required for the stack to hydrate.
2. The specifics of *Warm-Up* will depend on the application. In some applications, *start-up* is immediately followed by *Run*.
3. *Warm-Up* is only concerned with the stack current, voltage, and power. Fan and purge-valve control are described in *Run*, below.
4. In all cases, care must be taken to ensure that cells are not pulled into reversal. This can occur if too much current is drawn for the state that the stack is in.

Actuator	Cmd	Notes
Fan:	*See Run*	■ In Warm-Up, the fan and purge valve are controlled as described in *Run*.
Supply Valve:	Open	■ Keeping the coolant flow to a minimum allows the stack to warm-up quickly. The minimum airflow is defined as an air-stoich of greater than 20, as described in *Section 4.2.4, Page 51*, and enough airflow to dilute the anode purge, as described in *Section 5.5, Page 69*.
Purge Valve:	*See Run*	
Stack Contactor:	Closed	
Stack Current:	*See Below*	

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8.3.2 Warm-Up Time and Bridging Energy Requirements

1. The time required for the 1020ACS stack to reach steady-state performance and deliver rated power is a function of:

   A. **The difference between the stack temperature and the optimal operating temperature -** *See Section 4.3.2, Page 55 for more information stack temperature.*

   B. **The hydration state of the membrane -** This is dependant on ambient RH during start-up and the time since the most recent shutdown. *See Section 8.1.3, Page 100 for more information on the effect of ambient RH during standby.*

   C. **The condition of the catalyst –** Extended periods of non-operation can decrease catalyst performance. The performance can be improved using air-starves or current pulses. *See Section 8.6, Page 118 for more information on air starves and current pulses.*

2. Standby temperature partially influences the time required for the stack to go from *Standby* to a state where it can deliver rated power. A lower standby stack temperature increases the time required to reach rated power.

3. The figure below shows the thermal performance of a stack in the Ballard environmental chamber during start-up for a -5°C ambient temperature. The stack never reached the optimum temperature because the fan provides too much coolant flow at its minimum set-point.

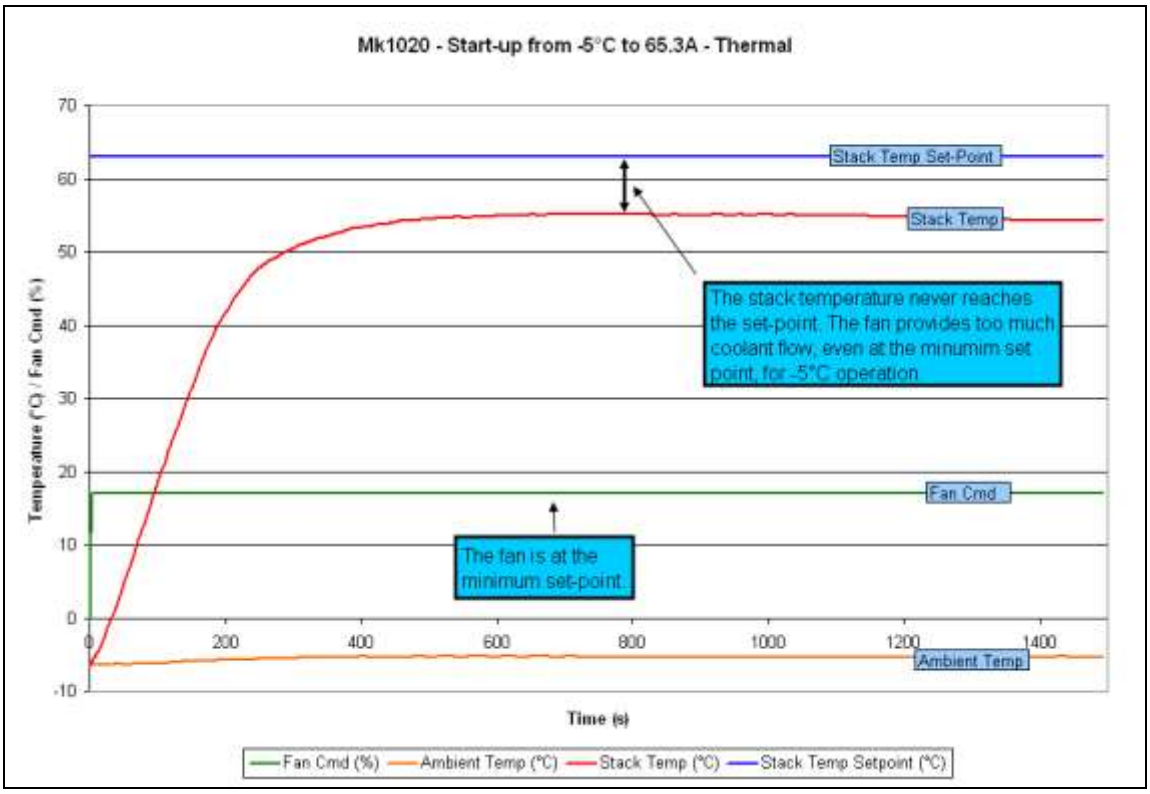


Figure 71: Thermal Performance during a Start-Up in -5°C Ambient

4. *More information on fan operation in cold ambient environments can be found in Section 8.4.4, Page 113.*

Warm-Up Time and Bridging Energy Requirements (*continued*)

5. After start-up, when the stack is warming up and cannot deliver rated power, the system still needs to be able to deliver rated power to the external load. In most applications, this difference in power is supplied by either batteries or super caps. *See Section 3.4, Page 41 for more information on energy storage in the system.*

6. The difference between the rated power and the actual power is referred to as bridging power. The total energy is referred to as bridging energy, as shown in the figure below.

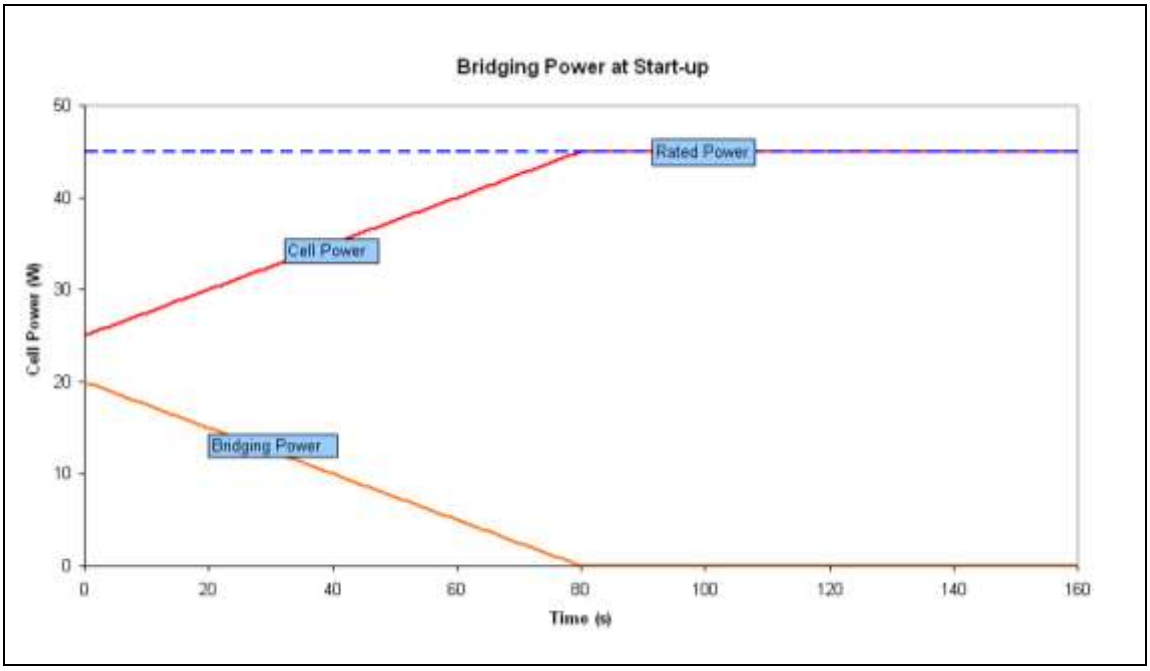


Figure 72: Bridging Power at Start-Up

7. The system should have enough bridging energy to meet the system output power requirements while the stack is warming up.

8. The below table shows a subset of the results from start-up testing performed in the Ballard environmental chamber. The testing showed that time to rated power and the bridging energy requirements were dependent on a multiple factors, including the ambient temperature. Throughout the start-up testing, the current was controlled as described in *Section 8.3.3, Page 108*. The current was limited by maintaining a minimum cell voltage of 0.50 V/cell.

Soak Temperature (°C)	Time to 80% Power (min:sec)	Time to 100% Power (min:sec)	Bridging Energy (J/cell)
20	0:20	1:35	340
0	1:00	2:51	1080
-10	1:20	3:47	1660

Figure 73: Start-Up Time and Bridging Energy Requirements

9. The system bridging energy capability should be much larger than the values specified in the table. The testing showed a large variability in start-up times and bridging energy requirements. Also, degradation of the bridging energy supply should be over life should be taken into account.

10. Ballard recommends performing system start-up tests over a wide range of ambient conditions to ensure that the system bridging energy capability is sufficient for the application.

8.3.3 Warm-Up – Stack Current

1. During *Warm-Up*, the stack current is controlled to warm up and hydrate the stack in a safe manner.
2. To improve the performance of the catalyst, the stack can be air-starved or current pulsed:
   - **A. Air-Starve –** Performing an air-starve during *Warm-Up* can quickly generate heat as well as improve stack performance. *See Section 8.6, Page 118 for more information on air-starves.*
   - **B. Current Pulse -** Performing several current pulses during *Warm-Up* can improve stack performance. Current pulses are better than air-starves for minimizing bridging energy requirements. *See Section 8.6, Page 118 for more information on current pulses.*
3. **Current Ramp** –Ramp-rate is defined as the change in current over the change in time, as shown in the figure below. The figure below illustrates a current ramp-up rate of 1 A/s.

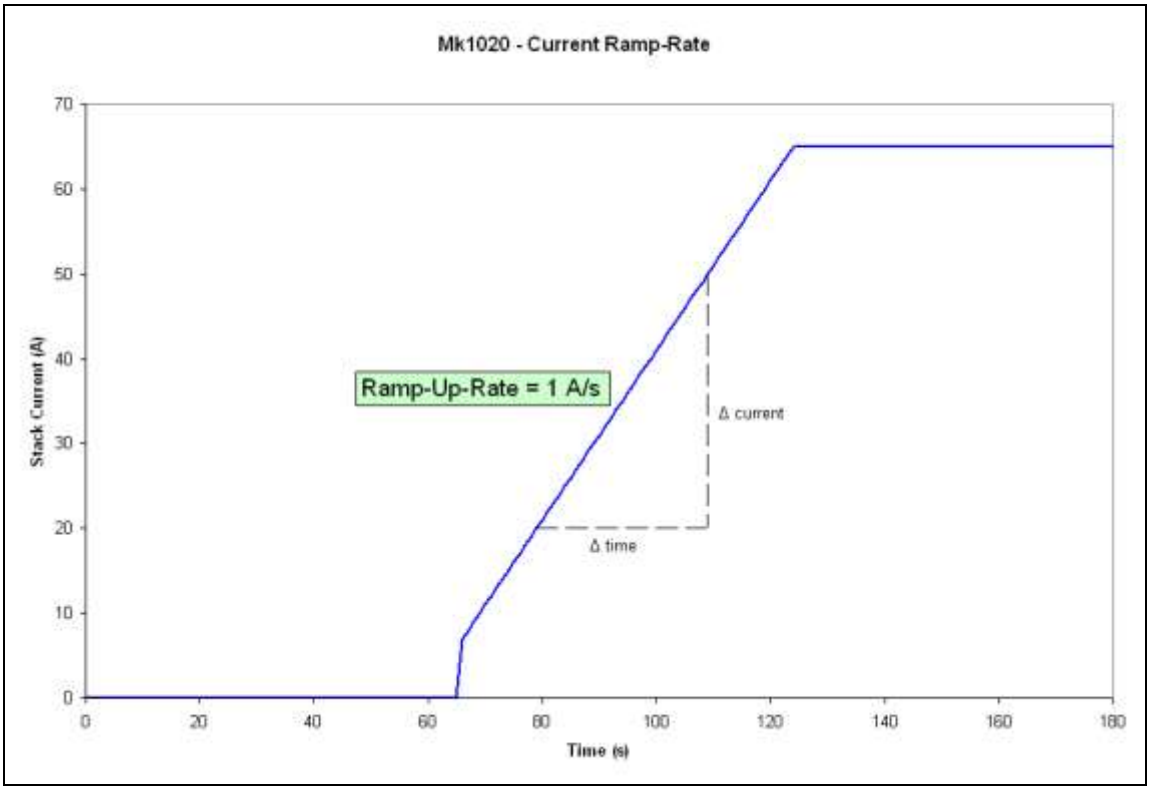


Figure 74: Current Ramp

Warm-Up – Stack Current (*continued*)

4. Control Current Ramp Based on Stack Voltage – Drawing as much load as possible during *Warm-Up* will optimize both warm up and hydration of the stack. Heat generation and water production by the fuel cell stack are proportional to current output. The stack will also produce more heat at lower stack voltage.

A. As much current as possible should be drawn while maintaining the minimum stack voltage ≥ 0.5V/cell.

B. During start-up from cold ambient situations, it can be advantageous to operate the stack below the peak power point. The current can be limited using a ≥0.3 V/cell minimum average cell voltage to generate heat more quickly. Care should be taken that cells are not pulled into reversal. *See Section 6.2.5, Page 80 for more information on peak power.*

C. In all situations, the stack current should never exceed the 75 A maximum stack current. Minimum stack voltage should also conform to requirement defined in Section 8.7.2 – Minimum Stack Voltage Alarm.

D. The figure below is an example of maintaining an average cell voltage of 0.5V/cell during *Warm-Up*. The plot is taken from start-up testing performed at Ballard using a Ballard test station. The load bank in the test station does not have the capability to maintain the stack voltage at exactly 0.5 V/cell, but the plot does show that the current is limited based on cell voltage.

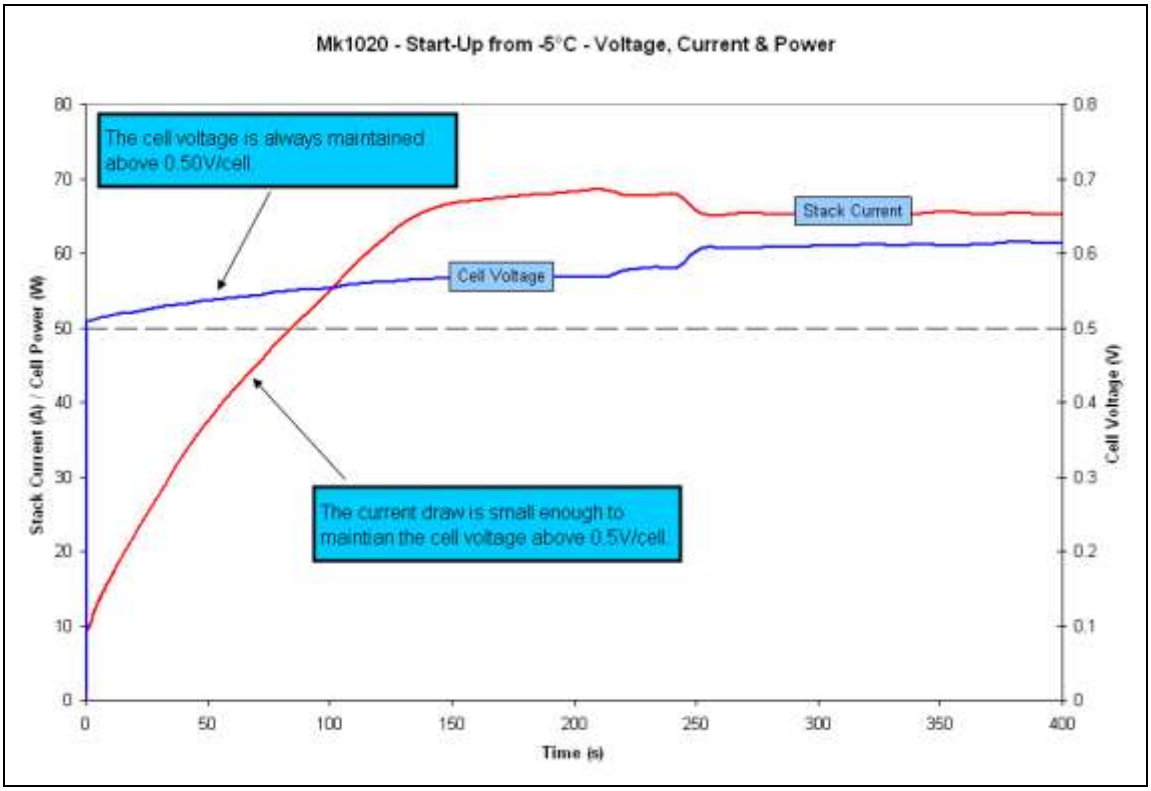


Figure 75: Stack Voltage Limited to 0.5 V During Start-Up

8.3.4 Warm-Up – Hot/Dry Conditions

1. When the stack is started from hot ambient conditions, the stack temperature will initially be higher than the optimum temperature recommended for the instantaneous current output of the stack. *For example, a stack started from an ambient temperature of 46°C will be running at a higher temperature than optimum until current output has increased to >38 A.* Until the stack temperature is at or below the optimum temperature for instantaneous current output, the stack will not reach optimum hydration and performance will be degraded.
2. If the stack is also starting from dry ambient conditions, initial stack performance will be limited by stack dehydration. This will limit the amount of current that can be applied to the stack while maintaining an acceptable voltage.
3. To maximize performance after a start-up from hot/dry conditions, it is necessary to ramp stack current as quickly as possible. A rapid increase in current output will allow the stack to operate at a temperature lower than optimum temperature, which will hydrate the stack and improve performance. This is possible because the thermal mass of the stack will limit stack temperature rise after initial start-up. A slow current ramp rate will prevent the stack from hydrating sufficiently during warm-up, resulting in a long warm-up period and degraded stack performance. If your application includes a critical requirement for maximum performance under hot/dry conditions, please contact Ballard Applications Engineering for further discussions. The figure below compares the effect of fast and slow current ramp rates during a hot/dry warm-up:

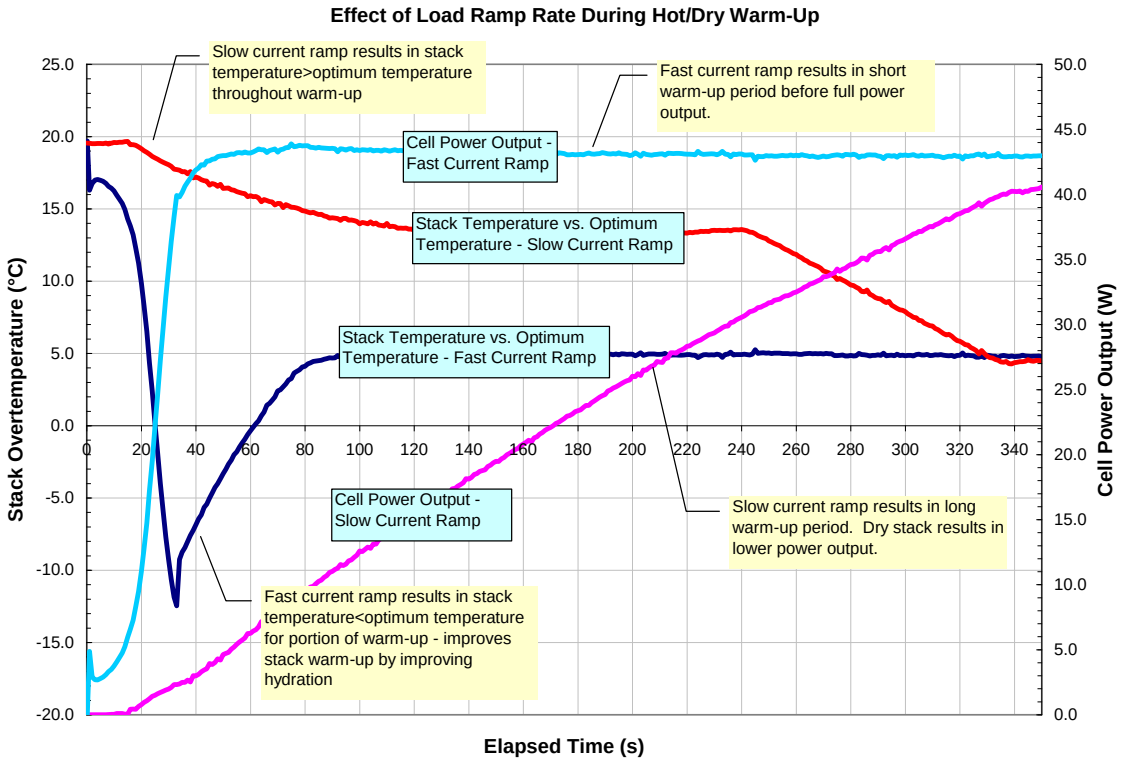


Figure 76 – Effect of Load Ramp Rate During Hot/Dry Warm-Up

8.4 Run

In *Run*, the stack is delivering rated power to the system.

8.4.1 Run Procedure

Actuator	Cmd	Notes
Fan:	*See Below*	
Supply Valve:	Open	
Purge Valve:	*See Below*	
Stack Contactor:	Closed	
Stack Current:	*See Below*	

8.4.2 Run Procedure - Stack Current

1. The stack current can range between the max and min stack current (0 to 75 A), as described in *Section 6.1.1, Page 75*.
2. All Ballard life testing is performed at 65.3 A. Running the stack above 65.3 A for extended periods will increase the performance degradation rate. At higher currents, the stack runs at a higher temperature, and performance degradation is accelerated with higher temperatures. Similarly, the performance degradation rate will decrease at lower currents. *See Section 6.6, Page 88 for tips on how to maximize stack life.*
3. Individual cell voltages or the stack voltage should be monitored to ensure that increases in stack current do not drive cells into reversal. *See Section 8.7.2, Page 127 for more information on detecting cell reversals by monitoring the stack voltage.*
4. In some applications, limiting the current "ramp-rate" can be advantageous to allow the stack temperature, and other stack properties, to better keep up with the stack current. *See Section 8.3.3, Page 108 for more information on ramp rate.* During *Run*, both the "Ramp-Up" and the "Ramp-Down" can be limited.
5. Air-Starves and current pulses can be performed during *Run* to increase stack performance.

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8.4.3 Run Procedure - Purge Valve

1. During *Run*, the purge valve should be operated to purge the stack as described in *Section 5.3.4, Page 67.*
2. The figure below is an example of the purge strategy. If the pure interval is 2300 Amp·Seconds,

$$\textit{Purge Interval at a constant } 65.3A, \quad \frac{2300A \cdot s}{65.3A} = 35s, \quad \textit{at } 7.3A, \quad \frac{2300A \cdot s}{7.3A} = 315s$$

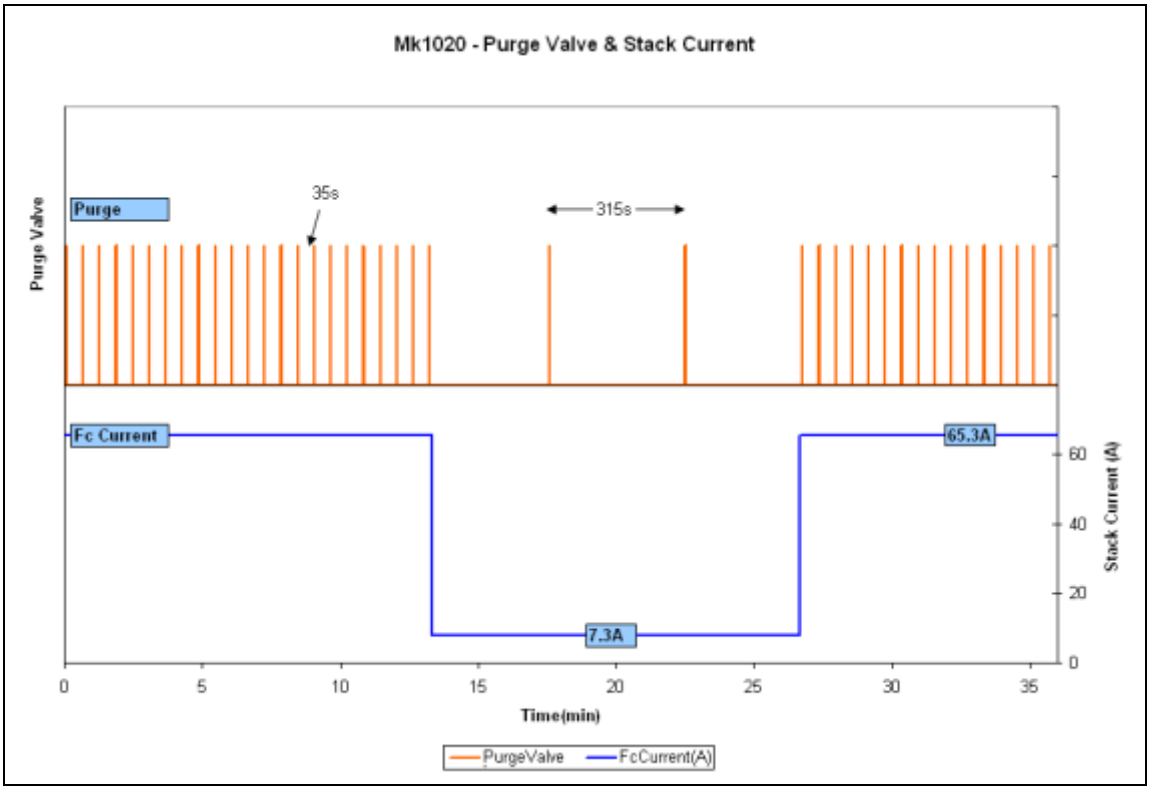


Figure 77: 1020ACS Purge Valve and Stack Current – 56 cell Stack

See Section 5.0, Page 61 for more information on the fuel stream.

8.4.4 Run Procedure - Fan

1. During *Run*, the fan should be operated to provide the proper amount of coolant flow to maintain the stack temperature (T_{stack}) at the stack optimum temperature (T_{opt}). If the stack cooling system cannot deliver the required flow, the stack temperature should be maintained between the max and min stack temperatures by adjusting the stack current. *See Section 4.3.2, Page 55 for more information on optimum stack temperature.*
2. The fan should always be operating at or above the minimum fan set-point to provide oxidant for the fuel cell reaction. The min-fan-set-point is the minimum control signal required to start the fan and keep it running.
3. In cold ambient temperatures, the airflow provided by the fan at the minimum fan set-point might overcool the stack. To reduce the airflow, but still maintain enough flow for the fuel cell reaction, it is possible to:
   - **E. Pulse the fan** – The fan command can be toggled between zero and the min-fan-set-point. The fan's momentum should keep it turning after the control signal is dropped to zero, and the min-fan-set-point will get the fan started again if it is spinning too slowly.
   - **F. Use a smaller secondary fan** – In addition to the main fan, a secondary fan with a smaller minimum flow can be used. The secondary fan will not be able to provide airflow for high ambient temperature, high current operation, but it will have enough turndown to provide the required airflow in low ambient temperature, low current operation.
4. The figure below is an example of the fan maintaining the stack temperature at T_{opt}.

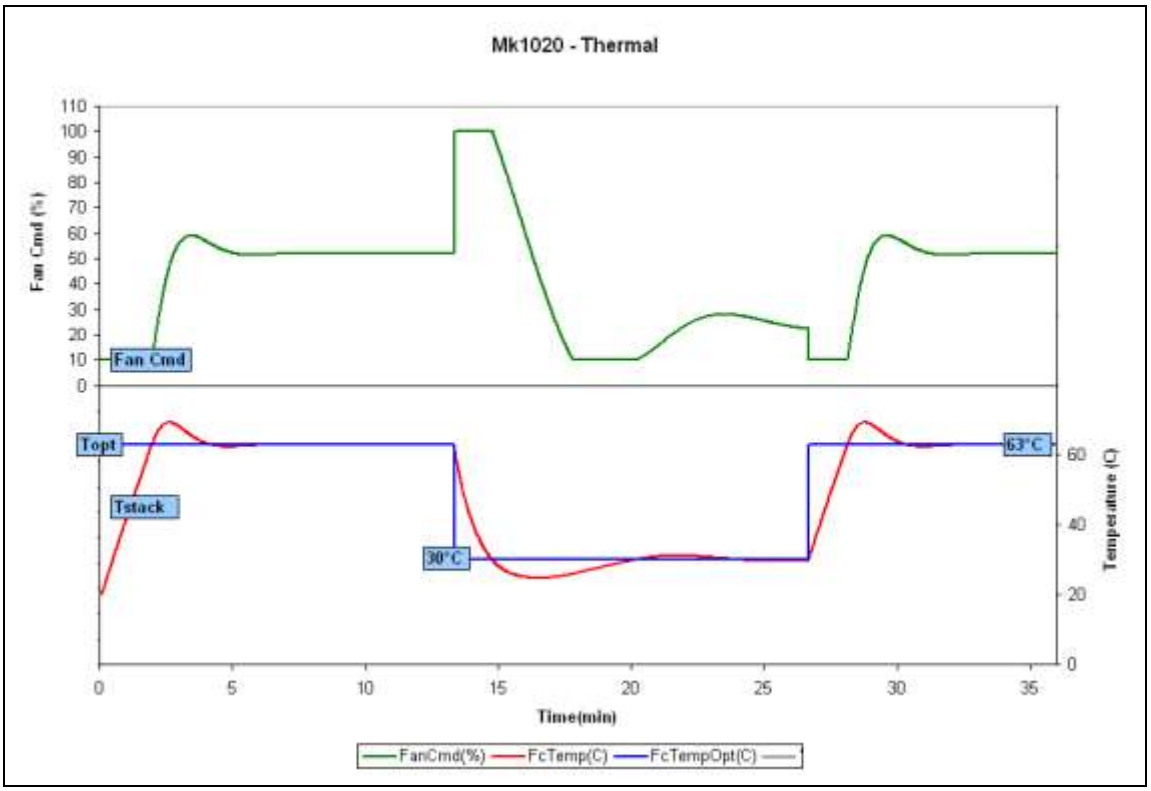


Figure 78: 1020ACS Thermal – 56 cell stack

See Section 4.3, Page 54 for more information on the coolant stream.

8.5 Shutdown

Shutdown takes the stack from *Run* and puts it into *Standby.*

8.5.1 Shutdown Procedure

Actuator	Cmd	Notes
Fan:	*See Below*	
Supply Valve:	Closed	
Purge Valve:	Closed	
Stack Contactor:	Open	
Stack Current:	0	

8.5.2 Shutdown – After Shutdown Anode Pressure

1. After *Shutdown*, the anode pressure reduces as the hydrogen is consumed. The H_2 gas is consumed through two mechanisms: 1) H_2 diffusing across the membrane and reacting with air on the cathode or 2) air diffusing across the membrane and reacting with H_2 on the anode.
2. As the H_2 gas is consumed, the anode pressure will drop from the 0.36 barg operating pressure down to an anode pressure as low as -0.80 barg. After all the H_2 gas is consumed, air diffuses into the membrane and the pressure equilibrates at atmospheric pressure.

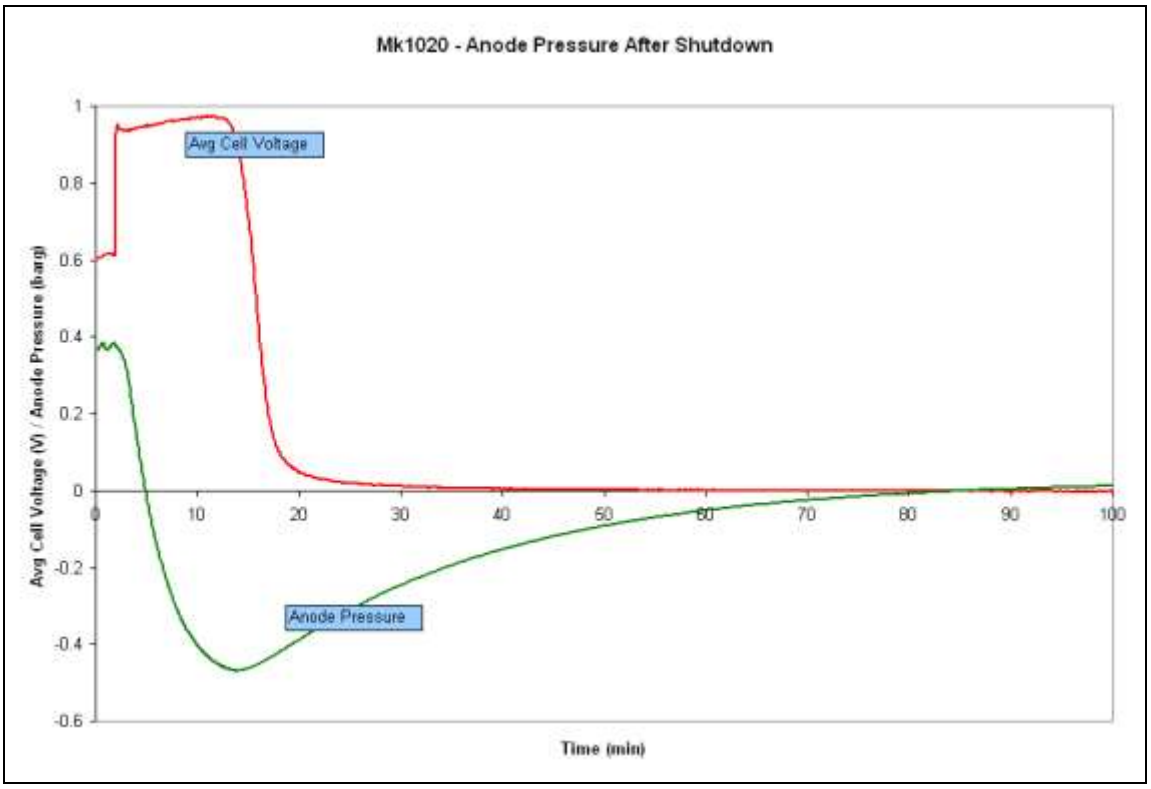


Figure 79: The Anode Pressure After a Shutdown

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8.5.3 Shutdown – After Shutdown Cell Voltage

1. After *Shutdown*, H_2 gas is still present on the anode. Since no load is being drawn, the stack will go to OCV, or ~1 V/cell.
2. The H_2 gas is consumed through two mechanisms: 1) H_2 diffusing across the membrane and reacting with air on the cathode or 2) air diffusing across the membrane and reacting with H_2 on the anode.
3. Once all of the H_2 gas has been consumed, air slowly diffuses through the membrane until the anode has the same composition and pressure as the ambient air.
4. Depending on the system set-up, and if there are leaks in the stack or anode plumbing, it will take 15 to 25 minutes for all the H_2 gas to be consumed and the stack voltage to drop to close to 0 V.
5. A bleed-down resistor or the power electronics should ***NOT*** be used to bleed off the cell voltage after shutdown.
6. Fuel leaks or plumbing leaks can speed up the bleed down process. Plumbing leaks should always be avoided, and fuel leaks will increase as the stack ages. *See Section 5.7, Page 72 for more information on fuel leaks.*
7. Mixing air and hydrogen can lead to corrosion currents that cause irreversible performance loss, as described in *Section 7.0, Page 92*. Plumbing leaks and fuel leaks will increase the severity of the corrosion currents.
8. The figure below is an example of cell voltages during a shutdown.

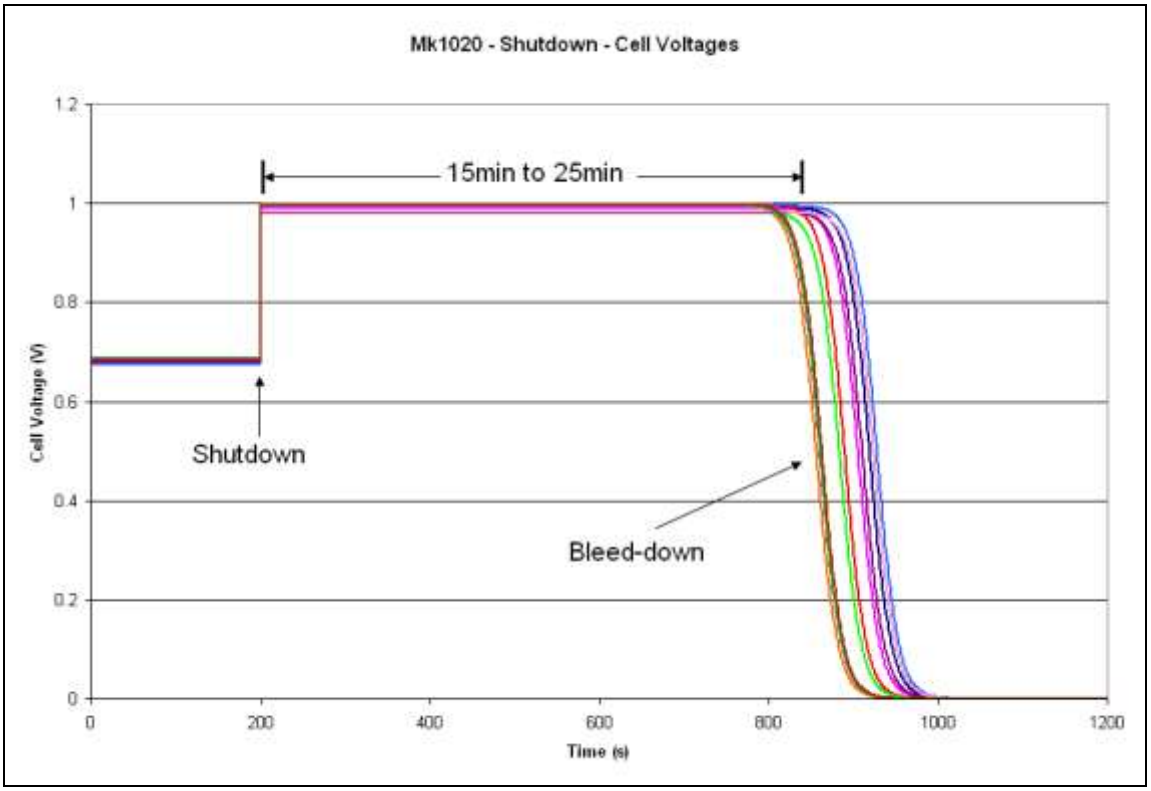


Figure 80: Cell Voltages During Shutdown

8.5.4 Shutdown – Supply Valve

1. The supply valve must be able to open and seal against:
   - **A.** An *upstream* pressure of ~0.36 barg.
   - **B.** A *downstream* pressure ranging from -0.80 barg to 0.36 barg.
2. The upstream pressure is the pressure-regulator outlet pressure. The downstream pressure is the stack anode pressure, which can range from the pressure-regulator outlet pressure to the anode-after-shutdown pressure. *See Section 5.6.3, Page 70 for more information on supply valves.*
3. If the supply valve leaks, a continuous stream of H_2 gas will leak into the anode even after the stack has been shutdown. The continuous presence of H_2 gas on the anode will result in corrosion currents, permanently damaging the stack. *See Section 7.6, Page 96 for carbon corrosion currents caused by a leaky supply valve.*
4. The presence of a leaky supply valve can be detected if the stack voltage does not eventually drop to 0V after a shutdown. *See Section 8.7.3, Page 128 for more information on leaky supply valve alarms.*

8.5.5 Shutdown – Purge Valve

1. The supply valve must be able to open and seal against:
   - **A.** An *upstream* pressure ranging from -0.80 barg to 0.36 barg.
   - **B.** A *downstream* pressure of 0 barg.
2. The upstream pressure is the stack anode pressure, which can range from the pressure-regulator outlet pressure to the anode-after-shutdown pressure. The downstream pressure is atmospheric pressure. *See Section 5.6.4, Page 71 for more information on purge valves.*
3. If the purge valve leaks, the vacuum created by the consumption of the H_2 gas after shutdown will suck air into the anode until the anode pressure equilibrates with atmospheric pressure. The presence of a pocket of air in an otherwise air-filled anode will cause corrosion currents, permanently damaging the stack. *See Section 7.5, Page 96 for carbon corrosion currents caused by a leaky purge valve.*
4. If the stack voltage drops too quickly after a shutdown, this could indicate a leaky purge valve. This could also indicate a fuel leak, a plumbing leak, or current being drawn from the stack after shutdown. *See Section 8.7.4, Page 128 for more information on leaky purge valve alarms.*

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8.5.6 Shutdown – Fan

1. The stack fan should be commanded to 80 to 100% of total flow with the objective of force cooling the stack. When setting appropriate fan speed, performance and fan durability criteria should be taken into account.
2. During this forced cooling, the fan should be commanded to turn off after the stack temperature has reached 20°C or the ambient temperature if it is greater than 20°C. A time limit on the forced cooling can also be set, as this procedure should only take 3 to 5 minutes after shutdown.
3. The forced cooling of the stack is done in order to minimize dry-out after shutdown of the stack. This improves robustness for subsequent start-ups.

WARNING **After shutdown, there is still voltage on the stack. The stack should be treated as being electrically live even after the reactant flow has been stopped.**

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8.6 Air-Starves and Current Pulses

Air-starves and current pulses are used to improve stack performance by improving catalyst performance. Air-starves and current pulses are recommended if:

1. The application includes significant periods of storage or standby.
2. The application requires the stack to deliver rated power in the first few minutes of operation.
3. The application requires operation of the stack in environments with significant air contamination or extended periods of operation at high cell voltage (>0.7 V/cell.)

8.6.1 Air-Starves and Current Pulses - Theory

1. Performance of the fuel cell depends on the chemical activity of the surface of the catalyst layer. The cathode catalyst layer surface activity can be decreased after adsorbing contaminants from the ambient environment, or by formation of a layer of platinum oxides. This results in a loss of fuel cell performance. The platinum oxide layer is formed if current is not drawn from the stack for a significant period of time, or if the stack is operated at high cell voltages (>0.7 V/cell) for extended periods.
2. The performance loss due to platinum oxide layer formation and some types of contaminants is recoverable by lowering the cell voltages below 0.6 V, causing a reduction reaction on the surface of the cathode catalyst. This removes the catalyst oxide layer and/or reducible contaminants and "reactivates" the catalyst. Lowering the cell voltage can be practically achieved by either running the stack at a high current density or by briefly starving the stack of oxygen.
3. Oxygen-starving the stack can be achieved in two ways:
   A. **Air-Starve -** Removing or reducing the air flow while continuing to draw current from the stack.
   B. **Current Pulse -** Drawing a brief, very high load from the stack. This is usually accomplished by briefly short-circuiting the stack. The high current does not allow O_2 gas enough time to diffuse to the catalyst layer, briefly air-starving the stack. A current pulse is a mini air-starve. Because the current pulse is brief, the stack will not fuel-starve because there is enough high-pressure H_2 to complete the anode reaction.
4. During an air-starve or a current pulse, the stack does not deliver any electrical energy. However, heat is created. This will increase the stack cooling flow requirement which might be challenging to the cooling system at extreme conditions.
5. Oxygen-starving the stack should not cause damage the stack. On the anode, the H_2 gas splits into protons and electrons. The protons pass through the proton-exchange-membrane and the electrons travel through the external circuit. However, on the cathode, since there is no oxygen, the protons and electrons react with PtO_X to form water, removing the oxide layer. If PtO_X is not available, the protons recombine to form H_2 gas. Unlink fuel-starvation, carbon material in the cell does not react and is not damaged. *See Section 7.7, Page 97 for more information on fuel starvation.* The air starve reaction is shown schematically in the figure below.

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Cathode Reaction

$$PtO_x + 4H^+ + 4e^- \rightarrow Pt + H_2O$$

or

$$4H^+ + 4e^- \rightarrow H_2$$

Anode Reaction

$$2H_2 \rightarrow 4H^+ + 4e^-$$

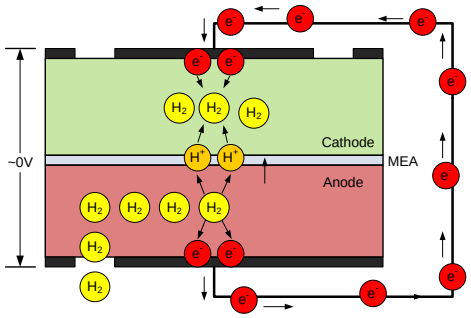


Figure 81: Air-Starve Reaction

8.6.2 Air-Starves and Current Pulses – Notes

1. Air-starves and current pulses improve stack performance. Appropriate situations to air-starve or pulse the stack are:
   - **A. After an extended period of stack inactivity -** Air-starves and current pulses can reduce the cathode catalyst oxide layer and increase the stack performance. This decreases the time to bring the stack up to rated power after an extended period of inactivity. *See Section 6.5.1.2, Page 85 for more information on non-operational-performance-loss (NOPL).*
   - **B. After a contamination event -** If the cathode is contaminated, an air-starve or current pulse might be able to reduce and remove the contaminant, recovering some or all of the performance loss.
   - **C. After Start-Up –** Air-starving or current pulsing the stack after start-up can temporarily improve performance, reducing the bridging energy requirements.
2. Excessive air-starving or current pulsing may lead to a reduction in stack life.
3. Air starves and current pulses generate heat and can increase the temperature of the fuel cell stack. Care should be taken to ensure that stack temperature does not exceed the operating limits outlined in *Section 4.3.2*.
4. Customers should evaluate their current pulsing strategy to ensure that it delivers a net performance benefit. The stack is not producing power during the current pulse. Pulse strategy efficiency should be determined by integrating total power production during operation, including time when stack producing no power output during a pulse and the improved performance between pulses. In some cases, long, frequent current pulses can lead to a net decrease to overall power output and/or fuel efficiency.
5. Optimum current pulse duration and frequency depend on the application and environment. Highly contaminated environments may lead to a long/frequent pulse strategy as optimum, whereas lab conditions may favour short pulses at longer intervals.

8.6.3 Air-Starves and Current Pulses – Pros and Cons

Air-Starve

Advantages

1. More effective than current-pulsing.
2. No additional hardware required if the system uses power electronics that are capable of drawing current close to a 0V input voltage.
3. The risk of pulling cells into reversal is less with than air-starve than a current pulse.

Disadvantages

1. Requires 15 s to 20 s, and during this time, the stack can deliver limited power, increasing bridging energy requirements.
2. The power electronics must be capable of drawing current at ~0 V for to be effective.
3. A power-resistor can be used, but it is not as effective as power electronics because the current reduces with voltage. *See Section 3.4.4.1, Page 45.*

Current Pulse

Advantages

1. Requires 100 ms to 1500 ms per pulse, resulting in minimized energy storage requirements.
2. If the system uses power electronics, and if the power electronics are capable of an extremely high current, low voltage pulse, no additional hardware is required.

Disadvantages

1. Not as effective as air-starving.
2. A current pulse board may be required. See *Section 3.4.4.2, Page 45 for more information on current pulse boards.*
3. There is a higher risk of pulling cells into reversal with a current pulse compared to an air-starve.

WARNING

Current pulses should not be attempted unless the stack is adequately hydrated due to the risk of cell reversals. Under worst-case conditions, these reversals can result in severe stack damage.

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8.6.4 Air-Starve - Procedure

1. This procedure outlines how to air-starve the stack by using the power electronics to draw a load from the stack when the fan is turned off. For this procedure to be successful, the power electronics must be able to draw a load with the stack voltage near 0 V.
2. The stack can be air-starved if the stack is in *Warm-Up* or *Run*, and the stack should be run for a period of time before an air-starve to ensure that the membrane is adequately hydrated. Air-starves are most effective when the membrane is hydrated.
3. Low stack voltage and low cell voltage alarms need to be disabled during an air-starve. *See Section 8.7, Page 126 for more information on stack alarms.*
4. After an air-starve or current pulse, there is an immediate jump in performance. Over time, the performance gain will reduce, but the performance will remain higher than before the air starve. The figure below shows stack performance before and after an air starve.

Step	Actuator	Cmd	Notes
1	**Fan:** **Supply Valve:** **Purge Valve:** **Stack Contactor:** **Stack Current:**	| *Same as Warm-Up or Run* |	■ Air-starves can be performed in either Warm-Up or *Run.* ■ The stack should be run for a period of time before an air-starve to ensure adequate membrane hydration.
	Start air-starve…		
2	**Fan:** **Supply Valve:** **Purge Valve:** **Stack Contactor:** **Stack Current:**	0% Open Closed Closed 15 A	■ Turn off the fan and wait for it to stop spinning. ■ The stack current should be set to at least 15 A. ■ After the fan has stop spinning, the oxygen is then totally consumed, and the cell voltages will drop.
	After the average cell voltage has dropped to ~0 V…		
3	**Fan:** **Supply Valve:** **Purge Valve:** **Stack Contactor:** **Stack Current:**	0% Open Closed Closed 15 A	■ Once the cell voltage has dropped to ~0 V, maintain the 15 A current for 10 s. The stack current output will decrease during the air-starve and the stack temperature will increase due to the lack of coolant flow. Do not allow the stack temp to exceed T_{max}.
	After 10 seconds…		
4	**Fan:** **Supply Valve:** **Purge Valve:** **Stack Contactor:** **Stack Current:**	| *Same as Warm-Up or Run* |	■ After the air-starve is finished, turn the fan back on.
	Transition to back to Warm-Up or Run		

8.6.5 Air-Starve - Figures

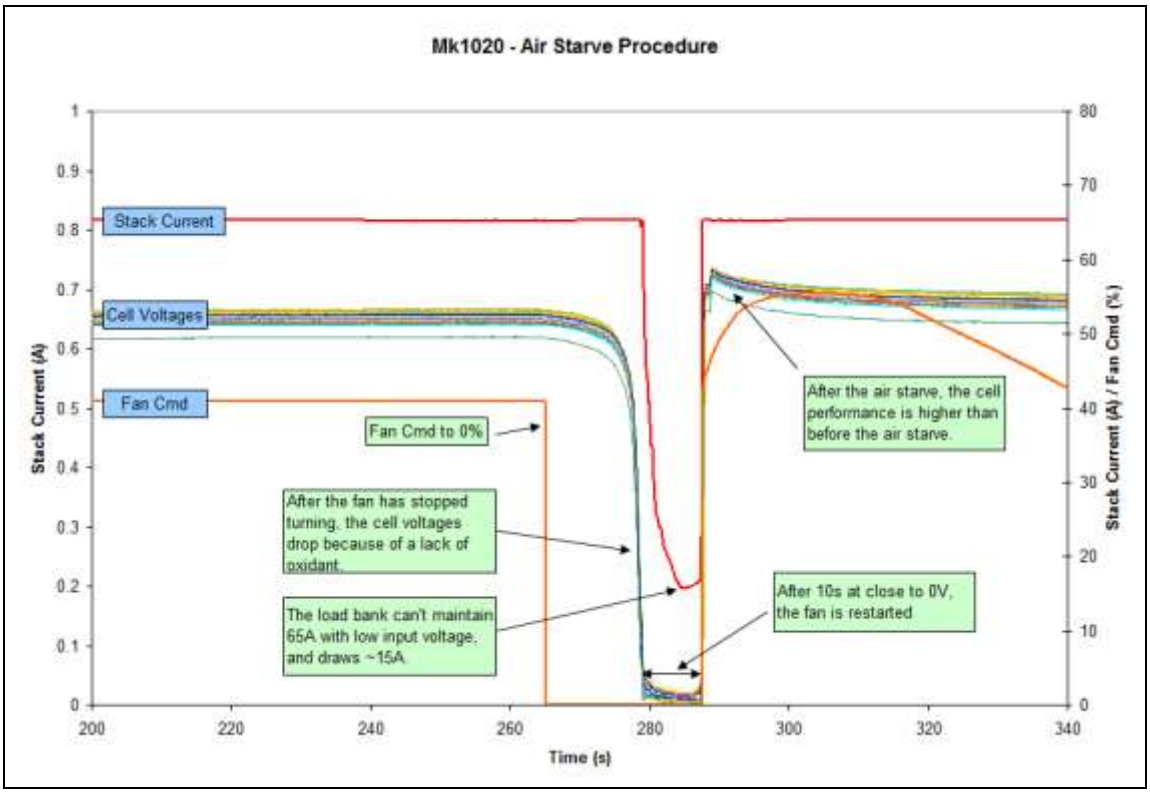


Figure 82: Air Starve Procedure

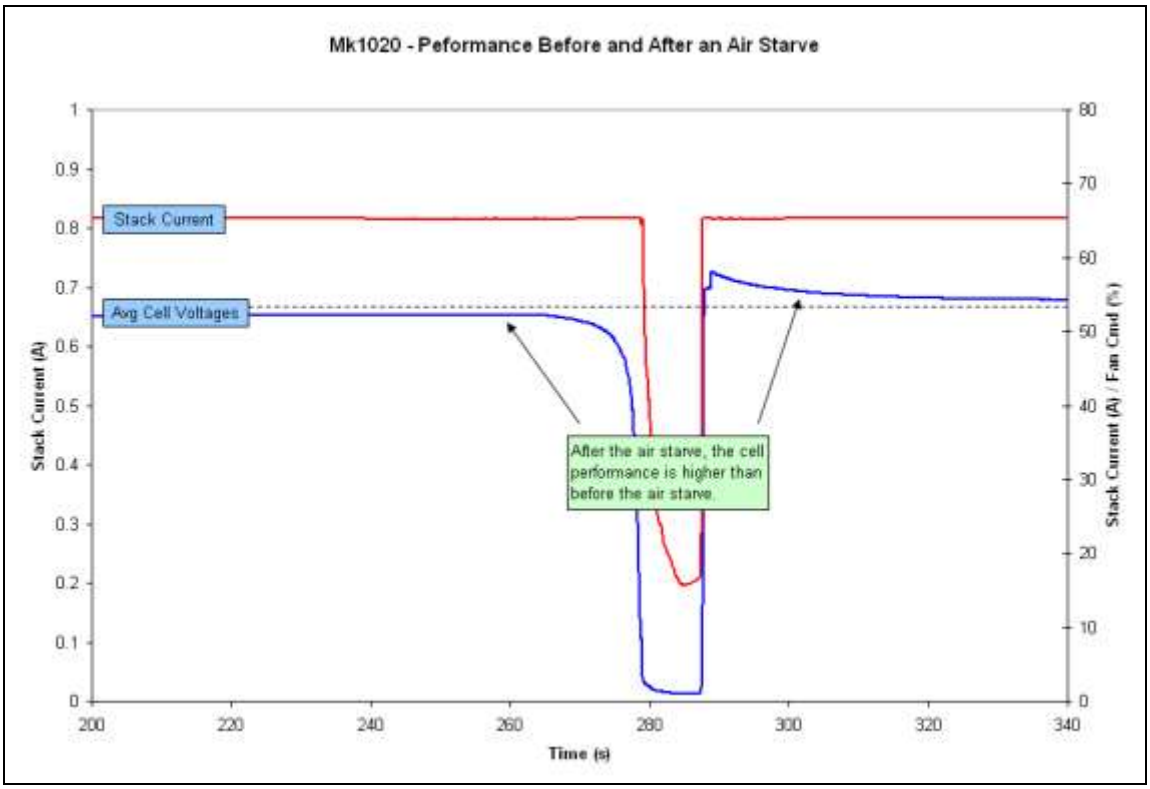


Figure 83: Stack Performance Before and After an Air-Starve

WARNING

Current pulses should not be attempted unless the stack is adequately hydrated due to the risk of cell reversals. Under worst-case conditions, these reversals can result in severe stack damage.

8.6.6 Current Pulse - Procedure

1. This procedure outlines how to current pulse the stack using a current pulse board. The current pulse board briefly shorts the stack, resulting in a large, brief stack current. *More information on the current pulse board can be found in Section 3.4.4.2, Page 45.* Contact Ballard for more information regarding current pulse boards and current pulse board designs.
2. Ballard has experience with current pulses meeting the specifications outlined in the figure below. A sample current pulse profile is also shown in the figure.
3. Current pulsing duration should be at a minimum of 100 ms, but should not exceed 1500 ms. Longer pulses are proportionally more effective.
4. The stack will generally reach rated power after about 3 to 5 current pulses. The first current pulse normally recovers about 70 to 80% of the performance loss.
5. The stack can be current pulsed if the stack is in *Warm-Up* or *Run*.
6. The stack should be run for a period of time greater than 500 Amp·Sec before current-pulsing to ensure the membrane is adequately hydrated.
7. Current pulses should not be applied to the stack if the current stack temperature exceeds the optimum temperature for stack current output. This is required to ensure that the stack is adequately hydrated. *See Section 4.3.2 - Coolant Flow – Optimum Stack Temperature for more information on optimum temperature*.
8. Low stack voltage, low cell voltage, and high stack current alarms will need to be disabled during a current pulse. *See Section 8.7, Page 126 for more information on stack alarms.*

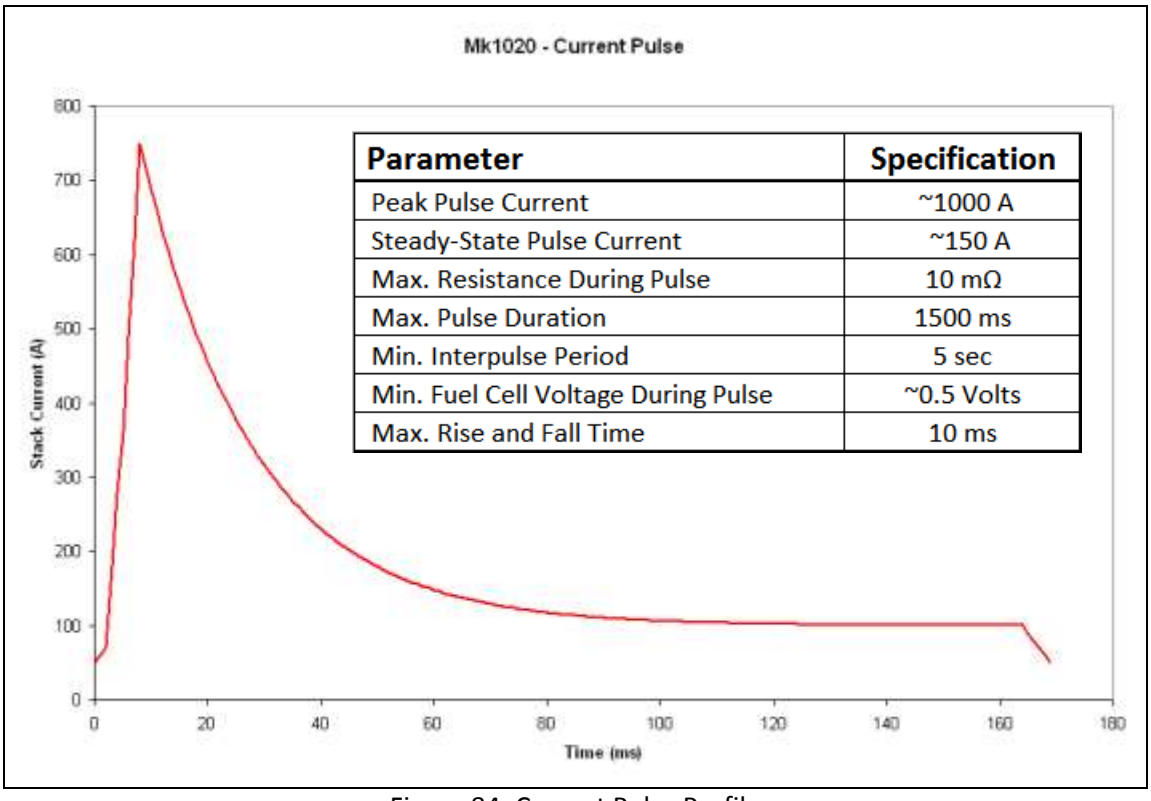
Step	Actuator	Cmd	Notes
1	**Fan:** **Supply Valve:** **Purge Valve:** **Stack Contactor:** **Stack Current:** **Pulse Board:**	| *Same as Warm-Up or Run* | Off	▪ Current pulses can be performed in either *Warm-Up* or *Run.* ▪ The stack should be run for greater than 500 Amp·Sec before a current pulse to ensure adequate membrane hydration.
	Start current pulse…		
2	**Fan:** **Supply Valve:** **Purge Valve:** **Stack Contactor:** **Stack Current:**	*Run* Open Closed Closed 0A	▪ Short the stack with the current pulse board. ▪ Care needs to be taken to ensure the pulse board does not adversely interact with the power electronics or other system components.

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	Pulse Board:	On
	After 100 ms – 1500 ms	
3	**Fan:** **Supply Valve:** **Purge Valve:** **Stack Contactor:** **Stack Current:** **Pulse Board:**	| *Same as* *Warm-Up* *or Run* | Off
	Transition to back to Warm-Up or Run	

WARNING **Current pulses should not be attempted unless the stack is adequately hydrated due to the risk of cell reversals. Under worst-case conditions, these reversals can result in severe stack damage.**

8.6.7 Current Pulse - Figures



Parameter	Specification
Peak Pulse Current	~1000 A
Steady-State Pulse Current	~150 A
Max. Resistance During Pulse	10 mΩ
Max. Pulse Duration	1500 ms
Min. Interpulse Period	5 sec
Min. Fuel Cell Voltage During Pulse	~0.5 Volts
Max. Rise and Fall Time	10 ms

Figure 84: Current Pulse Profile

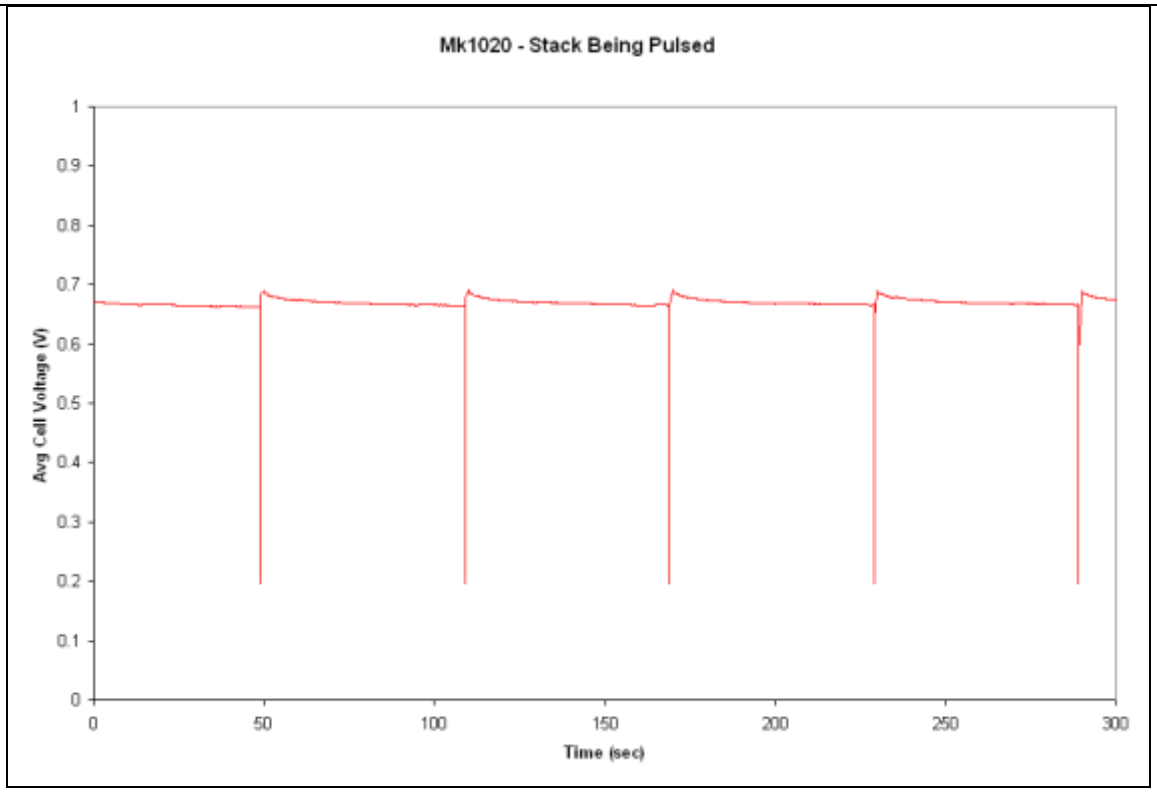


Figure 85: Stack Being Pulsed

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8.7 Stack Alarms

1. The stack shutdown alarm values are presented below. If the stack enters an alarm condition, executing a shutdown is necessary to avoid severe stack damage and irreversible performance degradation.
2. Additional system alarm values (low H_2 storage pressure, fan failure, etc.) will be required to safely operate the system.

8.7.1 Stack Alarm Values – Temp, Pressure, Current, Voltage

Alarm	Type	Value	Delay	Notes
Stack Temperature	Max	75°C	5 s	■ *See Note 1* ■ *See Section 4.3.2, Page 55*
	Max	*See Note 1*	60 s	
	Min	-10°C	5 s	
	Min	*See Note 1*	60 s	
Stack Thermistor	Differential	10°C	5 s	■ *See Note 2*
Anode Pressure	Max	0.69 barg	1 s	■ *See Section 5.2.1, Page 62*
	Max	0.56 barg	5 s	
	Min	0.16 barg	5 s	
	Min	0.07 barg	1 s	
Stack Current	Max	78 A	2 s	■ *See Note 3* ■ *See Section 6.1.1, Page 75*
	Min	-3 A	2 s	
Stack Voltage	Max	1.05 V/cell	1 s	■ *See Section 6.1.2, Page 75 for more information on stack voltage*
	Min	*(See Section 8.7.2)*		
Cell Voltage	Min	0.1 V	1 s	■ *See Note 4* ■ *See Section 6.1.2, Page 75*
	Min	0 V	0 s	

Note 1: Use the max and min values from *Section 4.3.2, Page 55* for the 60 second alarm values.

Note 2: Only applicable if a flex circuit two thermistor assembly is installed on stack. Use the absolute value of the temperature difference between the two thermistors.

Note 3: The stack current alarms are set to the current range, plus a buffer for sensor error. Applying a negative current to the stack will cause irreversible damage to the stack.

Note 4: Only applicable if a CVM is installed in the system. Cell voltage alarms are used to prevent irreversible damage caused by cell reversals, and potentially stack fires. CVMs are recommended for test systems and initial prototypes.

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8.7.2 Stack Alarm Values – Low Stack Voltage

Threshold: The stack voltage is greater than 10 V below the expected stack voltage for a given load

Delay: 1 s

1. In a system without a CVM, the stack voltage can be used to indirectly detect the presence of a deep cell reversal.
2. The stack voltage is compared to the expected stack voltage for a given load. If the stack voltage is more than 10 V below the expected curve, the stack should be shutdown after a 1 s delay.
3. The figure below is an example of the alarm threshold for a 56 cell stack.
4. Although not as effective or sensitive as using a CVM, the stack voltage can be used to detect significant reversals. Test data to date has shown that reversals of -10 V or greater are required for plate-to-plate shorting and cell reversals.
5. If the ambient RH is very low, during start-up the stack performance might be depressed enough to trigger the alarm. In this situation, none of the cells would be in reversal, but the summation of the performance loss experienced by each cell would be greater than 10 V.
6. This technique is less sensitive for high cell-count stacks compared to low cell-count stacks.

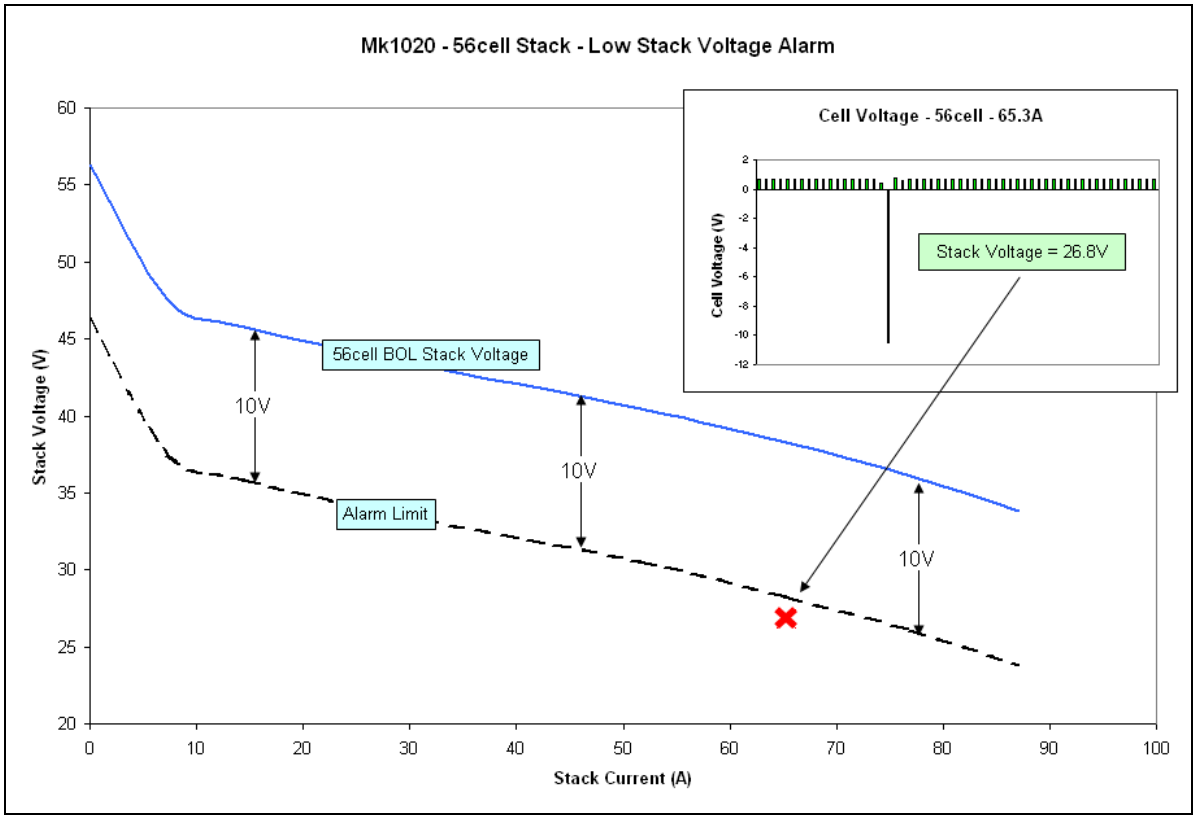


Figure 86: Low Stack Voltage Alarm for a 56 cell Stack

8.7.3 Stack Alarm Values – Leaky Supply Valve

Threshold: Average Cell Voltage greater than 0.1 V

Delay: 45 minutes

1. 45 minutes after a shutdown, the H_2 gas in the anode should be consumed and air should have diffused into the anode.
2. If there is still voltage on the stack after a shutdown, this is an indication of the supply valve leaking H_2 gas from upstream of the stack into the anode. The continuous stream of hydrogen into an otherwise air-filled anode causes permanent damage to the stack. *See Section 7.6, Page 96 for more information on performance degradation caused by a leaky supply valve.*
3. If this alarm is active, the stack can be safely operated. However, service personal should be notified to fix the problem.

8.7.4 Stack Alarm Values – Leaky Purge Valve

1. If the stack voltage drops too quickly after a shutdown, this could indicate a leaky purge valve. However, this could also indicate a fuel leak, a plumbing leak, or current being drawn from the stack after shutdown.
2. Anode leaks are detected using the fuel leak alarm, and service personal will need to diagnose if the leak is in the stack, plumbing, or purge valve. *See Section 5.7.6, Page 74 for more information on fuel leak detection.*
3. Air leaking into an otherwise H_2 filled anode causes permanent damage to the stack. *See Section7.5, Page 96 for more information on performance degradation caused by a leaky purge valve.*
4. If this alarm is active, the stack can be safely operated. However, service personal should be notified to fix the problem.

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8.8 Stack Variables to Log

1. Ballard recommends that the system should be capable of logging stack variables when the stack is operating. Logging stack data is useful because it enables the system integrator to determine how the stack is performing in the application.
2. Additionally, logging data is one of the requirements for a customer to receive an extended warranty on the stack. Contact Ballard for more information.

There are three classes of stack variables that can be logged.

3. **Basic –** Basic variables require minimal memory allocation. Logging basic variables is straight forward to implement.
4. **Mission Averages –** Stack variables are averaged between a start-up and a shutdown and the averages are logged. Calculating mission averages and logging multiple missions is more difficult to implement than storing basic variables, and the memory requirements are much larger than for basic variables.
5. **Run Time Data –** Stack variables are logged continuously at fixed logging period. Memory requirements for logging fixed variables are many orders of magnitude larger than for basic variables or mission averages.

All systems should log *Basic* variables. Some applications will require that *Mission Averages* and *Run Time Data* also be logged.

8.8.1 Stack Variables to Log – Basic

Basic variables require a single, permanent memory location for each variable.

Variable to Log	Threshold	Notes
1. Total Stack Hours	**Value:** Stack Current **Threshold:** Greater than 5 A	▪ Time should be added to *Total Stack Hours* when the stack current is above 5 A.
2. Total Starts	**Value:** Stack Current **Threshold:** Greater than 5 A	▪ A *Start* occurs if the stack current increases through 5 A.
3. Total Air-Air Starts	**Value1:** Stack Current **Threshold1:** Greater than 5 A **Value2:** Avg. Cell Voltage **Threshold2:** Less than 0.1 V/cell	▪ An *Air-Air Start* requires that the stack current increases through 5 A ***AND*** the average cell voltage be less 0.1 V/cell.

Note: Thresholds values can be modified if required by the application. 5 A was chosen because the stack current must pass through 5 A when transitioning from *Standby* to *Run*.

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8.8.2 Stack Variables to Log – Mission Averages

Mission Averages require multiple, permanent memory location for each variable. Each of the variables is logged between every start-up and shutdown. After the shutdown, the variables are average and stored in memory.

Variables to Log	
	1. Average Stack Current
	2. Average Cell Voltage
	3. Average Stack Temperature
	4. Average Ambient Temperature
	5. Average Anode Pressure

8.8.3 Stack Variables to Log – Run Time Data

Run-Time Data requires permanent memory for each variable for each time step

Variables to Log	
	1. Stack Current
	2. Stack Voltage
	3. Individual Cell Voltages
	4. Stack Temperature
	5. Ambient Temperature
	6. Anode Pressure
	7. Fan Command
	8. Supply Valve
	9. Purge Valve

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9.0 Stack Sizing

1. Stack sizing instructions and information will be presented in future revisions of the manual.

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10.0 Shipping and Storage

10.1 Preparation for Storage

1. The 1020ACS does not require any special operational procedures for storage. A normal shutdown, as described in *Section 8.5, Page 114*, is all that is required.
2. The standard shipping container can be used to protect the stack from dust and contaminates during storage.

10.2 Shipping and Storage Ambient Environment

Ambient Temperature Range	**1.** The 1020ACS stack was designed for a shipping and storage temperature range of –40°C to +70°C.
Freeze-Thaw Cycles	**1.** Testing at the Ballard lab has shown that the 1020ACS stack is able to withstand at least 473 freeze thaw cycles without irreversible performance loss or an increase in the fuel leak rate. More cycles may be possible. **2.** Each freeze-thaw cycles consisted of 2 hours at 20°C, 1 hour to transition to -40°C, 2 hrs at -40°C, and 1 hour to transition back up to 20°C. **3.** After multiple freeze-thaw cycles, performance recover techniques such as air starves and current pulses may be required to remove any Non-Operating Performance Loss (NOPL). *See Section 8.6, Page 118 for more information on air staves and current pulses, and Section 6.5.1.2, Page 85 for more information on NOPL.*
High Ambient Temperature Storage Limits	**1.** Storage at high temperatures will accelerate Non-Operating Performance Loss (NOPL). If the 1020ACS stack is stored at higher temperatures for longer than the durations shown below, special start-up procedures will be required.
Ambient Humidity Range	**1.** The 1020ACS stack was designed for a shipping and storage ambient humidity range of 0% RH to 100% RH non-condensing.

Storage Temp Range	Maximum Storage Duration
3°C to 30°C	2 years
30°C to 50°C	1 month
50°C to 70°C	3 days

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10.3 Shipping and Storage Shock and Vibration

1. Designed to withstand normal shipping shock and vibration in standard Ballard packaging. *See Section 2.4.9, Page 26 for more information on shock and vibration.*

10.4 Shipping and Storage Air Quality

1. When the stack is shipped or stored, the air quality should meet the specification presented in Section 4.2.7, Page 53.

Note: Some jurisdictions require that goods be fumigated before shipment or upon arrival at the destination. If a 1020ACS stack is fumigated, severe stack damage can occur. Stacks should never be fumigated, either when installed in a system or in the original packaging.

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11.0 Troubleshooting

Troubleshooting is broken up into three sections:

1. **Diagnostic Tools** - The collection of measurements and inspections used to troubleshoot issues with the stack and system.
2. **Troubleshooting** - Using the *diagnostic tools* to diagnose problems with the stack.
3. **Contacting Ballard for Further Assistance** - If the user cannot diagnose the problem with the stack, Ballard can be contacted for further assistance.

11.1 Diagnostic Tools

11.1.1 Visual Inspection

1. Examine the stack to ensure that fuel, electrical, and sensor connections have all been made properly.
2. Smell the stack. A smell like vinegar or curing silicon is normal. Any other strong smells may be a sign that the stack has been exposed to contamination.
3. Examine the stack, looking for the following defects:

A. Cracked plates (The damage in this specific example was caused by a forklift during shipping)	 Figure 87: Stack Damage
B. Plates Out of Alignment (The damage in this specific example was caused by the stack being dropped on a concrete floor)	 Figure 88: Cells Out of Alignment

Visual Inspection (*continued*)

C. Heat Damage to the Edge of the MEA

(The damage in this specific example was caused by cell reversals or over-temperature)



Figure 89: Heat Damage to MEAs

D. Burn Marks on the Bus Terminals

(The damage in this specific example was caused by installing the cable lug on the wrong side of the bus plate)



Figure 90: Burn Marks on the Bus Plates

E. Burn Marks on the Plates

(The damage in this specific example was caused by a severe and prolonged over-temperature)

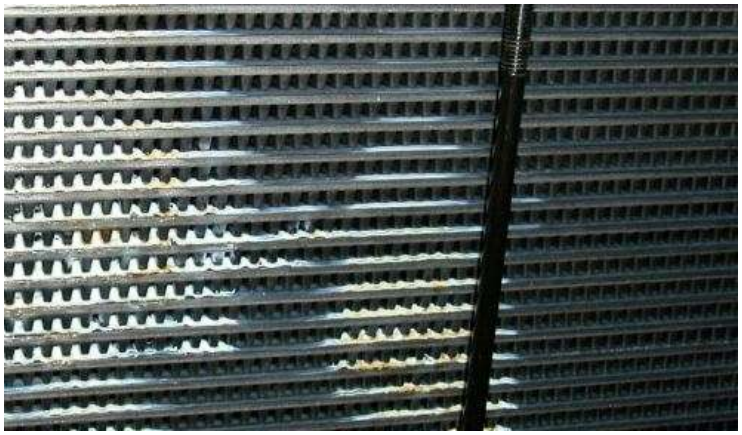


Figure 91: Stack Damage

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11.1.2 Measure Stack Voltage

1. Measure the stack voltage using a voltmeter. The voltage should be measured on the gold plated sides of the bus plates, as shown in the figure below.



Figure 92: Measure Stack Voltage



A large voltage (~1.0 V/cell) exists between the positive and negative bus plates. When measuring the stack voltage, do not electrically short the bus plates or contact any electrically-live stack or system components.

11.1.3 Measuring Stack Current

1. Measure the stack current using a current clamp, as shown in the below figure.



Figure 93: Measure Stack Current



Large voltages (~1.0 V/cell) exist within the stack. When measuring the stack current, do not contact any electrically-live stack or system components.

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11.1.4 Measuring Cell Voltage

1. Manually measuring cell voltages is only required if there is not a Cell-Voltage-Monitor (CVM). *See Section 3.4.2.4, Page 43.* A CVM is strongly recommended for laboratory systems.
2. Measure the individual cell voltages using a voltmeter by measuring the voltage across the bi-polar plates on either side of the cell, as shown in below figure.
3. Make sure that the placement of the multi-meter probes does not cause an electrical plate-to-plate short.



Figure 94: Measure Cell Voltage



Large voltages (~1.0 V/cell) exist within the stack. When measuring the cell voltages, do not contact any electrically-live stack or system components and do not cause a plate to plate short with the voltmeter probes.

11.1.5 Check Fuel Supply

1. Check that the fuel supply is industrial Grade 3.5 hydrogen (99.95% pure). If not, refer to *Section 5.2.7 Page 64* for more information on fuel quality.
2. Verify that the supply pressure is within the pressure-regulator inlet pressure range.
3. Ensure that the supply valve is operating properly.
4. Ensure the anode pressure is close to 0.36 barg (5 psig), as described in *Section 5.2.1, Page 62*.
5. Ensure that with the purge valve and supply valve open, H_2 gas flows through the stack.

11.1.6 Check Fuel Purge

1. Ensure the purge valve is operating properly.
2. With a hydrogen filled anode, verify that the *purge volume* is 20 mL/cell for a 200 ms purge, as described in *Section 5.3.4, Page 67*. The purge volume can be measured as described in *Section 5.3.3, Page 66*.
3. Verify that the *purge interval* is 2300 Amp·Sec, as described in *Section 5.3.4, Page 67.*

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11.1.7 Check Air Supply

1. Verify that the fan is operating properly.

11.1.8 Check Stack Thermistor(s)

1. Verify that the temperature measured by the stack thermistor corresponds to the temperature measured by an external thermocouple. *See Section 3.2.4, Page 37 for more information on the stack thermistor.*
2. The thermocouple must be stainless steel with a maximum 1/16” sheath, as shown in the figure below.
3. The thermocouple must be straight. The stack can be damaged if a bent thermocouple is inserted into the stack.
4. Place a piece of tape 53mm from the tip of the thermocouple, as shown in the picture below.
5. Insert the thermocouple into the air-inlet side of the stack, into an air channel just above the position of the stack thermistor. Insert the thermocouple to the depth of the tape, as shown in the figure below.
6. Compare the thermocouple temperature with the stack thermistor temperature at 2 or more temperatures (ambient temperature and operating temperature).

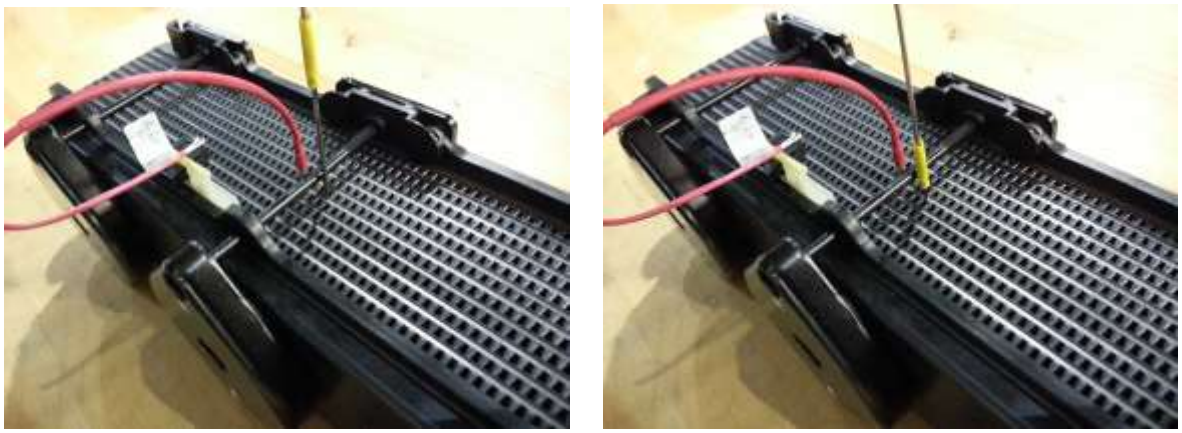


Figure 95: Check Stack Thermistor

11.1.9 Measure Stack Temperature Distribution

1. Verify that the coolant flow is distributed uniformly across the stack by measuring the temperature of the stack air-outlet face. The expected stack temperature is described in *Section 4.3.3, Page 56.*
2. To measure the stack air-outlet face temperature, use the same technique used to verify the stack thermistor temperature, described in *Section 11.1.8, Page 138*, above. The thermocouple should be inserted into air channels in all regions of the stack to create a temperature map of the stack air-outlet face temperature.

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11.1.10 Stack Leak Test

1. The Ballard lab leak test set-up is shown schematically in the figure below.
   - **A. Nitrogen Gas Supply -** Using air instead of N_2 gas will also work
   - **B. Pressure Regulator** -Set to 7.5 psig
   - **C. Pressure Sensor**
   - **D. Mass Flow Meter -** 0 to 50 sccm
   - **E. Ball Valves**
   - **F. Plug** - Plugs the stack anode outlet

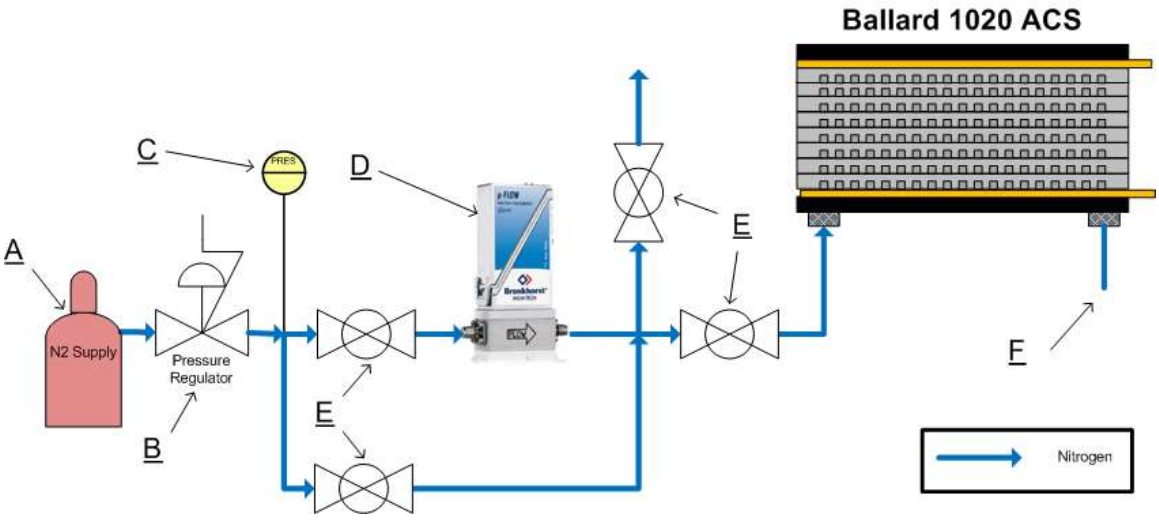


Figure 96: Ballard Lab Leak Test Set-Up

2. The bypass around the mass flow meter allows the stack to be quickly filled with N_2 gas. The mass flow meter restricts flow, which would result in a significant amount of time to fill the stack if N_2 gas can only flow through the mass flow meter.
3. The ball-valve before the stack and the vent ball-valve allow the N_2 gas pressure and flow to be verified without damaging the stack.
4. After the stack is filled with N_2 at 7.5 psig, the leak test procedure calls for:
   1) **Bypass Valve** - Closed
   2) **MFM Valve** - Open
   3) **Vent Valve** - Closed
   4) **Stack Valve** - Open
5. With the valves in the positions described above, any gas that leaks out of the stack must first flow through the mass flow meter, measuring the stack leak rate. The mass flow meter should be given enough time to reach steady-state before a measurement is taken.
6. The stack fuel leak specification can be found in *Section 5.7, Page 72*.

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11.2 Troubleshooting

11.2.1 Problem: Low Stack Voltage at 0A Stack Current

At 0 A stack current, the stack voltage should be ~1.0 V/cell. If this is not the case, the following diagnostics should be performed:

1. **Visual Inspection -** Check for stack damage and system connections.
2. **Check Air Supply -** Air on the cathode is required to produce a stack voltage.
3. **Check Fuel Supply -** H_2 gas on the anode is required to produce a stack voltage.
4. **Measure Stack Voltage -** Verify the stack voltage with a multi-meter to ensure the system data-acquisition is functioning properly.
5. **Measure Stack Current -** Verify that current is not being drawn from the stack using a current clamp. A small current will significantly reduce the stack voltage.
6. **Measure Cell Voltages** - Measure the individual cell voltages to see if they are all ~1.0 V/cell. A cell with a low cell voltage indicates a problem with that specific cell.

11.2.2 Problem: Stack Voltage Crashes When Current is Drawn

If the stack voltage crashes when a small current is drawn, the following diagnostics should be performed:

1. Verify the stack voltage is ~1.0 V/cell at 0 A stack current. If this is not the case, see *Section 11.2.1, Page 140*, above.
2. **Visual Inspection -** Check for stack damage and system connections.
3. **Check Air Supply -** Air on the cathode is required to produce a stack voltage when a load is drawn.
4. **Check Fuel Supply -** H_2 gas on the anode is required to produce a stack voltage when a load is drawn.
5. **Measure Stack Voltage -** Verify the stack voltage with a multi-meter to ensure the system data-acquisition is functioning properly.
6. **Measure Stack Current -** Verify the stack current with a current clamp.
7. **Check Fuel Purge -** Impurities on the anode can cause the stack voltage to crash when a load is drawn.
8. **Measure Cell Voltages** – Measure individual cell voltages to see if specific cell voltages are crashing.
9. **Stack Leak Test** – Fuel leaks can cause the stack to crash when a load is drawn.

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11.2.3 Problem: Stack Performance Unstable

After the stack has warmed-up and run for a period of time, if the current is constant, the stack voltage should change by 1% to 2%. If this is not the case, the following diagnostics should be performed:

1. **Visual Inspection -** Check for stack damage and system connections.
2. **Check Air Supply –** Sufficient air on the cathode is required for stable operation.
3. **Check Fuel Supply -** H_2 gas on the anode is required for stable operation.
4. **Check Fuel Purge –** Either a reduce purge volume or an increased purge interval can result in unstable performance.
5. **Measure Cell Voltages** – With the stack under load, measure individual cell voltages to see if specific cell voltages are unstable or if the problem is uniform across the stack.
6. **Check Stack Thermistor** - Improper stack temperature measurement can lead to unstable operation.
7. **Check Stack Temperature Distribution** – Uneven stack temperature, resulting from non-uniform coolant flow, can lead to unstable stack performance.
8. **Stack Leak Test** – Fuel leaks can cause unstable stack performance.

11.2.4 Problem: Low Stack Performance

The stack performance specified in *Section 6.2.1, Page 76* should be achievable in a customer's system with a BOL stack. If this is not the case, the following diagnostics should be performed:

1. **Visual Inspection -** Check for stack damage and system connections.
2. **Check Air Supply -** Sufficient air on the cathode is required.
3. **Check Fuel Supply -** H_2 gas on the anode is required.
4. **Check Fuel Purge –** Either a reduce purge volume or an increased purge interval can result in reduced performance.
5. **Measure Cell Voltages** - Measure the individual cell voltages to see if there are low cells that cause the reduction in the overall stack voltage.
6. **Check Stack Thermistor** - Improper stack temperature measurement causes non-optimal stack temperature which can lead to reduced stack performance.
7. **Check Stack Temperature Distribution** – Uneven stack temperature, resulting from non-uniform coolant flow, can lead to reduced stack performance.
8. **Stack Leak Test** – Fuel leaks can cause a reduction in stack performance.
9. Stack performance should be measured in the same manner as it is in the Ballard lab. The procedure to generate the polarization curve is described in *Section 6.2.3, Page 78*.
10. **Air-Starve or Current Pulse the Stack** - Catalyst contamination or Non-Operational-Performance-Loss (NOPL) can cause reduced stack performance. *See Section 6.5.1.2, Page 85 for more information on NOPL.* The stack should be current pulsed or air-starved, as described in *Section 8.6, Page 118* to improve performance.

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11.3 Contacting Ballard for Further Assistance

11.3.1 Contacting Ballard

This information presented in “Troubleshooting” (Section *11.2, Page 140,* above), should enable the user to identify simple technical issues with the Ballard 1020ACS fuel cell stack and/or its installation in a system. If further technical assistance is required, please contact Ballard at:

www.ballard.com - “Customer Support” link
(www.ballard.com/members)

11.3.2 Required Stack Information

Before contacting Ballard, the following information should be compiled to assist Ballard with diagnosing the stack issue.

1. Stack part number & serial number
2. Date that the stack was commissioned on site
3. Total run time & start-up cycles
4. Type of system
5. Type of fuel supply
6. Any uncontrolled shutdowns prior to failure

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Appendix A: Useful Fuel Cell Equations

A.1 References

At Ballard, the system-integration reference text book used is:

Fuel Cell Systems Explained by James Larminie and Andrew Dicks (ISBN: 978-0-470-84857-9)

A.2 Overall Fuel Cell Equations

A.2.1 Overall Fuel Cell Equations

Overall fuel cell reaction

$$2H_2 + O_2 \rightarrow 2H_2O \quad \text{(E.1)}$$

Anode reaction

$$2H_2 + \rightarrow 4H^+ + 4e^- \quad \text{(E.2)}$$

Cathode reaction

$$O_2 + 4e^- + 4H^+ \rightarrow 2H_2O \quad \text{(E.3)}$$

Looking at the anode reaction, 2 electrons are transferred for every H_2 molecule. Looking at the cathode reaction, 4 electrons are transfer for every O_2 molecule. Therefore,

$$e^- = 2F \times moles\ of\ H_2 \quad \text{(E.4)}$$

$$e^- = 4F \times moles\ of\ O_2 \quad \text{(E.5)}$$

where $F = 96485\ ^{Coulombs}/_{mole}$. Dividing by time, and multiplying by the number of cells, n,

$$H_2\ Required = \frac{I \cdot n}{2 \cdot F}\ \left(^{moles}/_{s}\right) \quad \text{(E.6)}$$

$$O_2\ Required = \frac{I \cdot n}{4 \cdot F}\ \left(^{moles}/_{s}\right) \quad \text{(E.7)}$$

A.3 Hydrogen Equations

A.3.1 Hydrogen Density

$$P = \rho R T \quad \text{(E.8)}$$

where $R_{H_2} = 4124\ ^{J}/_{kg \cdot K}$

At Standard Conditions, $P = 101325\,Pa$, $T = 0°C$, $\rho_{H_2} = 0.090\ ^{kg}/_{m^3}$

At Anode Outlet Conditions, $P = 101325\,Pa$, $T = 20°C$, $\rho_{H_2} = 0.084\ ^{kg}/_{m^3}$

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A.3.2 Hydrogen Reacted

H_2 *molar mass* $= 2.0158 \times 10^{-3}\ {}^{kg}/_{mole}$

$$H_2\ Reacted = \frac{\left(2.0158 \times 10^{-3}\ {}^{kg}/_{mole}\right)}{2 \cdot \left(96485\ {}^{C}/_{mole}\right)} \cdot I \cdot n \qquad \left({}^{kg}/_{s}\right) \qquad \text{(E.9)}$$

$$H_2\ Reacted = \left(1.04 \times 10^{-8}\ {}^{kg}/_{s}\ {}^{1}/_{A}\ {}^{1}/_{cell}\right) \cdot I \cdot n \qquad \left({}^{kg}/_{s}\right) \qquad \text{(E.9)}$$

$$H_2\ Reacted = \frac{\left(2.0158 \times 10^{-3}\ {}^{kg}/_{mole}\right) \cdot \left(1000\ {}^{L}/_{m^3}\right) \cdot \left(60\ {}^{\min}/_{s}\right)}{2 \cdot \left(96485\ {}^{C}/_{mole}\right) \cdot \left(0.090\ {}^{kg}/_{m^3}\right)} \cdot I \cdot n \qquad (slpm) \qquad \text{(E.9)}$$

$$H_2\ Reacted = \left(0.00696\ slpm\ {}^{1}/_{A}\ {}^{1}/_{cell}\right) \cdot I \cdot n \qquad (slpm) \qquad \text{(E.9)}$$

A.3.3 Hydrogen Purged

$$H_2\ Purged\ per\ purge = n \cdot \left(Purge\ {}^{mL}/_{cell}\right) \cdot \left(0.084\ {}^{kg}/_{m^3}\right) \cdot \left(10^{-6}\ {}^{m^3}/_{mL}\right) \qquad (kg) \qquad \text{(E.10)}$$

A.3.4 Hydrogen Utilization

$$H_2\ Utilization = \frac{H_2\ Reacted}{H_2\ Reacted + H_2\ Purged} \qquad (fraction) \qquad \text{(E.11)}$$

Using $2300\ A{\cdot}s$ and $20\ {}^{mL}/_{cell}$ per purge,

$$H_2\ Reacted\ in\ 2300\ A{\cdot}s = \left(1.04 \times 10^{-8}\ {}^{kg}/_{s}\ {}^{1}/_{A}\ {}^{1}/_{cell}\right) \cdot 2300\ A{\cdot}s \cdot n \qquad (kg) \qquad \text{(E.12)}$$

$$H_2\ Reacted\ in\ 2300\ A{\cdot}s = n \cdot \left(2.39 \times 10^{-5}\ kg\ {}^{1}/_{cell}\right) \qquad (kg) \qquad \text{(E.12)}$$

$$H_2\ Purged\ in\ 2300\ A{\cdot}s = n \cdot \left(1.68 \times 10^{-6}\ kg\ {}^{1}/_{cell}\right) \qquad (kg) \qquad \text{(E.13)}$$

$$H_2\ Utilization = 93.4\% \qquad \text{(E.14)}$$

A.3.5 Stack Thermodynamic Efficiency

$$H_2\ Efficiency = \frac{Stack\ Energy\ Delivered}{H_2\ Consumed} \quad \text{(E.15)}$$

$$Instantaneous\ H_2\ Efficiency = \frac{Stack\ Voltage \cdot Stack\ Current}{H_2\ Reacted + H_2\ Purged} \quad \left(kW{\cdot}hrs/kg\right) \quad \text{(E.16)}$$

$$Overall\ H_2\ Efficiency = \frac{\int (Stack\ Voltage \cdot Stack\ Current)\,dt}{\int (H_2\ Reacted + H_2\ Purged)\,dt} \quad \left(kW{\cdot}hrs/kg\right) \quad \text{(E.17)}$$

A.3.6 Instantaneous Efficiency

Using a $2300\ amp{\cdot}sec$, $20\ mL/cell$ purge, and a $0.683\ V$ cell voltage, over a $2300\ amp{\cdot}sec$ interval,

$$H_2\ Efficiency = \frac{n \cdot (0.683\,V) \cdot (2300\ amp{\cdot}sec)}{\left[n \cdot \left(24.0 \times 10^{-6}\,kg\right) + n \cdot \left(1.7 \times 10^{-6}\,kg\right)\right]} \frac{1}{3600\ s/hr} \frac{1}{1000\ W/kW} = 17.0\ kW{\cdot}hrs/kg \quad \text{(E.18)}$$

A.3.7 Efficiency as a Fraction

Using the lower heating value (LHV), the total energy of H_2 gas is $33.3\ kW{\cdot}hrs/kg$. Therefore,

$$H_2\ Efficiency = \frac{H_2\ Efficiency\ \left(kW{\cdot}hrs/kg\right)}{33.3\ kW{\cdot}hrs/kg} \quad (fraction) \quad \text{(E.19)}$$

A.4 Air Equations

A.4.1 Air Density

$P = \rho R T$ (E.8) where $R_{Air} = 286.9\ J/kg{\cdot}K$

At Standard Conditions, $P = 101325\ Pa$, $T = 0°C$, $\rho_{Air} = 1.29\ kg/m^3$

A.4.2 Oxygen Reacted / Air Required

$$Air\ molar\ mass = 28.97 \times 10^{-3}\ {}^{kg}/_{mole}$$

$$O_2\ \ fraction\ in\ air = 0.21$$

$$Air\ \ Required = \frac{\left(28.97 \times 10^{-3}\ {}^{kg}/_{mole}\right)}{(0.21) \cdot 4 \cdot \left(96485\ {}^{C}/_{mole}\right)} \cdot I \cdot n \qquad \left({}^{kg}/_{s}\right) \quad \text{(E.20)}$$

$$Air\ \ Required = \left(3.57 \times 10^{-7}\ {}^{kg}/_{s}\ {}^{1}/_{A}\ {}^{1}/_{cell}\right) \cdot I \cdot n \qquad \left({}^{kg}/_{s}\right) \quad \text{(E.20)}$$

$$Air\ \ Required = \frac{\left(28.97 \times 10^{-3}\ {}^{kg}/_{mole}\right) \cdot \left(1000\ {}^{L}/_{m^3}\right) \cdot \left(60\ {}^{\min}/_{s}\right)}{(0.21) \cdot 4 \cdot \left(96485\ {}^{C}/_{mole}\right) \cdot \left(1.29\ {}^{kg}/_{m^3}\right)} \cdot I \cdot n \qquad (slpm) \quad \text{(E.20)}$$

$$Air\ \ Required = \left(0.0166\ slpm\ {}^{1}/_{A}\ {}^{1}/_{cell}\right) \cdot I \cdot n \qquad (slpm) \quad \text{(E.20)}$$

A.4.3 Air Stoichiometry

$$Air\ \ Stoich = \frac{Required\ Air\ Flow}{Total\ Air\ \ Flow} \qquad \text{(E.21)}$$

A.4.4 Air Properties (at 1atm)

Temp (°C)	ρ (kg/m^3)	C_p (kJ/kg·K)	k (W/m·K)
0	1.293	1.005	0.0243
20	1.205	1.005	0.0257
40	1.127	1.005	0.0271
60	1.067	1.009	0.0285
80	1.000	1.009	0.0299
100	0.946	1.009	0.0314

A.5 Coolant Equations

A.5.1 Coolant Equation

$$Stack\ \ Heat\ Production = n \cdot I \cdot \left(1.253 - V_{cell}\right) \qquad (W) \quad \text{(E.22)}$$

$$Stack\ \ Heat\ Removed\ by\ Air = \frac{\dot{m} \cdot C_p \cdot \left(T_{stack} - T_{amb}\right)}{1 + 0.403 \cdot \dot{m} \cdot C_p} \qquad (W) \quad \text{(E.23)}$$

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Appendix B: Thermistor Temperature vs. Resistance Curve

Temp. (°C)	Resistance (kΩ)
-40	4397
-35	3089
-30	2197
-25	1582
-20	1151
-15	846.6
-10	629.0
-5	471.6
0	357.0
5	272.5
10	209.7
15	162.7
20	127.1
25	100.0
30	79.22
35	63.17
40	50.68
45	40.90
50	33.20
55	27.09
60	22.22
65	18.32
70	15.18
75	12.64
80	10.57
85	8.873
90	7.481
95	6.337
100	5.384
105	4.594
110	3.934
115	3.380
120	2.916
125	2.522

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